

6-5

1.  If $0 \leq b \leq 2$, for what value of b is $\int_0^b \cos(e^x) dx$ a minimum?

(A) 0

(B) 0.452

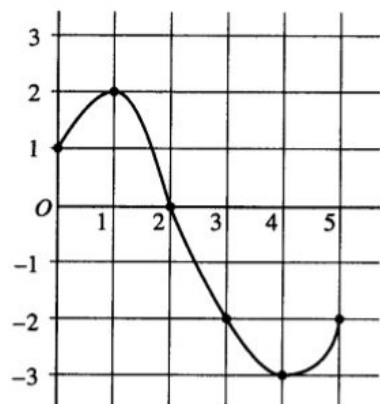
(C) 1.145

(D) 1.550

(E) 2



6-5

Graph of f

Let f be a function whose domain is the closed interval $[0, 5]$. The graph of f is shown above.

Let $h(x) = \int_0^{\frac{x}{2}+3} f(t) dt$.

2. At what x is $h(x)$ a minimum? Show the analysis that leads to your conclusion.

Part C

1 point is earned for correctly identifying the minimum is not in interior of domain

h' is positive, then negative, so minimum is at an endpoint

1 point is earned for correctly stating $h(-6) = 0$

$$h(-6) = \int_0^0 f(t) dt = 0$$

1 point is earned for correctly stating $h(4)$ is negative

$$h(4) = \int_0^5 f(t) dt < 0$$

1 point is earned for the correct answer, $x = 4$

Since the area below the axis is greater than the area above the axis therefore minimum at $x = 4$

Note: Domain $[0, 5]$, max $1/4$, $1 - 0 - 0 - 0$

No interior critical point in student's domain, max $1/4$, $0 - 0 - 0 - 1$



0	1	2	3	4
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6-5

The student response earns four of the following points:

1 point is earned for correctly identifying the minimum is not in interior of domain

h' is positive, then negative, so minimum is at an endpoint

1 point is earned for correctly stating $h(-6) = 0$

$$h(-6) = \int_0^0 f(t)dt = 0$$

1 point is earned for correctly stating $h(4)$ is negative

$$h(4) = \int_0^5 f(t)dt < 0$$

1 point is earned for the correct answer, $x = 4$

Since the area below the axis is greater than the area above the axis therefore minimum at $x = 4$

Note: Domain $[0,5]$, max $1/4$, $1 - 0 - 0 - 0$

No interior critical point in student's domain, max $1/4$, $0 - 0 - 0 - 1$

6-5

3. Find the domain of h .
-

Part A

2 points are earned for the correct endpoints, where one point is earned for the first endpoint and one point is earned for the second endpoint

$< -1 >$ uses strict inequality

$$0 \leq \frac{x}{2} + 3 \leq 5$$

$$-6 \leq x \leq 4$$



0	1	2
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The student response earns two of the following points:

2 points are earned for the correct endpoints, where one point is earned for the first endpoint and one point is earned for the second endpoint

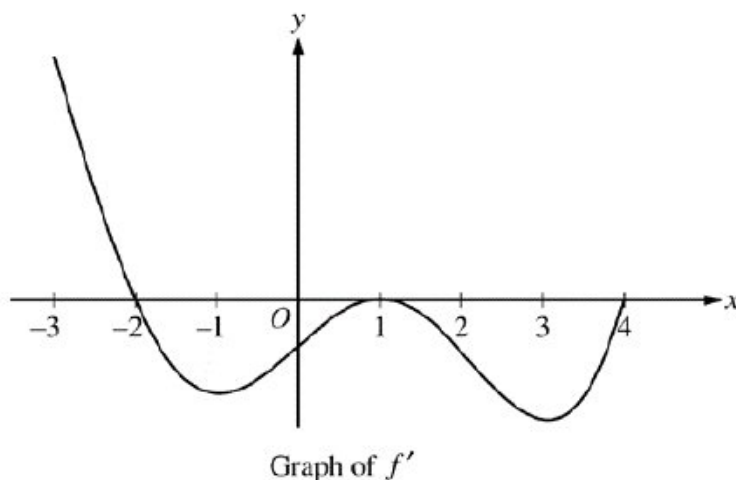
$< -1 >$ uses strict inequality

$$0 \leq \frac{x}{2} + 3 \leq 5$$

$$-6 \leq x \leq 4$$

6-5

4.



The figure above shows the graph of f' , the derivative of a twice-differentiable function f , on the interval $[-3, 4]$. The graph of f' has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x -axis and the graph of f' on the intervals $[-2, 1]$ and $[1, 4]$ are 9 and 12, respectively.

- Find all x -coordinates at which f has a relative maximum. Give a reason for your answer.
- On what open intervals contained in $-3 < x < 4$ is the graph of f both concave down and decreasing? Give a reason for your answer.
- Find the x -coordinates of all points of inflection for the graph of f . Give a reason for your answer.
- Given that $f(1) = 3$, write an expression for $f(x)$ that involves an integral. Find $f(4)$ and $f(-2)$.

Part A

1 point is earned for identifying $x = -2$

1 point is earned for answer with reason

$f'(x) = 0$ at $x = -2$, $x = 1$, and $x = 4$.

$f'(x)$ changes from positive to negative at $x = -2$.

Therefore, f has a relative maximum at $x = -2$.



0	1	2
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The student response earns all of the following points:

1 point is earned for identifying $x = -2$

1 point is earned for answer with reason

$f'(x) = 0$ at $x = -2$, $x = 1$, and $x = 4$.

6-5

$f(x)$ changes from positive to negative at $x = -2$.

Therefore, f has a relative maximum at $x = -2$.

Part B

1 point is earned for the intervals

1 point is earned for the reason

The graph of f is concave down and decreasing on the intervals $-2 < x < -1$ and $1 < x < 3$ because f' is decreasing and negative on these intervals.



0	1	2
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The student response earns all of the following points:

1 point is earned for the intervals

1 point is earned for the reason

The graph of f is concave down and decreasing on the intervals $-2 < x < -1$ and $1 < x < 3$ because f' is decreasing and negative on these intervals.

Part C

1 point is earned for identifying $x = -1, 1$, and 3

1 point is earned for the reason

The graph of f has a point of inflection at $x = -1$ and $x = 3$ because f' changes from decreasing to increasing at these points. The graph of f has a point of inflection at $x = 1$ because f' changes from increasing to decreasing at this point.



0	1	2
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The student response earns all of the following points:

1 point is earned for identifying $x = -1, 1$, and 3

1 point is earned for the reason

6-5

The graph of f has a point of inflection at $x = -1$ and $x = 3$ because f' changes from decreasing to increasing at these points. The graph of f has a point of inflection at $x = 1$ because f' changes from increasing to decreasing at this point.

Part D

1 point is earned for the integrand

1 point is earned for expressing $f(x)$

1 point is earned for $f(4)$ and $f(-2)$

$$f(x) = 3 + \int_1^x f'(t) dt$$

$$f(4) = 3 + \int_1^4 f'(t) dt = 3 + (-12) = -9$$

$$\begin{aligned} f(-2) &= 3 + \int_1^{-2} f'(t) dt = 3 - \int_{-2}^1 f'(t) dt \\ &= 3 - (-9) = 12 \end{aligned}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for expressing $f(x)$

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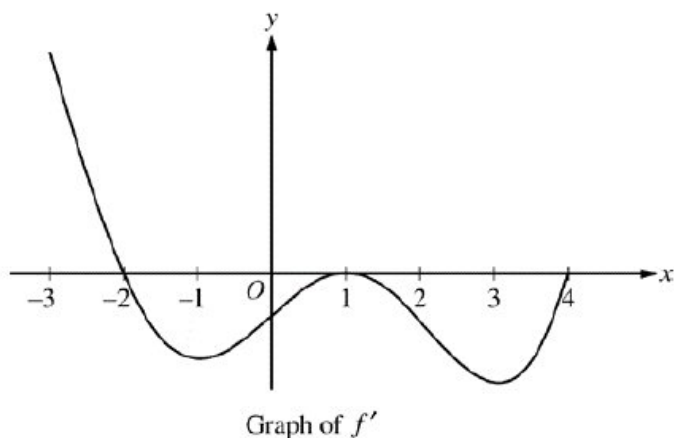
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6-5

5.



The figure above shows the graph of f' , the derivative of a twice-differentiable function f , on the interval $[-3, 4]$. The graph of f' has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x -axis and the graph of f' on the intervals $[-2, 1]$ and $[1, 4]$ are 9 and 12, respectively.

Given that $f(1) = 3$, write an expression for $f(x)$ that involves an integral. Find $f(4)$ and $f(-2)$.

Part D

1 point is earned for the integrand

1 point is earned for expressing $f(x)$

1 point is earned for $f(4)$ and $f(-2)$

$$f(x) = 3 + \int_1^x f'(t) dt$$

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$$= 3 - (-9) = 12$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the integrand

6-5

1 point is earned for expressing $f(x)$

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$$f(x) = 3 + \int_1^x f'(t) dt$$

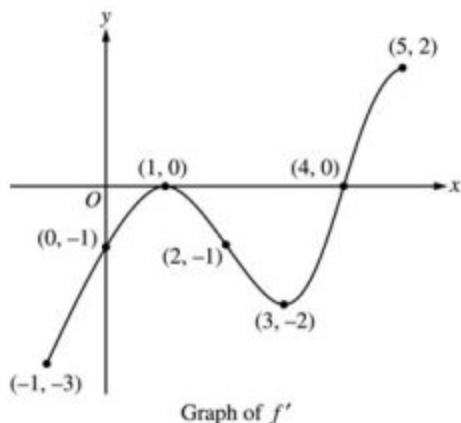
$$f(4) = 3 + \int_1^4 f'(t) dt = 3 + (-12) = -9$$

$$f(-2) = 3 + \int_1^{-2} f'(t) dt = 3 - \int_{-2}^1 f'(t) dt$$

$$= 3 - (-9) = 12$$

6-5

6.



The figure above shows the graph of f' , the derivative of the function f , on the closed interval $-1 \leq x \leq 5$. The graph of f' has horizontal tangent lines at $x=1$ and $x=3$. The function f is twice differentiable with $f(2)=6$.

At what value of x does f attain its absolute minimum value on the closed interval $-1 \leq x \leq 5$? At what value of x does f attain its absolute maximum value on the closed interval $-1 \leq x \leq 5$? Show the analysis that leads to your answers.

Part B

1 point is earned for indicating f decreases then increases

1 point is earned for eliminating $x=5$ for maximum

1 point is earned for absolute minimum at $x=4$

1 point is earned for absolute maximum at $x=-1$

The function f decreases from $x=-1$ to $x=4$, then increases from $x=4$ to $x=5$.

Therefore, the absolute minimum value for f is at $x=4$. The absolute maximum value must occur at $x=-1$ or at $x=5$.

$$f(5) - f(-1) = \int_{-1}^5 f(t) dt < 0$$

Since $f(5) < f(-1)$, the absolute maximum value occurs at $x=-1$.



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for indicating f decreases then increases

1 point is earned for eliminating $x=5$ for maximum

6-5

1 point is earned for absolute minimum at $x=4$

1 point is earned for absolute maximum at $x=-1$

The function f decreases from $x=-1$ to $x=4$, then increases from $x=4$ to $x=5$.

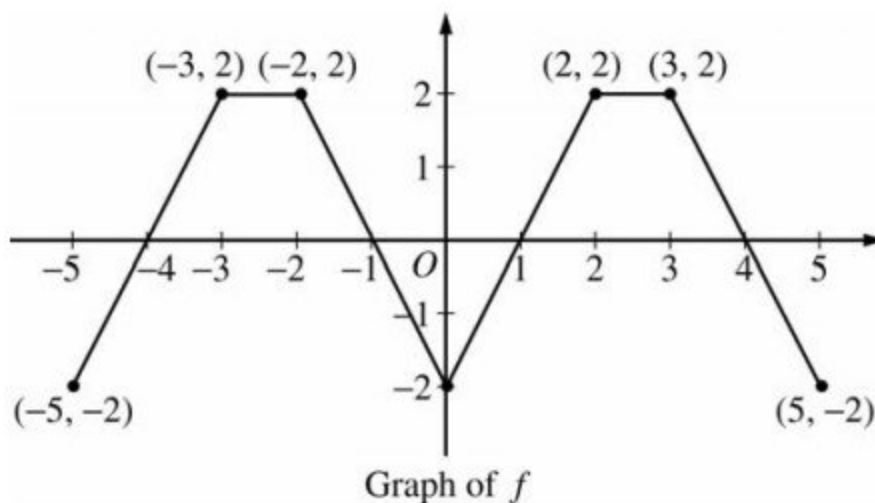
Therefore, the absolute minimum value for f is at $x=4$. The absolute maximum value must occur at $x=-1$ or at $x=5$.

$$f(5) - f(-1) = \int_{-1}^5 f(t) \, dt < 0$$

Since $f(5) < f(-1)$, the absolute maximum value occurs at $x=-1$.

6-5

7.



The graph of the function f shown above consists of six line segments. Let g be the function given by

$$g(x) = \int_0^x f(t) dt.$$

Suppose that f is defined for all real numbers x and is periodic with a period of length 5. The graph above shows two periods of f . Given that $g(5) = 2$, find $g(10)$ and write an equation for the line tangent to the graph of g at $x = 108$.

Part C

1 point is earned for: $g(10)$

1 point is earned for: $g(108)$

1 point is earned for: $g'(108)$

1 point is earned for the equation of tangent line

$g(0) = 0$ and the function values of f increase by 2 for every increase of 5 in x .

$$g(10) = 2g(5) = 4$$

$$\begin{aligned} g(108) &= \int_0^{105} f(t) dt + \int_{105}^{108} f(t) dt \\ &= 21g(5) + g(3) = 44 \end{aligned}$$

$$g'(108) = f(108) = f(3) = 2$$

An equation for the line tangent to the graph of g at $x = 108$ is $y - 44 = 2(x - 108)$.



0	1	2	3	4
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6-5

The student response earns all of the following points:

1 point is earned for: $g(10)$

1 point is earned for: $g(108)$

1 point is earned for: $g'(108)$

1 point is earned for the equation of tangent line

$g(0) = 0$ and the function values of f increase by 2 for every increase of 5 in x .

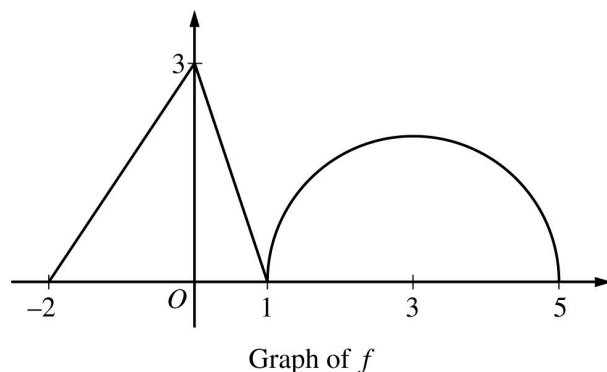
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$$g'(108) = f(108) = f(3) = 2$$

An equation for the line tangent to the graph of g at $x = 108$ is $y - 44 = 2(x - 108)$.

8.



The graph of the function f shown above consists of two line segments and a semicircle. Let g be defined

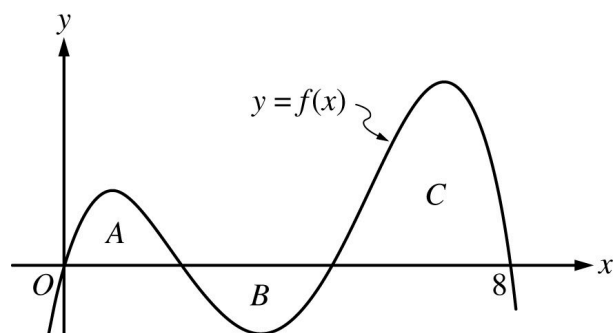
by $g(x) = \int_0^x f(t) dt$. What is the value of $g(5)$?

- (A) 0
- (B) $-1.5 + 2\pi$
- (C) 2π
- (D) $1.5 + 2\pi$
- (E) $4.5 + 2\pi$



6-5

9.



The regions A , B , and C in the figure above are bounded by the graph of the function f and the x -axis. The area of region A is 14, the area of region B is 16, and the area of region C is 50. What is the average value of f on the interval $[0, 8]$?

(A) 6



(B) 10

(C) $\frac{40}{3}$ (D) $\frac{80}{3}$

(E) 48

6-5

10. Let $F(x) = \int_1^{2x} \sqrt{t^2 + t} dt$.

Find the domain of F .

Part B

1 point is earned if indicates domain of integrand

1 point is earned if chooses component of domain of integrand containing 1 and finds all x such that $2x$ is in that component.

$$t^2 + t \geq 0;$$

$$(t \geq 0) \text{ or } (t \leq -1)$$

Since $1 \geq 0$, want $2x \geq 0$ Therefore $x \geq 0$



0	1	2
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The student response earns two of the following points:

1 point is earned if indicates domain of integrand

1 point is earned if chooses component of domain of integrand containing 1 and finds all x such that $2x$ is in that component.

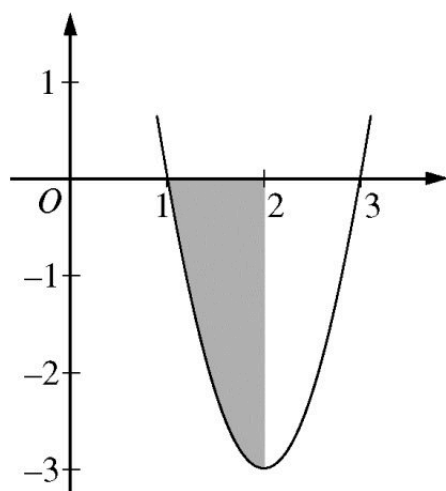
$$t^2 + t \geq 0;$$

$$(t \geq 0) \text{ or } (t \leq -1)$$

Since $1 \geq 0$, want $2x \geq 0$ Therefore $x \geq 0$

6-5

11.

Graph of f

The figure above shows the graph of the function f . If $g(x) = \int_1^x f(t) dt$ and the shaded region has an area of 2, what is the value of $g(2)$?

(A) -3

(B) -2

(C) 0

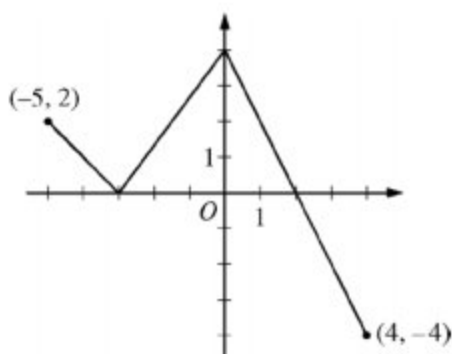
(D) 1

(E) 2



6-5

12.

Graph of f

The function f is defined on the closed interval $[-5, 4]$. The graph of f consists of three line segments and is shown in the figure above. Let g be the function defined by $g(x) = \int_{-3}^x f(t) dt$.

On what open intervals contained in $-5 < x < 4$ is the graph of g both increasing and concave down? Give a reason for your answer.

Part B

1 point is earned for the answer

1 point is earned for the reason

$g'(x) = f(x)$ The graph of g is increasing and concave down on the intervals $-5 < x < -3$ and $0 < x < 2$ because $g' = f$ is positive and decreasing on these intervals.



0	1	2
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The student response earns all of the following points:

1 point is earned for the answer

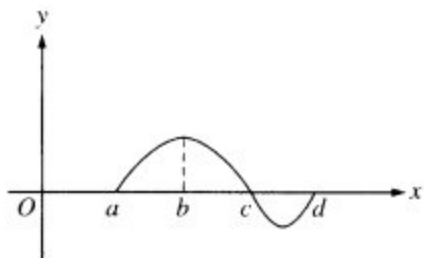
1 point is earned for the reason

$g'(x) = f(x)$

The graph of g is increasing and concave down on the intervals $-5 < x < -3$ and $0 < x < 2$ because $g' = f$ is positive and decreasing on these intervals.

6-5

13.



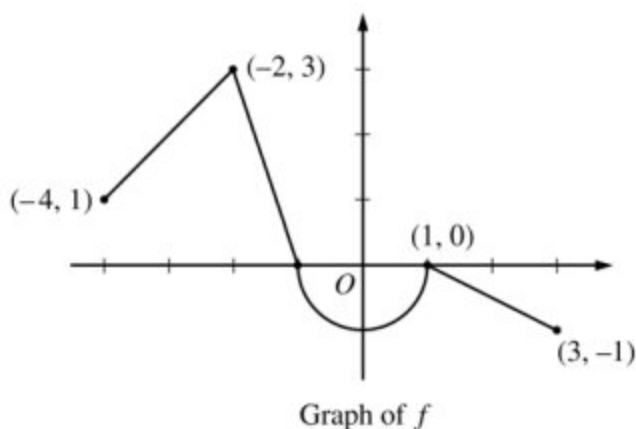
The graph of f is shown in the figure above. If $g(x) = \int_a^x f(t)dt$, for what value of x does $g(x)$ have a maximum?

- (A) a
- (B) b
- (C) c
- (D) d
- (E) It cannot be determined from the information given.



6-5

14.



Let f be the continuous function defined on $[-4, 3]$ whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let g be the function given by $g(x) = \int_1^x f(t) dt$.

Find the x -coordinate of each point at which the graph of g has a horizontal tangent line. For each Graph of f of these points, determine whether g has a relative minimum, relative maximum, or neither a minimum nor a maximum at the point. Justify your answers.

Part C

1 point is earned if considers $g'(x)=0$

1 point is earned for $x=-1$ and $x=1$

1 point is earned for answers with justification

The graph of g has a horizontal tangent line where

$g'(x)=f(x)=0$. This occurs at $x=-1$ and $x=1$. $g'(x)$ changes sign from positive to negative at $x=-1$. Therefore, g has a relative maximum at $x=-1$. $g'(x)$ does not change sign at $x=1$. Therefore, g has neither a relative maximum nor a relative minimum at $x=1$.



0	1	2	3
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The student response earns all of the following points:

1 point is earned if considers $g'(x)=0$

1 point is earned for $x=-1$ and $x=1$

1 point is earned for answers with justification

The graph of g has a horizontal tangent line where

$g'(x)=f(x)=0$. This occurs at $x=-1$ and $x=1$.

6-5

$g'(x)$ changes sign from positive to negative at $x=-1$. Therefore, g has a relative maximum at $x=-1$.

$g'(x)$ does not change sign at $x=1$. Therefore, g has neither a relative maximum nor a relative minimum at $x=1$.

15.

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t=0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t=0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.



For $20 \leq t \leq 25$, the function W that models the water temperature has first derivative given by $W'(t) = 0.4\sqrt{t} \cos(0.06t)$. Based on the model, what is the temperature of the water at time $t=25$?

Part D

1 point is earned for the integrand

1 point is earned for the answer

$$W(t) = 71.0 + \int_{20}^{25} W'(t) dt$$

$$= 71.0 + 2.043155 = 73.043$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the answer

$$W(t) = 71.0 + \int_{20}^{25} W'(t) dt$$

$$= 71.0 + 2.043155 = 73.043$$

6-5

16.  Which of the following is equal to $\int_0^{\pi} \sin x \, dx$?

(A) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \, dx$ ✓

(B) $\int_0^{\pi} \cos x \, dx$

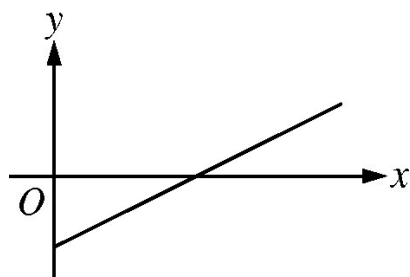
(C) $\int_{-\pi}^0 \sin x \, dx$

(D) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x \, dx$

(E) $\int_{\pi}^{2\pi} \sin x \, dx$

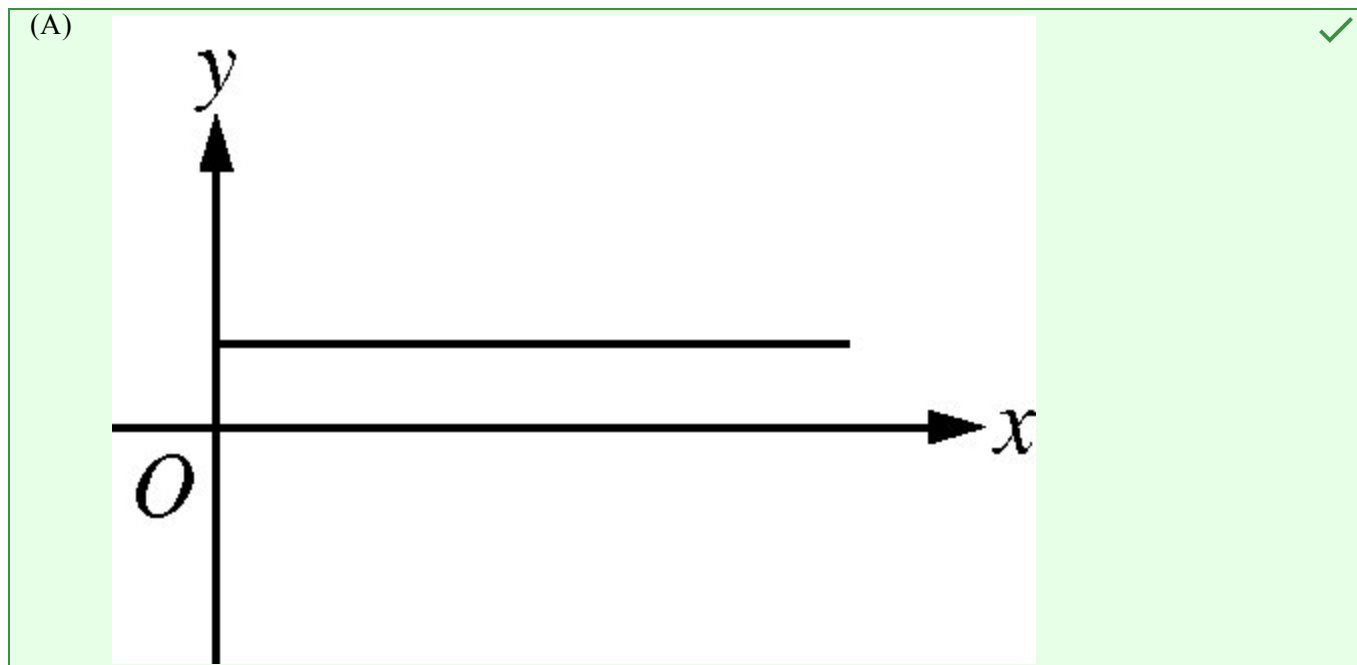
6-5

17.

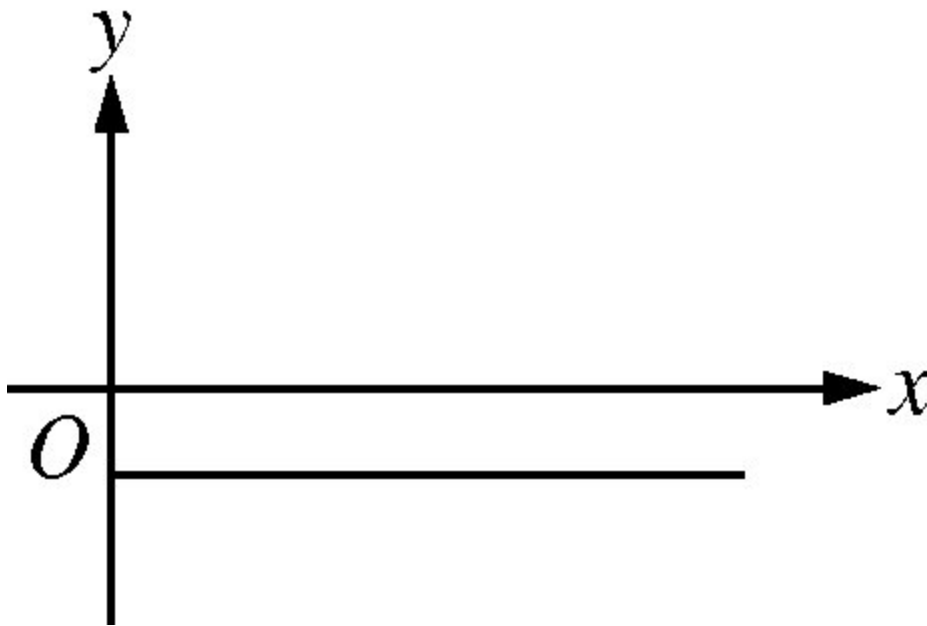
Graph of f

The figure above shows the graph of f . If $f(x) = \int_2^x g(t)dt$, which of the following could be the graph of $y = g(x)$?

6-5

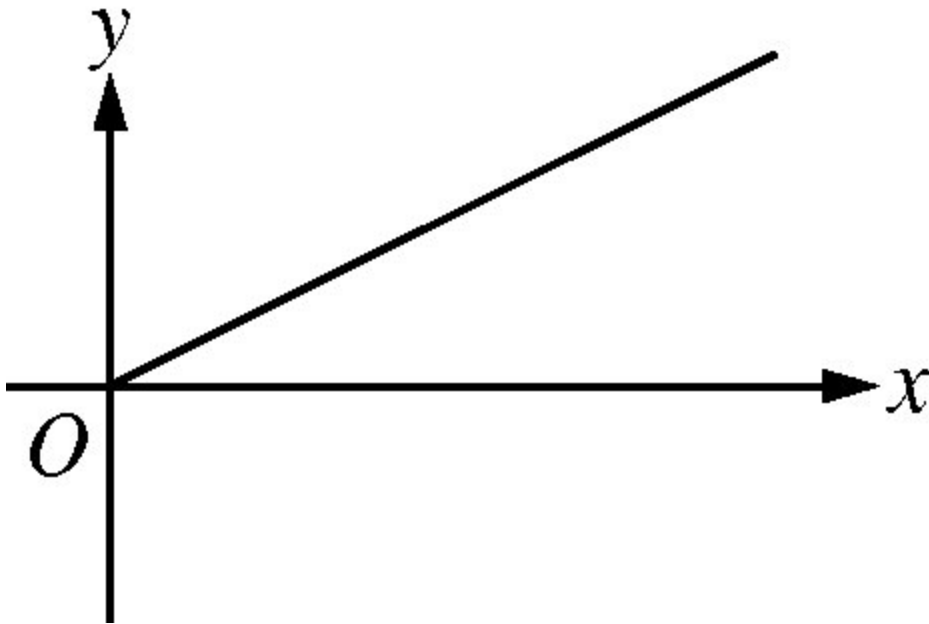


(B)

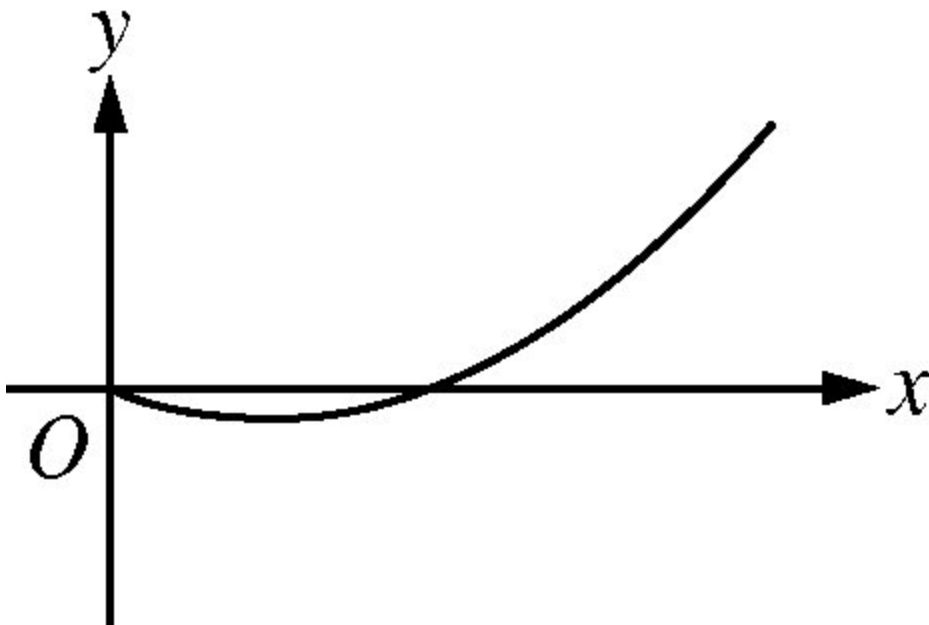


6-5

(C)

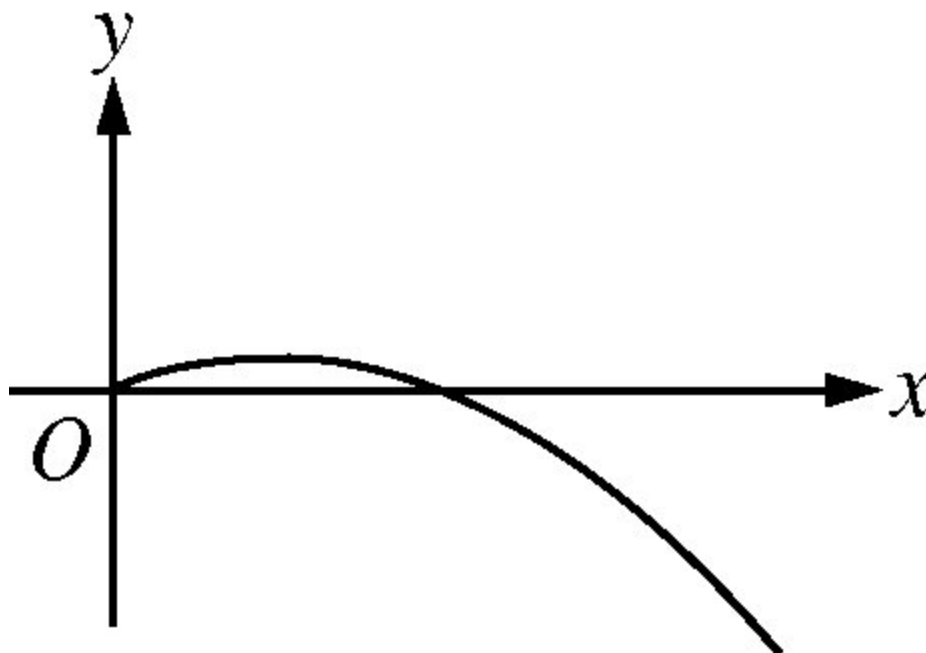


(D)

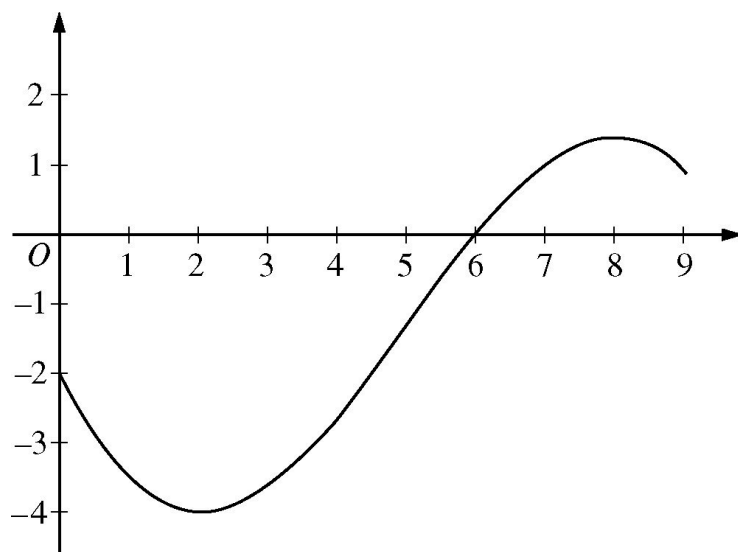


6-5

(E)



18.

Graph of f

The graph of a differentiable function f is shown above. If $h(x) = \int_0^x f(t) dt$, which of the following is true?

(A) $h(6) < h'(6) < h''(6)$



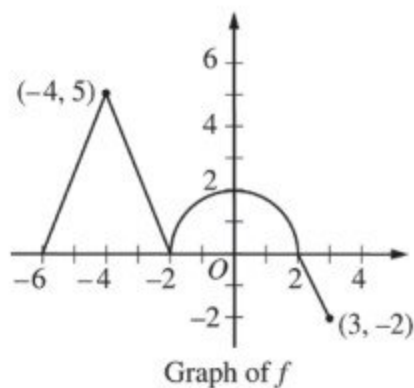
(B) $h(6) < h''(6) < h'(6)$

(C) $h'(6) < h(6) < h''(6)$

(D) $h''(6) < h(6) < h'(6)$

(E) $h''(6) < h'(6) < h(6)$

6-5



The graph of the continuous function f , consisting of three line segments and a semicircle, is shown above. Let g be the function given by $g(x) = \int_{-2}^x f(t) dt$.

19. Find $g'(0)$.

Part B

One point is earned for $g'(0)$

$$g'(0) = f(0) = 2$$



0

1

The student earns all of the following points:

One point is earned for $g'(0)$

$$g'(0) = f(0) = 2$$

6-5

20. Find $g(-6)$ and $g(3)$.**Part A**One point is earned for $g(-6)$ One point is earned for $g(3)$

$$g(-6) = \int_{-2}^{-6} f(t)dt = - \int_{-6}^{-2} f(t)dt = -\frac{1}{2} \cdot 4.5 = -10$$

$$g(3) = \int_{-2}^3 f(t)dt = \frac{1}{2}\pi \cdot 2^2 - \frac{1}{2} \cdot 1.2 = 2\pi - 1$$



0	1	2
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The student earns all of the following points:

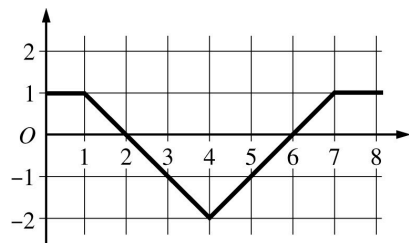
One point is earned for $g(-6)$ One point is earned for $g(3)$

$$g(-6) = \int_{-2}^{-6} f(t)dt = - \int_{-6}^{-2} f(t)dt = -\frac{1}{2} \cdot 4.5 = -10$$

$$g(3) = \int_{-2}^3 f(t)dt = \frac{1}{2}\pi \cdot 2^2 - \frac{1}{2} \cdot 1.2 = 2\pi - 1$$

6-5

21.

Graph of f

The graph of the function f in the figure above consists of four line segments. Let g be the function defined by $g(x) = \int_0^x f(t)dt$. Which of the following is an equation of the line tangent to the graph of g at $x = 5$?

- (A) $y + 1 = x - 5$
(B) $y - 2 = x - 5$
(C) $y - 2 = -1(x - 5)$
(D) $y + 2 = x - 5$
(E) $y + 2 = -1(x - 5)$



6-5

22. If $f'(x) > 0$ for all real numbers x and $\int_4^7 f(t)dt = 0$ which of the following could be a table of values for the function f ?

(A)

x	$f(x)$
4	-4
5	-3
7	0

(B)

x	$f(x)$
4	-4
5	-2
7	5



(C)

x	$f(x)$
4	-4
5	6
7	3

(D)

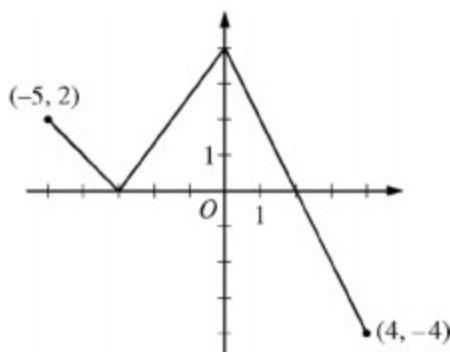
x	$f(x)$
4	0
5	0
7	0

(E)

x	$f(x)$
4	0
5	4
7	6

6-5

23.

Graph of f

The function f is defined on the closed interval $[-5, 4]$. The graph of f consists of three line segments and is shown in the figure above. Let g be the function defined by $g(x) = \int_{-3}^x f(t) dt$.

(a) Find $g(3)$.

(b) On what open intervals contained in $-5 < x < 4$ is the graph of g both increasing and concave down? Give a reason for your answer.

(c) The function h is defined by $h(x) = \frac{g(x)}{5x}$. Find $h'(3)$.

(d) The function p is defined by $p(x) = f(x^2 - x)$. Find the slope of the line tangent to the graph of p at the point where $x = -1$.

Part A

1 point is earned for the answer

$$g(3) = \int_{-3}^3 f(t) dt = 6 + 4 - 1 = 9$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$g(3) = \int_{-3}^3 f(t) dt = 6 + 4 - 1 = 9$$

6-5

Part B

1 point is earned for the answer

1 point is earned for the reason

$$g'(x) = f(x)$$

The graph of g is increasing and concave down on the intervals $-5 < x < -3$ and $0 < x < 2$ because $g' = f$ is positive and decreasing on these intervals.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the reason

$$g'(x) = f(x)$$

The graph of g is increasing and concave down on the intervals $-5 < x < -3$ and $0 < x < 2$ because $g' = f$ is positive and decreasing on these intervals.

Part C2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{5xg'(x) - g(x)5}{(5x)^2} = \frac{5xg'(x) - 5g(x)}{25x^2}$$

$$\begin{aligned} h'(3) &= \frac{(5)(3)g'(3) - 5g(3)}{25 \cdot 3^2} \\ &= \frac{15(-2) - 5(9)}{225} = \frac{-75}{225} = -\frac{1}{3} \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

6-52 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{5xg'(x) - g(x)5}{(5x)^2} = \frac{5xg'(x) - 5g(x)}{25x^2}$$

$$h'(3) = \frac{(5)(3)g'(3) - 5g(3)}{25 \cdot 3^2}$$

$$= \frac{15(-2) - 5(9)}{225} = \frac{-75}{225} = -\frac{1}{3}$$

Part D2 points are earned for $p'(x)$

1 point is earned for the answer

$$p'(x) = f'(x^2 - x)(2x - 1)$$

$$p'(-1) = f'(2)(-3) = (-2)(-3) = 6$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $p'(x)$

1 point is earned for the answer

$$p'(x) = f'(x^2 - x)(2x - 1)$$

$$p'(-1) = f'(2)(-3) = (-2)(-3) = 6$$

6-5

24. Let f be the function defined by $f(x) = x + \int_0^x \sin(t^2) dt$ for $0 < x < \sqrt{2\pi}$. At what value of x is the line tangent to the graph of f horizontal?

(A) $\sqrt{\frac{\pi}{2}}$

(B) $\sqrt{\pi}$

(C) $\sqrt{\frac{5\pi}{4}}$

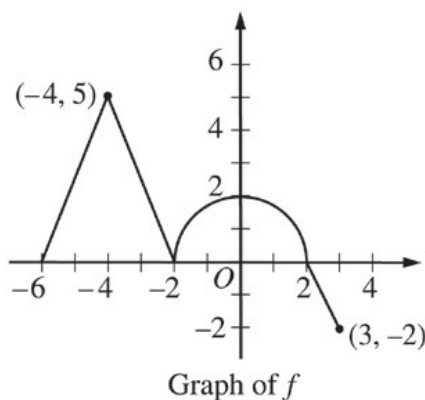
(D) $\sqrt{\frac{3\pi}{2}}$

**Answer D**

Correct. The line tangent to the graph of f will be horizontal at values of x for which $0 = f'(x) = 1 + \sin(x^2)$. In the interval $0 < x < 2\pi$, $\sin(x^2) = -1$ only at $x^2 = \frac{3\pi}{2}$, so $x = \sqrt{\frac{3\pi}{2}}$ is the only solution in the interval $0 < x < \sqrt{2\pi}$.

6-5

25.



The graph of the continuous function f , consisting of three line segments and a semicircle, is shown above. Let g be the function given by $g(x) = \int_{-2}^x f(t) dt$.

- Find $g(-6)$ and $g(3)$.
- Find $g'(0)$.
- Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a horizontal tangent. Determine whether g has a local maximum, a local minimum, or neither at each of these values. Justify your answers.
- Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a point of inflection. Explain your reasoning.

Part A

The response can earn up to 2 points:

1 point is earned for $g(-6)$

$$g(-6) = \int_{-2}^{-6} f(t) dt = - \int_{-6}^{-2} f(t) dt = -\frac{1}{2} \cdot 4 \cdot 5 = -10$$

1 point is earned for $g(3)$

$$g(3) = \int_{-2}^3 f(t) dt = \frac{1}{2}\pi \cdot 2^2 - \frac{1}{2} \cdot 1 \cdot 2 = 2\pi - 1$$



0	1	2
---	---	---

The response earns all of the following points:

1 point is earned for $g(-6)$

6-5

$$g(-6) = \int_{-2}^{-6} f(t) dt = - \int_{-6}^{-2} f(t) dt = -\frac{1}{2} \cdot 4 \cdot 5 = -10$$

1 point is earned for $g(3)$

$$g(3) = \int_{-2}^3 f(t) dt = \frac{1}{2}\pi \cdot 2^2 - \frac{1}{2} \cdot 1 \cdot 2 = 2\pi - 1$$

Part B

The response can earn up to 1 point:

1 point is earned for the correct evaluation of $g'(0)$

$$g'(0) = f(0) = 2$$



0	1
---	---

The response earns all of the following point:

1 point is earned for the correct evaluation of $g'(0)$

$$g'(0) = f(0) = 2$$

Part C

The response can earn up to 3 points:

1 point is earned for the horizontal tangent at $x = -2$ and $x = 2$

The graph of g has a horizontal tangent at $x = -2$ and $x = 2$ where $g'(x) = f(x) = 0$.

2 points are earned for correct answers with justifications

The graph of g has neither a local maximum nor a local minimum at $x = -2$ because $g'(x) = f(x)$ does not change sign at $x = -2$.

The graph of g has a local maximum at $x = 2$ because $g'(x) = f(x)$ changes sign from positive to negative at $x = 2$.



0	1	2	3
---	---	---	---

6-5

The response earns all of the following points:

1 point is earned for the horizontal tangent at $x = -2$ and $x = 2$

The graph of g has a horizontal tangent at $x = -2$ and $x = 2$ where $g'(x) = f(x) = 0$.

2 points are earned for correct answers with justifications

The graph of g has neither a local maximum nor a local minimum at $x = -2$ because $g'(x) = f(x)$ does not change sign at $x = -2$.

The graph of g has a local maximum at $x = 2$ because $g'(x) = f(x)$ changes sign from positive to negative at $x = 2$.

Part D

The response can earn up to 3 points:

2 points are earned for the correct values of x

1 point is earned for the explanation

The graph of g has a point of inflection at $x = -4$, $x = -2$, and $x = 0$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -4$ and $x = 0$, and changes from decreasing to increasing at $x = -2$.

OR

$g''(x) = f'(x)$ changes from positive to negative at $x = -4$ and $x = 0$, and changes from negative to positive at $x = -2$.



0	1	2	3
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The response earns all of the following points:

2 points are earned for the correct values of x

1 point is earned for the explanation

The graph of g has a point of inflection at $x = -4$, $x = -2$, and $x = 0$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -4$ and $x = 0$, and changes from decreasing to increasing at $x = -2$.

OR

$g''(x) = f'(x)$ changes from positive to negative at $x = -4$ and $x = 0$, and changes from negative to positive at $x = -2$.

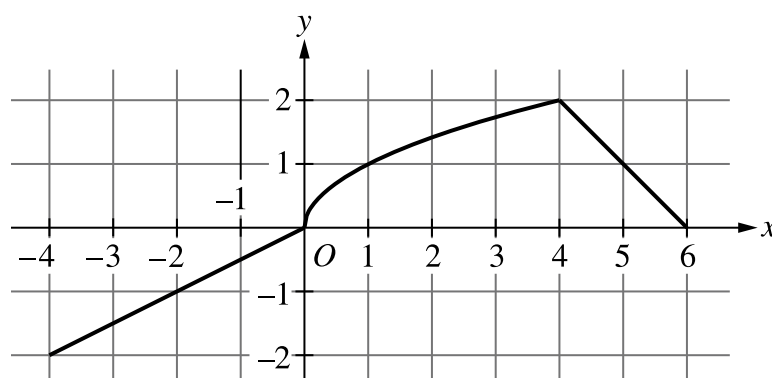
6-5

26. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the function g is shown on the interval $[-4, 6]$. The graph of g consists of two line segments and the portion of the graph of $y = \sqrt{x}$ for $0 \leq x \leq 4$. Let f be the function defined by

$$f(x) = g(x) + \int_{-4}^x g(t) \, dt.$$

(a) Find the value of $f(4)$.

(b) Show that f has a critical point at $x = -1$. Use the Second Derivative Test to determine whether f has a local minimum or a local maximum at $x = -1$. Show the work that leads to your answer.

(c) Find the value of $\lim_{x \rightarrow 4^-} \frac{g(x) - g(4)}{x - 4}$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response does not need to present $\int_{-4}^0 g(t) \, dt$ to earn the first point. The value of ± 4 may be determined directly from the graph.

The inclusion of $g(4)$ is considered in the third point.

6-5

The second point is earned by finding the antiderivative $\frac{2}{3}t^{3/2}$ or by determining the area under this portion of the parabola is $\frac{2}{3} \cdot 8$.

A response that finds an antiderivative of the form $k \cdot t^{3/2}$ where $k \neq 0$ and $k \neq \frac{2}{3}$ does not earn the second point but does earn the third point for the consistent answer of $-2 + k \cdot 8$.

The final answer does not need to be simplified. Any error in simplification does not earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $\pm$ value of $\int_{-4}^0 g(t) dt$
- ☐ Antiderivative of $\sqrt{t}$
- ☐ Answer

Solution:

$$\begin{aligned}
 f(4) &= g(4) + \int_{-4}^4 g(t) dt = 2 + \int_{-4}^0 g(t) dt + \int_0^4 g(t) dt \\
 &= 2 + (-4) + \int_0^4 \sqrt{t} dt = -2 + \left[\frac{2}{3} t^{3/2} \right]_{t=0}^{t=4} \\
 &= -2 + \frac{2}{3} \cdot 8 = \frac{10}{3}
 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is only earned for the statement $f'(x) = g'(x) + g(x)$ or If there is any equating of unequal quantities, or miscommunication, for example, the response does not earn the first point.

If the first point is not earned, the second point can only be earned for a response that presents

If the first point is earned, a response can earn the second point by stating that

To earn the third point, a response must correctly find or and go on to state that

The fourth point can be earned without any of the first 3 points, provided the response declares a nonzero value or sign of and makes a consistent Second Derivative Test argument based on that value or sign.

6-5



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ $f'(x) = g'(x) + g(x)$ OR $f'(-1) = g'(-1) + g(-1)$
- ☐ Shows $f'(-1) = 0$
- ☐ $f''(-1)$
- ☐ Answer with reason

Solution:

$$f'(x) = g'(x) + g(x)$$

$$f'(-1) = g'(-1) + g(-1) = \frac{1}{2} + \left(-\frac{1}{2}\right) = 0$$

Because $f'(-1) = 0$, f has a critical point at $x = -1$.

$$f''(x) = g''(x) + g'(x)$$

$$f''(-1) = g''(-1) + g'(-1) = 0 + \frac{1}{2}$$

Because $f'(-1) = 0$ and $f''(-1) > 0$, the Second Derivative Test indicates that f has a local minimum at $x = -1$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response does not need to state that $\sqrt{x}$ is differentiable.

A minimal response to earn both points is $\frac{1}{2\sqrt{4}}$.

Special case (using L'Hospital's Rule): For $0 \leq x \leq 4$, $g(x) = \sqrt{x}$.

$$\lim_{x \rightarrow 4^-} (\sqrt{x} - g(4)) = 2 - 2 = 0 \text{ and } \lim_{x \rightarrow 4^-} (x - 4) = 0.$$

Because $\sqrt{x}$ is continuous on the interval $[0, 4]$ and differentiable on the interval $(0, 4)$,

$\lim_{x \rightarrow 4^-} [g(x) - g(4)] = 2 - 2 = 0$, and $\lim_{x \rightarrow 4^-} [x - 4] = 4 - 4 = 0$, L'Hospital's Rule may be applied:

6-5

$$\lim_{x \rightarrow 4^-} \frac{g(x) - g(4)}{x - 4} = \lim_{x \rightarrow 4^-} \frac{g'(x)}{1} = \lim_{x \rightarrow 4^-} \frac{\frac{d}{dx} \sqrt{x}}{1} = \lim_{x \rightarrow 4^-} \frac{\frac{1}{2\sqrt{x}}}{1} = \frac{1}{2\sqrt{4}} = \frac{1}{4}.$$

- A response earns the first point by evaluating the limits of the numerator and denominator.
- A response that equates the limit to $\frac{0}{0}$ does not earn the first point but is eligible for the second point. A response may say that the limit is of indeterminate form $\frac{0}{0}$.
- The second point is earned by correctly applying L'Hospital's Rule (by presenting a limit of a ratio of derivatives) and finding the correct answer.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Uses $g(x) = \sqrt{x}$
- ☐ Answer

Solution:

For $0 \leq x \leq 4$, $g(x) = \sqrt{x}$.

Because $\sqrt{x}$ has a left-hand derivative at $x = 4$, $\lim_{x \rightarrow 4^-} \frac{g(x) - g(4)}{x - 4} = \left. \frac{d}{dx} \sqrt{x} \right|_{x=4} = \left. \frac{1}{2\sqrt{x}} \right|_{x=4} = \frac{1}{4}.$

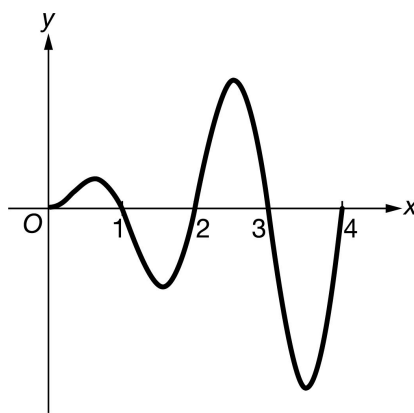
6-5

27. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f'

The figure above shows the graph of f' , the derivative of a twice-differentiable function f , on the closed interval $[0, 4]$. The areas of the regions bounded by the graph of f' and the x -axis on the intervals $[0, 1]$, $[1, 2]$, $[2, 3]$, and $[3, 4]$ are 2, 6, 10, and 14, respectively. The graph of f' has horizontal tangents at $x = 0.6$, $x = 1.6$, $x = 2.5$, and $x = 3.5$. It is known that $f(2) = 5$.

On what open intervals contained in $(0, 4)$ is the graph of f both decreasing and concave down? Give a reason for your answer.

Find the absolute minimum value of f on the interval $[0, 4]$. Justify your answer.

Evaluate $\int_0^4 f(x)f'(x) \, dx$.

The function g is defined by $g(x) = x^3 f(x)$. Find $g'(2)$. Show the work that leads to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

6-5

- ☐ answer
- ☐ reason

Solution:

The graph of f is decreasing and concave down on the intervals $(1, 1.6)$ and $(3, 3.5)$ because f' is negative and decreasing on these intervals.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The second point can be earned without earning the first point. The integrals that lead to the answer must be shown.

✓

0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ considers $x = 2$ as a candidate
- ☐ answer with justification

Solution:

The graph of f' changes from negative to positive only at $x = 2$.

$$f(2) = 5$$

$$f(0) = f(2) + \int_2^0 f'(x) dx = f(2) - \int_0^2 f'(x) dx = 5 - (2 - 6) = 9$$

$$f(4) = f(2) + \int_2^4 f'(x) dx = 5 + (10 - 14) = 1$$

On the interval $[0, 4]$, the absolute minimum value of f is $f(4) = 1$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The third point, for the answer, cannot be earned if no points are earned for the antiderivative.

6-5



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ antiderivative of the form $a(f(x))^2$
- ☐ earned the first point, and $a = \frac{1}{2}$
- ☐ answer

Solution:

$$\begin{aligned}\int_0^4 f(x)f'(x) \, dx &= \frac{1}{2}(f(x))^2 \Big|_{x=0}^{x=4} = \frac{1}{2}((f(4))^2 - (f(0))^2) \\ &= \frac{1}{2}(1^2 - 9^2) = -40\end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The second point requires a correct expression for the product rule; a correct numerical answer without supporting work will not earn the point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ product rule
- ☐ answer, with supporting work

Solution:

$$g'(x) = 3x^2 f(x) + x^3 f'(x)$$

$$g'(2) = 3 \cdot 2^2 f(2) + 2^3 f'(2) = 12 \cdot 5 + 8 \cdot 0 = 60$$

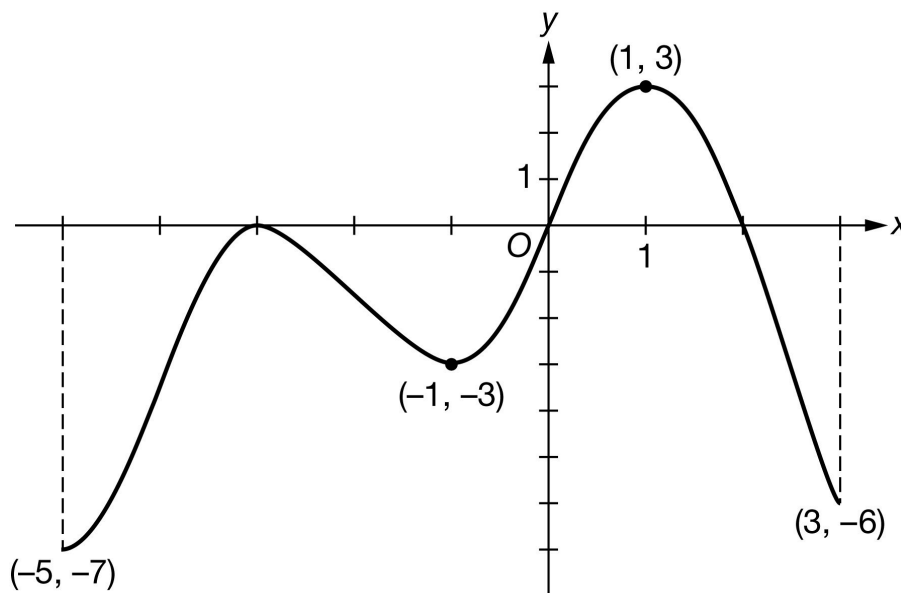
6-5

28. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The figure above shows the graph of h' , the derivative of a twice-differentiable function h , on the closed interval $-5 \leq x \leq 3$. The graph of h' has horizontal tangent lines at $x = -3$, $x = -1$, and $x = 1$. The areas of the regions bounded by the x -axis and the graph of h' on the intervals $[-5, -3]$, $[-3, 0]$, $[0, 2]$, and $[2, 3]$ are 7, 5, 4, and 3, respectively. The function h is defined for all real numbers and satisfies $h(1) = 12$ and $h(3) = 11$.

- Find the x -coordinate of each critical point of h on the open interval $-5 < x < 3$.
- Identify the critical point from part (a) which corresponds to a relative maximum of h . Give a reason for your answer.
- Find the value of h at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute maximum value of h on $-5 \leq x \leq 3$. Justify your answer.
- On what open intervals is h both increasing and concave down? Give a reason for your answer.
- Explain why there must be a value of b , for $1 < b < 3$, such that $h(b) = 11.5$.
- Find $\lim_{x \rightarrow 3^-} \frac{h(x) - 11}{x^2 - 9}$. Give a reason for your answer.

6-5

(h) The function p is defined by $p(x) = (h(x))^2$. Find $p'(1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = 0$, and $x = 2$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-5, 3)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-5, 3)$. The point is still earned provided the three critical points, $x = -3$, $x = 0$, and $x = 2$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(0, 0)$, $(2, 0)$ we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = 0$ at $x = -3$, $x = 0$, and $x = 2$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local maximum as 2 and give the correct reason of “ h' changes from positive to negative” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ h switches from increasing to decreasing at $x = 2$ ”
- Saying that “ $h'(2)$ changes signs from positive to negative”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of h' , the one given in the stem of the problem.

6-5

A response indicating that “ $h'(x)$ is positive for $x < 2$ and negative for $x > 2$ ” is allowed to give this statement as a reason to earn the point.

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

A relative maximum value of h occurs where $h'(x)$ changes from positive to negative values.

$x = 2$ is the only location on $-3 < x < 5$ for which this is true, so $x = 2$ is the only location of a relative maximum value of h on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -5$, $x = -3$, or $x = 0$ is selected as the location of the local maximum in part (b). Note that $h(-5) = 22$, $h(-3) = 15$, and $h(0) = 10$. No other x -coordinates will be read for consistency.

The response $h(2) = h(3) - (-3) = 14$ earns both points.

The response of both $11 - (-3) = 14$ and $h(2) = 14$ does not earn the first point, but earns the second point, provided no other work is presented.

6-5

Special Case: If a response uses $\int_1^2 h'(x) dx = 2$ then

· $h(2) = h(1) + \int_1^2 h'(x) dx = 12 + 2 = 14$ earns both points, because it's possible to find

$\int_1^2 h'(x) dx = 2$ using the areas and initial conditions given.

· any response using perceived symmetry about $x = 1$ can earn at most the first point, for example

$h(2) = h(1) + \int_1^2 h'(x) dx = 12 + \frac{1}{2} \cdot 4 = 14$ earns just the first point.

Special Case: If in part (b), a response picks $x = 1$ or $x = 3$ as the location of the maximum, then there is no work to do in part (c). This response will earn zero points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $h(2) = h(3) + \int_3^2 h'(x) dx$

☐ Answer

Solution:

$$h(2) = h(3) + \int_3^2 h'(x) dx$$

$$= h(3) - \int_2^3 h'(x) dx = 11 - (-3) = 14$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x - value.

For $x = 2$, the response must either state its y -value or reference it as a maximum.

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

6-5

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 1$), and thus the third point cannot be earned.

It is possible to not refer to $h(3) = 11$ by indicating that h' is negative on the interval $(2, 3)$, thus eliminating 11 as a possible absolute maximum.

A complete list of given or computable values for $h(x)$ is given in the table below

x	$h(x)$
-5	22
-3	15
0	10
1	12
2	14
3	11



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 2$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$$\begin{aligned}
 h(-5) &= h(3) + \int_3^{-5} h'(x) \, dx \\
 &= h(3) - \int_{-5}^3 h'(x) \, dx = 11 - (-7 - 5 + 4 - 3) = 22
 \end{aligned}$$

$$h(2) = 14$$

$$h(3) = 11$$

So the absolute maximum value of h on $-5 \leq x \leq 3$ is $h(-5) = 22$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-5

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[1, 2]$, $[1, 2)$, $(1, 2]$, and $(1, 2)$ all earn the first point.

The reason “ h' is decreasing” can be expressed by either “ h'' is negative” or “the slope of h' is negative.”

We will overlook the false implication that $h' > 0$ on $[1, 2]$.

The second point can be earned without earning the first point only if the given answer is a subset of $[1, 2]$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

h is increasing where h' is positive.

h is concave down where h' is decreasing.

h is both increasing and concave down on the open interval $(1, 2)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that h is continuous because h is differentiable or h is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[1, 3]$, or the point is not earned.

The y -values of h need to be mentioned by either referring to the numerical values of 11 and 12, or mentioning $h(1)$ and $h(3)$

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since h is differentiable it’s continuous and 11.5 is between 11 and 12 so such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.

6-5



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

h is differentiable. $\Rightarrow h$ is continuous.

$$h(3) = 11 < 11.5 < 12 = h(1)$$

By the Intermediate Value Theorem, there must be a value of b , for $1 < b < 3$, such that $h(b) = 11.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of h' or h .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $h(3) - 11 = 0$ and $3^2 - 9 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $h(3) = 11$ and $3^2 = 9$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{h(3)-11}{3^2-9} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{0}{0}$ earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

6-5

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{h'(x)}{2x} = \frac{-6}{6}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{0}{0}$ and $\lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} = \frac{h'(x)}{2x}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $-\frac{6}{6}$ or its numeric equivalent.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h is twice differentiable. $\Rightarrow$ both h' and h are continuous. $\Rightarrow \lim_{x \rightarrow 3^-} h(x) = h(3)$ and $\lim_{x \rightarrow 3^-} h'(x) = h'(3)$.

$$\lim_{x \rightarrow 3^-} (h(x) - 11) = h(3) - 11 = 11 - 11 = 0$$

$$\lim_{x \rightarrow 3^-} (x^2 - 9) = 3^2 - 9 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 3^-} \frac{h(x)-11}{x^2-9} &= \lim_{x \rightarrow 3^-} \frac{h'(x)}{2x} \\ &= \frac{h'(3)}{2 \cdot 3} = \frac{-6}{6} = -1 \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $2 \cdot h(x)$ or $2 \cdot h(1)$.

6-5

The chain rule point can be earned by seeing an expression involving h but not h' times $h'(1)$.

All three points will be awarded for seeing only $2 \cdot 12 \cdot 3$, but $24 \cdot 3$, $2 \cdot 36$, or $6 \cdot 12$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$p'(x) = 2h(x) \cdot h'(x)$$

$$p'(1) = 2 \cdot h(1) \cdot h'(1) = 2 \cdot 12 \cdot 3 = 72$$

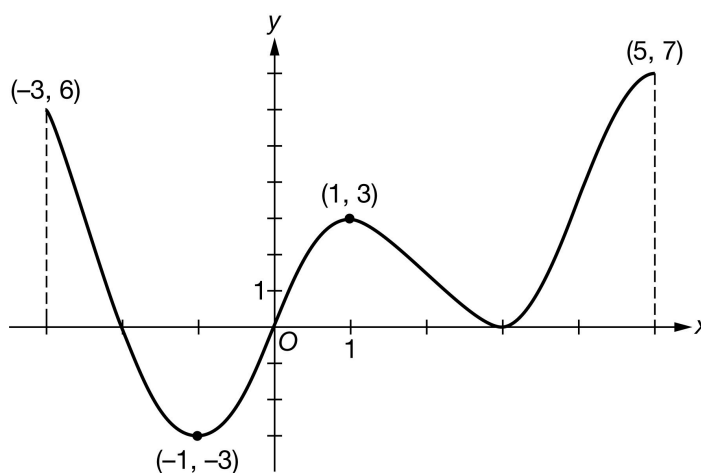
6-5

29. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The figure above shows the graph of g' , the derivative of a twice-differentiable function g , on the closed interval $-3 \leq x \leq 5$. The graph of g' has horizontal tangent lines at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x -axis and the graph of g' on the intervals $[-3, -2]$, $[-2, 0]$, $[0, 3]$, and $[3, 5]$ are 3, 4, 5, and 7, respectively. The function g is defined for all real numbers and satisfies $g(1) = 3$ and $g(5) = 13$.

- Find the x -coordinate of each critical point of g on the open interval $-3 < x < 5$.
- Identify the critical point from part (a) which corresponds to a relative maximum of g . Give a reason for your answer.
- Find the value of g at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute maximum value of g on $-3 \leq x \leq 5$. Justify your answer.
- On what open intervals is g both decreasing and concave up? Give a reason for your answer.
- Explain why there must be a value of a , for $1 < a < 5$, such that $g(a) = 10$.
- Find $\lim_{x \rightarrow 5^-} \frac{g(x) - 13}{3x - 15}$. Give a reason for your answer.
- The function r is defined by $r(x) = (g(x))^3$. Find $r'(1)$. Show the computations that lead to your answer.

Part A

6-5

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -2$, $x = 0$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 5)$. The point is still earned provided the three critical points, $x = -2$, $x = 0$, and $x = 3$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-2, 0)$, $(0, 0)$, $(3, 0)$, we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = 0$ at $x = -2$, $x = 0$, and $x = 3$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local maximum as -2 and give the correct reason of “ g' changes from positive to negative” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ g switches from increasing to decreasing at $x = -2$ ”
- Saying that “ $g'(-2)$ changes signs from positive to negative”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of g' , the one given in the stem of the problem.

A response indicating that “ $g'(x)$ is positive for $x < -2$ and negative for $x > -2$ ” is allowed to give this statement as a reason to earn the point.

6-5

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

A relative maximum value of g occurs where $g'(x)$ changes from positive to negative values.

$x = -2$ is the only location on $-3 < x < 5$ for which this is true, so $x = -2$ is the only location of a relative maximum value of g on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -3$, $x = 0$, or $x = 3$ is selected as the location of the local maximum in part (b). Note that $g(-3) = 2$, $g(0) = 1$, and $g(3) = 6$. No other x -coordinates will be read for consistency.

The response $g(-2) = g(5) - (-4 + 5 + 7) = 5$ earns both points.

The response of both $13 - 8 = 5$ and $g(-2) = 5$ does not earn the first point, but earns the second point, provided no other work is presented.

The string of equalities $g(-2) = g(5) - \int_{-2}^5 g'(x) dx = 13 - 8 = 5$ earns both points.

6-5

Special Case: If in part (b), a response picks $x = 1$ or $x = 5$ as the location of the maximum, then there is no work to do in part (c). This response will earn zero points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $g(-2) = g(5) + \int_5^{-2} g'(x) dx$

☐ Answer

Solution:

$$\begin{aligned}
 g(-2) &= g(5) + \int_5^{-2} g'(x) dx \\
 &= g(5) - \int_{-2}^5 g'(x) dx = 13 - (7 + 5 - 4) = 5
 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x -value.

For $x = -2$, the response must either state its y -value or reference it as a maximum.

For the endpoints, the y -values must be presented but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 1$), and thus the third point cannot be earned.

It is possible to not refer to $g(-3) = 2$ by indicating that g' is positive on the interval $(-3, -2)$, thus eliminating 2 as a possible absolute maximum.

A complete list of given or computable values for $g(x)$ is given in the table below.

6-5

x	$g(x)$
-3	2
-2	5
0	1
1	3
3	6
5	13



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = -2$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$$\begin{aligned}
 g(-3) &= g(5) + \int_5^{-3} g'(x) dx \\
 &= g(5) - \int_{-3}^5 g'(x) dx = 13 - (7 + 5 - 4 + 3) = 2
 \end{aligned}$$

$$g(-2) = 5$$

$$g(5) = 13$$

So the absolute maximum value of g on $-3 \leq x \leq 5$ is $g(5) = 13$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[-1, 0]$, $[-1, 0)$, $(-1, 0]$, $(-1, 0)$ all earn the first point.

The reason “ g' is increasing” can be expressed by either “ g'' is positive” or “the slope of g' is positive.”

We will overlook the false implication that $g' < 0$ on $[-1, 0]$.

The second point can be earned without earning the first point only if the given answer is a subset of $[-1, 0]$.

6-5



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

g is decreasing where g' is negative.

g is concave up where g' is increasing.

g is both decreasing and concave up on the open interval $(-1, 0)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that g is continuous because g is differentiable or g is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[1, 5]$, or the point is not earned.

The y -values of g need to be mentioned by either referring to the numerical values of 3 and 13, or mentioning $g(1)$ and $g(5)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since g is differentiable it’s continuous and 10 is between 3 and 13 so such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

6-5

g is differentiable. $\Rightarrow g$ is continuous.

$$g(1) = 3 < 10 < 13 = g(5)$$

By the Intermediate Value Theorem, there must be a value of a , for $1 < a < 5$, such that $g(a) = 10$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of g' or g .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $g(5) - 13 = 0$ and $3 \cdot 5 - 15 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $g(5) = 13$ and $3 \cdot 5 = 15$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{g(5)-13}{3 \cdot 5-15} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{g'(x)}{3} = \frac{7}{3}$, will earn the second point, but not the third point; note that even just one of two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{0}{0}$ and $\lim_{x \rightarrow 5^-} \frac{g(x)-13}{3x-15} = \frac{g'(x)}{3}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{7}{3}$, or its numeric equivalent.

6-5



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g is twice differentiable. $\Rightarrow$ both g' and g are continuous. $\Rightarrow \lim_{x \rightarrow 5^-} g(x) = g(5)$ and $\lim_{x \rightarrow 5^-} g'(x) = g'(5)$.

$$\lim_{x \rightarrow 5^-} (g(x) - 13) = g(5) - 13 = 13 - 13 = 0$$

$$\lim_{x \rightarrow 5^-} (3x - 15) = 3 \cdot 5 - 15 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 5^-} \frac{g(x) - 13}{3x - 15} &= \lim_{x \rightarrow 5^-} \frac{g'(x)}{3} \\ &= \frac{g'(5)}{3} = \frac{7}{3} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $3 \cdot (g(x))^2$ or $3 \cdot (g(1))^2$.

The chain rule point can be earned by seeing an expression involving g but not g' times $g'(1)$.

All three points will be awarded for seeing only $3 \cdot 3^2 \cdot 3$, but $3 \cdot 9 \cdot 3$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

6-5

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$r'(x) = 3(g(x))^2 \cdot g'(x)$$

$$r'(1) = 3(g(1))^2 \cdot g'(1) = 3 \cdot 3^2 \cdot 3 = 81$$

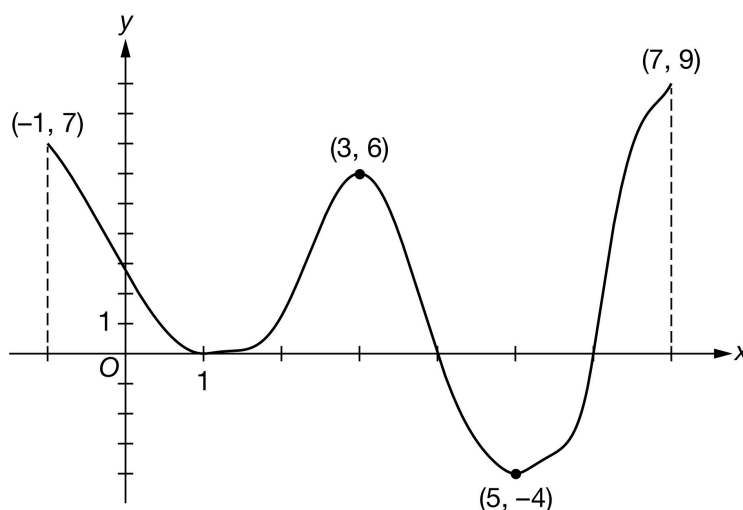
6-5

30. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The figure above shows the graph of h' , the derivative of a twice-differentiable function h , on the closed interval $-1 \leq x \leq 7$. The graph of h' has horizontal tangent lines at $x = 1$, $x = 3$, and $x = 5$. The areas of the regions bounded by the x -axis and the graph of h' on the intervals $[-1, 1]$, $[1, 4]$, $[4, 6]$, and $[6, 7]$ are 6, 8, 5.6, and 6, respectively. The function h is defined for all real numbers and satisfies $h(5) = 5$ and $h(7) = 8$.

- Find the x -coordinate of each critical point of h on the open interval $-1 < x < 7$.
- Identify the critical point from part (a) which corresponds to a relative minimum of h . Give a reason for your answer.
- Find the value of h at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute minimum value of h on $-1 \leq x \leq 7$. Justify your answer.
- On what open intervals is h both increasing and concave up? Give a reason for your answer.
- Explain why there must be a value of p , for $5 < p < 7$, such that $h(p) = 6.5$.
- Find $\lim_{x \rightarrow 7^-} \frac{h(x) - 8}{3x - 21}$. Give a reason for your answer.
- The function a is defined by $a(x) = (h(x))^2$. Find $a'(5)$. Show the computations that lead to your answer.

6-5

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = 1$, $x = 4$, and $x = 6$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-1, 7)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-1, 7)$. The point is still earned provided the three critical points, $x = 1$, $x = 4$, and $x = 6$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(1, 0)$, $(4, 0)$, $(6, 0)$, we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = 0$ at $x = 1$, $x = 4$, and $x = 6$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local minimum as 6 and give the correct reason of “ h' changes from negative to positive” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ h switches from decreasing to increasing at $x = 6$.”
- Saying that “ $h'(6)$ changes signs from negative to positive”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of h' , the one given in the stem of the problem.

A response indicating that “ $h'(x)$ is negative for $x < 6$ and positive for $x > 6$ ” is allowed to give this statement as a reason to earn the point.

6-5

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

A relative minimum value of h occurs where $h'(x)$ changes from negative to positive values.

$x = 6$ is the only location on $-1 < x < 7$ for which this is true, so $x = 6$ is the only location of a relative minimum value of h on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -1$, $x = 1$, or $x = 4$ is selected as the location of the local minimum in part (b). Note that $h(-1) = -6.4$, $h(1) = -0.4$, and $h(4) = 7.6$. No other x -coordinates will be read for consistency.

The response $h(6) = h(7) - 6 = 2$ earns both points.

The responses of both $8 - 6 = 2$ and $h(6) = 2$ do not earn the first point, but earn the second point, provided no other work is presented.

The string of equalities $h(6) = h(7) - \int_6^7 h'(x) dx = 8 - 6 = 2$ earns both points.

6-5

Special Case: If in part (b), a response picks $x = 5$ or $x = 7$ as the location of the minimum, then there is no work to do in part (c). This response will earn zero points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $h(6) = h(7) + \int_7^6 h'(x) dx$
- ☐ Answer

Solution:

$$\begin{aligned}
 h(6) &= h(7) + \int_7^6 h'(x) dx \\
 &= h(7) - \int_6^7 h'(x) dx = 8 - 6 = 2
 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x -value.

For $x = 6$, the response must either state its y -value or reference it as a minimum.

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 5$), and thus the third point cannot be earned.

It is possible to not refer to $h(7) = 8$ by indicating that h' is positive on the interval $(6, 7)$, thus eliminating 8 as a possible absolute minimum.

A complete list of given or computable values for $h(x)$ is given in the table below

6-5

x	$h(x)$
-1	-6.4
1	-0.4
4	7.6
5	5
6	2
7	8



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 6$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$$h(-1) = h(7) + \int_7^{-1} h'(x) dx$$

$$= h(7) - \int_{-1}^7 h'(x) dx = 8 - (6 + 8 - 5.6 + 6) = -6.4$$

$$h(6) = 2$$

$$h(7) = 8$$

Therefore, the absolute minimum value of h on $-1 \leq x \leq 7$ is $h(-1) = -6.4$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, for example “(1, 3] and [6, 7)” earns the first point.

The reason “ h' is increasing” can be expressed by either “ h'' is positive” or “the slope of h' is positive.”

We will overlook the false implication that $h' > 0$ on $[1, 3]$ and $[6, 7]$.

6-5

The second point can be earned without earning the first point only if the given answer is a subset of $[1, 3] \cup [6, 7]$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

h is increasing where h' is positive.

h is concave up where h' is increasing.

h is both increasing and concave up on the open intervals $(1, 3)$ and $(6, 7)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that h is continuous because h is differentiable or h is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[5, 7]$, or the point is not earned.

The y -values of h need to be mentioned by either referring to the numerical values of 5 and 8, or mentioning $h(5)$ and $h(7)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since h is differentiable it’s continuous and 6.5 is between 5 and 8 such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation using Intermediate Value Theorem

6-5

Solution:

h is differentiable. $\Rightarrow h$ is continuous.

$$h(5) = 5 < 6.5 < 8 = h(7)$$

By the Intermediate Value Theorem, there must be a value of p , for $5 < p < 7$, such that $h(p) = 6.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of h' or h .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $h(7) - 8 = 0$ and $3 \cdot 7 - 21 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $h(7) = 8$ and $3 \cdot 7 = 21$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{h(7)-8}{3 \cdot 7-21} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{h'(x)}{3} = \frac{9}{3}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{0}{0}$ and

$\lim_{x \rightarrow 7^-} \frac{h(x)-8}{3x-21} = \frac{h'(x)}{3}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

6-5

The third point is earned only for the answer of 3, or its numeric equivalent.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h is twice differentiable. $\Rightarrow$ both h' and h are continuous. $\Rightarrow \lim_{x \rightarrow 7^-} h(x) = h(7)$ and $\lim_{x \rightarrow 7^-} h'(x) = h'(7)$.

$$\lim_{x \rightarrow 7^-} (h(x) - 8) = h(7) - 8 = 8 - 8 = 0$$

$$\lim_{x \rightarrow 7^-} (3x - 21) = 3 \cdot 7 - 21 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \rightarrow 7^-} \frac{h(x) - 8}{3x - 21} = \lim_{x \rightarrow 7^-} \frac{h'(x)}{3}$$

$$= \frac{h'(7)}{3} = \frac{9}{3} = 3$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The Power rule point can be earned by just seeing $2 \cdot h(x)$ or $2 \cdot h(5)$.

The chain rule point can be earned by seeing an expression involving h but not h' times $h'(5)$.

All three points will be awarded for seeing only $2 \cdot 5 \cdot (-4)$, but $10 \cdot (-4)$, $2 \cdot (-20)$, or $-8 \cdot 5$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

6-5

The student response accurately includes **all three** of the criteria below.

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$a'(x) = 2h(x) \cdot h'(x)$$

$$a'(5) = 2 \cdot h(5) \cdot h'(5) = 2 \cdot 5 \cdot (-4) = -40$$

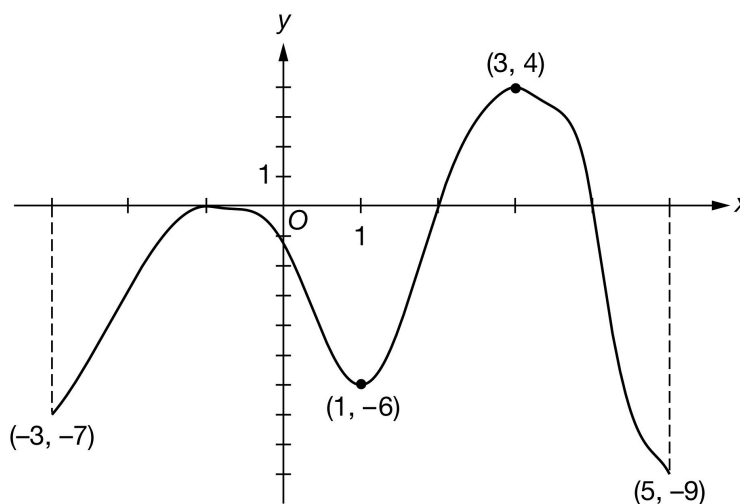
6-5

31. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f'

The figure above shows the graph of f' , the derivative of a twice-differentiable function f , on the closed interval $-3 \leq x \leq 5$. The graph of f' has horizontal tangent lines at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x -axis and the graph of f' on the intervals $[-3, -1]$, $[-1, 2]$, $[2, 4]$, and $[4, 5]$ are 6, 8, 5.6, and 6, respectively. The function f is defined for all real numbers and satisfies $f(3) = 10$ and $f(5) = 7$.

- Find the x -coordinate of each critical point of f on the open interval $-3 < x < 5$.
- Identify the critical point from part (a) which corresponds to a relative minimum of f . Give a reason for your answer.
- Find the value of f at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute minimum value of f on $-3 \leq x \leq 5$. Justify your answer.
- On what open intervals is f both decreasing and concave down? Give a reason for your answer.
- Explain why there must be a value of d , for $3 < d < 5$, such that $f(d) = 8.5$.
- Find $\lim_{x \rightarrow 5^-} \frac{2f(x) - 14}{x^2 - 25}$. Give a reason for your answer.
- The function m is defined by $m(x) = (f(x))^2$. Find $m'(3)$. Show the computations that lead to your answer.

6-5

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -1$, $x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 5)$. The point is still earned provided the three critical points, $x = -1$, $x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(2, 0)$, $(4, 0)$, we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$f'(x) = 0$ at $x = -1$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local minimum as 2 and give the correct reason of “ f' changes from negative to positive” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ f switches from decreasing to increasing at $x = 2$ ”
- Saying that “ $f'(2)$ changes signs from negative to positive”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of f' , the one given in the stem of the problem.

A response indicating that “ $f'(x)$ is negative for $x < 2$ and positive for $x > 2$ ” is allowed to give this statement as a reason to earn the point.

6-5

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

A relative minimum value of f occurs where $f'(x)$ changes from negative to positive values.

$x = 2$ is the only location on $-3 < x < 5$ for which this is true, so $x = 2$ is the only location of a relative minimum value of f on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -3$, $x = -1$, or $x = 4$ is selected as the location of the local minimum in part (b). Note that $f(-3) = 21.4$, $f(-1) = 15.4$, and $f(4) = 13$. No other x -coordinates will be read for consistency.

The response $f(2) = f(5) - (5.6 - 6) = 7.4$ earns both points.

The responses of both $7 - (-0.4) = 7.4$ and $f(2) = 7.4$ do not earn the first point, but earn the second point, provided no other work is presented.

The string of equalities $f(2) = f(5) - \int_2^5 f'(x) dx = 7 - (-0.4) = 7.4$ earns both points.

6-5

Special Case: If in part (b), a response picks $x = 3$ or $x = 5$ as the location of the minimum, then there is no work to do in part (c). This response will earn zero points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $f(2) = f(5) + \int_5^2 f'(x) dx$
- ☐ Answer

Solution:

$$\begin{aligned} f(2) &= f(5) + \int_5^2 f'(x) dx = f(5) - \int_2^5 f'(x) dx \\ &= f(5) - \int_2^4 f'(x) dx - \int_4^5 f'(x) dx = 7 - 5.6 - (-6) = 7.4 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x -value.

For $x = 2$, the response must either state its y -value or reference it as a minimum.

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 3$), and thus the third point cannot be earned.

It is possible to not refer to $f(-3) = 21.4$ by indicating that f' is negative on the interval $(-3, -1)$, thus eliminating 21.4 as a possible absolute minimum.

A complete list of given or computable values for $f(x)$ is given in the table below.

6-5

x	$f(x)$
-3	21.4
-1	15.4
2	7.4
3	10
4	13
5	7



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 2$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$$f(-3) = f(5) + \int_5^{-3} f'(x) dx$$

$$= f(5) - \int_{-3}^5 f'(x) dx = 7 - (-6 + 5.6 - 8 - 6) = 21.4$$

$$f(2) = 7.4$$

$$f(5) = 7$$

So the absolute minimum value of f on $-3 \leq x \leq 5$ is

$$f(5) = 7.$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, for example “ $(-1, 1]$ and $[4, 5)$ ” earns the first point.

The reason “ f' is decreasing” can be expressed by either “ f'' is negative” or “the slope of f' is negative.”

We will overlook the false implication that $f' < 0$ on $[-1, 1]$ and $[4, 5]$.

6-5

The second point can be earned without earning the first point only if the given answer is a subset of $[-1, 1] \cup [4, 5]$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

f is decreasing where f' is negative.

f is concave down where f' is decreasing.

f is both decreasing and concave down on the open intervals $(-1, 1)$ and $(4, 5)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that f is continuous because f is differentiable or f is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[3, 5]$, or the point is not earned.

The y -values of f need to be mentioned by either referring to the numerical values of 7 and 10, or mentioning $f(3)$ and $f(5)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since f is differentiable it’s continuous and 8.5 is between 7 and 10 so such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation using Intermediate Value Theorem

6-5

Solution:

f is differentiable. $\Rightarrow f$ is continuous.

$$f(5) = 7 < 8.5 < 10 = f(3)$$

By the Intermediate Value Theorem, there must be a value of d , for $3 < d < 5$, such that $f(d) = 8.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of f' or f .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $2f(5) - 14 = 0$ and $5^2 - 25 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $2f(5) = 14$ and $5^2 = 25$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{2f(5)-14}{5^2-25} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{2f'(x)}{2x} = \frac{-9}{5}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{0}{0}$ and

$\lim_{x \rightarrow 5^-} \frac{2f(x)-14}{x^2-25} = \frac{2f'(x)}{2x}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

6-5

The third point is earned only for the answer of $-\frac{9}{5}$, or its numeric equivalent.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

f is twice differentiable. $\Rightarrow$ both f' and f are continuous. $\Rightarrow \lim_{x \rightarrow 5^-} f(x) = f(5)$ and $\lim_{x \rightarrow 5^-} f'(x) = f'(5)$.

$$\lim_{x \rightarrow 5^-} (2f(x) - 14) = 2 \cdot f(5) - 14 = 14 - 14 = 0$$

$$\lim_{x \rightarrow 5^-} (x^2 - 25) = 5^2 - 25 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 5^-} \frac{2f(x) - 14}{x^2 - 25} &= \lim_{x \rightarrow 5^-} \frac{2f'(x)}{2x} \\ &= \frac{2f'(5)}{2 \cdot 5} = \frac{-9}{5} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $2 \cdot f(x)$ or $2 \cdot f(3)$.

The chain rule point can be earned by seeing an expression involving f but not f' times $f'(3)$.

All three points will be awarded for seeing only $2 \cdot 10 \cdot 4$, but $20 \cdot 4$, $2 \cdot 40$, or $8 \cdot 10$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

6-5

The student response accurately includes **all three** of the criteria below.

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$m'(x) = 2f(x) \cdot f'(x)$$

$$m'(3) = 2 \cdot f(3) \cdot f'(3) = 2 \cdot 10 \cdot 4 = 80$$

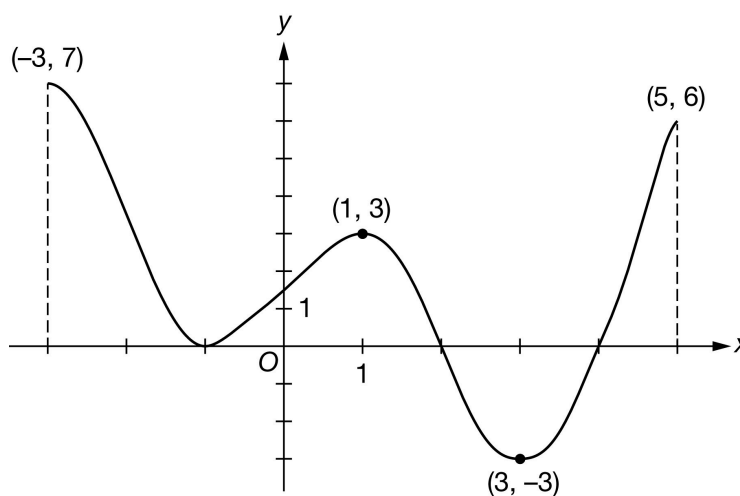
6-5

32. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The figure above shows the graph of g' , the derivative of a twice-differentiable function g , on the closed interval $-3 \leq x \leq 5$. The graph of g' has horizontal tangent lines at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x -axis and the graph of g' on the intervals $[-3, -1]$, $[-1, 2]$, $[2, 4]$, and $[4, 5]$ are 7, 5, 4, and 3, respectively. The function g is defined for all real numbers and satisfies $g(3) = 8$ and $g(5) = 9$.

- Find the x -coordinate of each critical point of g on the open interval $-3 < x < 5$.
- Identify the critical point from part (a) which corresponds to a relative minimum of g . Give a reason for your answer.
- Find the value of g at the x -coordinate identified in part (b). Show the computations that lead to your answer.
- Find the absolute minimum value of g on $-3 \leq x \leq 5$. Justify your answer.
- On what open intervals is g both decreasing and concave down? Give a reason for your answer.
- Explain why there must be a value of r , for $3 < r < 5$, such that $g(r) = 8.5$.
- Find $\lim_{x \rightarrow 5^-} \frac{g(x) - 9}{x^2 - 25}$. Give a reason for your answer.
- The function f is defined by $f(x) = (g(x))^3$. Find $f'(3)$. Show the computations that lead to your answer.

6-5

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -1$, $x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 5)$. The point is still earned provided the three critical points, $x = -1$, $x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(2, 0)$, $(4, 0)$, we will read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = 0$ at $x = -1$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local minimum as 4 and give the correct reason of “ g' changes from negative to positive” to earn the point.

Insufficient reasons, which do not earn the point, include:

- Saying that “ g switches from decreasing to increasing at $x = 4$ ”
- Saying that “ $g'(4)$ changes signs from negative to positive”

If the response talks about “the graph” without specifying which graph, we will interpret “the graph” here (and throughout the rest of the problem) as being the graph of g' , the one given in the stem of the problem.

A response indicating that “ $g'(x)$ is negative for $x < 4$ and positive for $x > 4$ ” is allowed to give this statement as a reason to earn the point.

6-5

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 4$ is the only location on $-3 < x < 5$ for which this is true, so $x = 4$ is the only location of a relative minimum value of g on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -3$, $x = -1$, or $x = 2$ is selected as the location of the local minimum in part (b). Note that $g(-3) = -2$, $g(-1) = 5$, and $g(2) = 10$. No other x -coordinates will be read for consistency.

The response $g(4) = g(5) - 3 = 6$ earns both points.

The responses of both $9 - 3 = 6$ and $g(4) = 6$ do not earn the first point, but earns the second point, provided no other work is presented.

Special Case: If a response uses $\int_4^3 g'(x) dx = 2$ then

$g(4) = g(3) - \int_4^3 g'(x) dx = 8 - 2 = 6$ earns both points, because it's possible to find $\int_4^3 g'(x) dx = 2$ using the areas and initial conditions given.

6-5

any response using perceived symmetry about $x = 3$ can earn at most the first point, for example

$$g(4) = g(3) - \int_4^3 g'(x) dx = 8 - \frac{1}{2} \cdot 4 = 6 \text{ earns just the first point.}$$

Special Case: If in part (b), a response picks $x = 3$ or $x = 5$ as the location of the minimum, then there is no work to do in part (c). This response will earn zero points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $g(4) = g(5) + \int_5^4 g'(x) dx$

☐ Answer

Solution:

$$\begin{aligned} g(4) &= g(5) + \int_5^4 g'(x) dx \\ &= g(5) - \int_4^5 g'(x) dx = 9 - 3 = 6 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x -value.

For $x = 4$, the response must either state its y -value or reference it as a minimum.

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 3$), and thus the third point cannot be earned.

It is possible to not refer to $g(5) = 9$ by indicating that g' is positive on the interval $(4, 5)$, thus eliminating 9 as a possible absolute minimum.

A complete list of given or computable values for $g(x)$ is given in the table below.

6-5

x	$g(x)$
-3	-2
1	5
2	10
3	8
4	6
5	9



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 4$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$$\begin{aligned}
 g(-3) &= g(5) + \int_5^{-3} g'(x) \, dx \\
 &= g(5) - \int_{-3}^5 g'(x) \, dx = 9 - (7 + 5 - 4 + 3) = -2
 \end{aligned}$$

$$g(4) = 6$$

$$g(5) = 9$$

So the absolute minimum value of g on $-3 \leq x \leq 5$ is $g(-3) = -2$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[2, 3]$, $[2, 3)$, $(2, 3]$, and $(2, 3)$ all earn the first point.

The reason “ g' is decreasing” can be expressed by either “ g'' is negative” or “the slope of g' is negative.”

We will overlook the false implication that $g' < 0$ on $[2, 3]$.

The second point can be earned without earning the first point only if the given answer is a subset of $[2, 3]$.

6-5



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

g is decreasing where g' is negative.

g is concave down where g' is decreasing.

g is both decreasing and concave down on the open interval $(2, 3)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that g is continuous because g is differentiable or g is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[3, 5]$, or the point is not earned.

The y -values of g need to be mentioned by either referring to the numerical values of 8 and 9, or mentioning $g(3)$ and $g(5)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since g is differentiable it’s continuous and 8.5 is between 8 and 9 so such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.



0	1
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The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

6-5

g is differentiable. $\Rightarrow g$ is continuous.

$$g(3) = 8 < 8.5 < 9 = g(5)$$

By the Intermediate Value Theorem, there must be a value of r , for $3 < r < 5$, such that $g(r) = 8.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of g' or g .

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $g(5) - 9 = 0$ and $5^2 - 25 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $g(5) = 9$ and $5^2 = 25$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{g(5)-9}{5^2-25} \rightarrow \frac{0}{0}$ earns the point.

· The false equation $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{g'(x)}{2x} = \frac{3}{5}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{0}{0}$ and $\lim_{x \rightarrow 5^-} \frac{g(x)-9}{x^2-25} = \frac{g'(x)}{2x}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{3}{5}$, or its numeric equivalent.

6-5



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g is twice differentiable. $\Rightarrow$ that both g' and g are continuous. $\Rightarrow \lim_{x \rightarrow 5^-} g(x) = g(5)$ and $\lim_{x \rightarrow 5^-} g'(x) = g'(5)$.

$$\lim_{x \rightarrow 5^-} (g(x) - 9) = g(5) - 9 = 9 - 9 = 0$$

$$\lim_{x \rightarrow 5^-} (x^2 - 25) = 5^2 - 25 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 5^-} \frac{g(x) - 9}{x^2 - 25} &= \lim_{x \rightarrow 5^-} \frac{g'(x)}{2x} \\ &= \frac{g'(5)}{2 \cdot 5} = \frac{6}{10} = \frac{3}{5} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $3 \cdot (g(x))^2$ or $3 \cdot (g(3))^2$.

The chain rule point can be earned by seeing an expression involving g but not g' times $g'(3)$.

All three points will be awarded for seeing only $3 \cdot 8^2 \cdot (-3)$, but $3 \cdot 64 \cdot (-3)$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

6-5

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$f'(x) = 3(g(x))^2 \cdot g'(x)$$

$$f'(3) = 3(g(3))^2 \cdot g'(3) = 3 \cdot 8^2 \cdot (-3) = -576$$

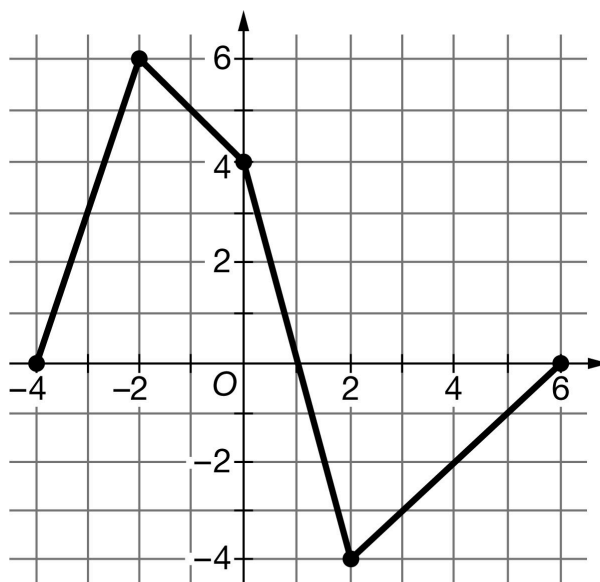
6-5

33. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

Let f be a continuous function defined on the closed interval $-4 \leq x \leq 6$. The graph of f , consisting of four line segments, is shown above. Let G be the function defined by $G(x) = \int_0^x f(t) \, dt$.

- On what open intervals is the graph of G concave up? Give a reason for your answer.
- Let P be the function defined by $P(x) = G(x) \cdot f(x)$. Find $P'(3)$.
- Find $\lim_{x \rightarrow 2} \frac{G(x)}{x^2 - 2x}$.
- Find the average rate of change of G on the interval $[-4, 2]$. Does the Mean Value Theorem guarantee a value c , $-4 < c < 2$, for which $G'(c)$ is equal to this average rate of change? Justify your answer.

Global

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This “global point” can be earned in any one part. Expressions that show this connection and therefore earn this point include: $G' = f$, $G'(x) = f(x)$, $G''(x) = f'(x)$ in part (a), $G'(3) = f(3)$ in part (b), or $G'(2) = f(2)$

6-5

in part (c).



0	1
---	---

The student response accurately includes the criteria below.

☐ $G'(x) = f(x)$

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Intervals may also include one or both endpoints.



0	1
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The student response accurately includes the criteria below.

☐ Answer with reason
Solution:

$$G'(x) = f(x)$$

The graph of G is concave up for $-4 < x < -2$ and $2 < x < 6$, because $G' = f$ is increasing on these intervals.**Part B**

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the correct application of the product rule in terms of x or in the evaluation of $P'(3)$. Once earned, this point cannot be lost.

The second point is earned by correctly evaluating $G(3) = -3.5$, $G'(3) = -3$, or $f(3) = -3$.

To be eligible to earn the third point a response must have earned the first two points.

Simplification of the numerical value is not required to earn the third point.

6-5



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ $G(3)$ or $G'(3)$
- ☐ Answer

Solution:

$$P'(x) = G(x) \cdot f'(x) + f(x) \cdot G'(x)$$

$$P'(3) = G(3) \cdot f'(3) + f(3) \cdot G'(3)$$

Substituting $G(3) = \int_0^3 f(t) dt = -3.5$ and $G'(3) = f(3) = -3$ into the above expression for $P'(3)$ gives the following:

$$P'(3) = -3.5 \cdot 1 + (-3) \cdot (-3) = 5.5$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point the response must show $\lim_{x \rightarrow 2} (x^2 - 2x) = 0$ and $\lim_{x \rightarrow 2} G(x) = 0$ and must show a ratio of the two derivatives, $G'(x)$ and $2x - 2$. The ratio may be shown as evaluations of the derivatives at $x = 2$, such as $\frac{G'(2)}{2}$.

To earn the second point the response must evaluate correctly with appropriate limit notation. In particular the response must include $\lim_{x \rightarrow 2} \frac{G'(x)}{2x-2}$ or $\lim_{x \rightarrow 2} \frac{f(x)}{2x-2}$.

With any linkage errors (such as $\frac{G'(x)}{2x-2} = \frac{f(2)}{2}$), the response does not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Uses L'Hospital's Rule

6-5

☐ Answer with justification
Solution:

$$\lim_{x \rightarrow 2} (x^2 - 2x) = 0$$

Because G is continuous for $-4 \leq x \leq 6$, $\lim_{x \rightarrow 2} G(x) = \int_0^2 f(t) dt = 0$.

Therefore, the limit $\lim_{x \rightarrow 2} \frac{G(x)}{x^2 - 2x}$ is an indeterminate form of type $\frac{0}{0}$.

Using L'Hospital's Rule,

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{G(x)}{x^2 - 2x} &= \lim_{x \rightarrow 2} \frac{G'(x)}{2x - 2} \\ &= \lim_{x \rightarrow 2} \frac{f(x)}{2x - 2} = \frac{f(2)}{2} = \frac{-4}{2} = -2 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must present at least a difference and a quotient and a correct evaluation. For

example, $\frac{0+16}{6}$ or $\frac{G(2)-G(-4)}{6} = \frac{16}{6}$.

Simplification of the numerical value is not required to earn the first point, but any simplification must be correct.

The second point can be earned without the first point if the response has the correct setup but an incorrect or no evaluation of the average rate of change. The Mean Value Theorem need not be explicitly stated provided the conditions and conclusion are stated.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Average rate of change
- ☐ Answer with justification

Solution:

$$G(2) = \int_0^2 f(t) dt = 0 \text{ and } G(-4) = \int_0^{-4} f(t) dt = -16$$

6-5

$$\text{Average rate of change} = \frac{G(2) - G(-4)}{2 - (-4)} = \frac{0 - (-16)}{6} = \frac{8}{3}$$

Yes, $G'(x) = f(x)$ so G is differentiable on $(-4, 2)$ and continuous on $[-4, 2]$. Therefore, the Mean Value Theorem applies and guarantees a value c , $-4 < c < 2$, such that $G'(c) = \frac{8}{3}$.

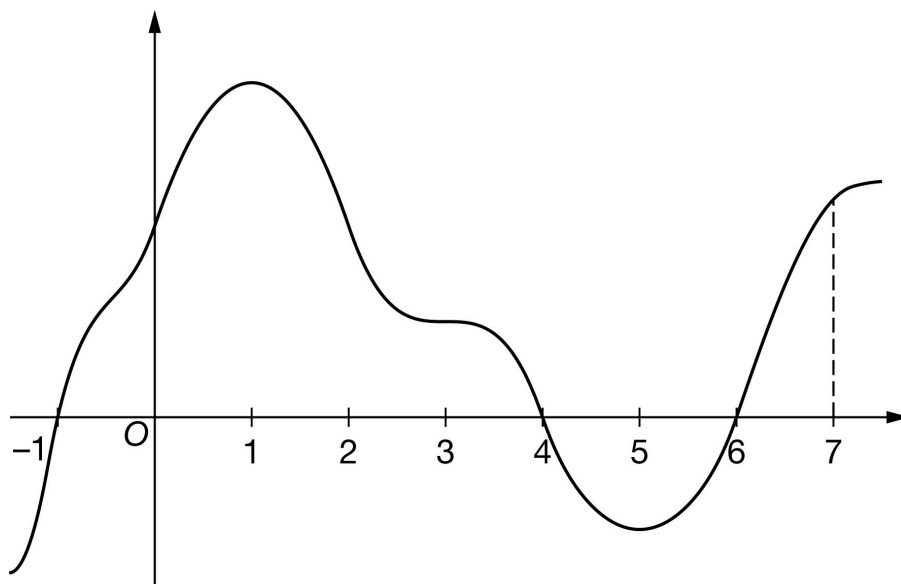
6-5

34. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The graph of the differentiable function f with domain $-1.5 \leq x \leq 7.5$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of f on the intervals $[-1, 4]$, $[4, 6]$, and $[6, 7]$ are 56, 10, and 8, respectively. The graph of f has horizontal tangents at $x = 1$, $x = 3$, and $x = 5$.

Let h be the function defined by $h(x) = \int_6^x f(t) \, dt$ for $-1.5 \leq x \leq 7.5$.

- Find the x -coordinate of each critical point of h on the interval $-1.5 < x < 7.5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- Find the x -coordinate of each point of inflection for the graph of h on the interval $-1.5 < x < 7.5$. Justify your answer.
- Find the value of $h(7)$.
- Find the value of $h(-1)$. Show the computations that lead to your answer.
- Must there exist a value of d , for $-1 < d < 4$, such that $h'(d)$ is equal to the average rate of change of h over the interval $-1 \leq x \leq 4$? Justify your answer.
- Find $\lim_{x \rightarrow 7} \frac{h(x)}{4x}$. Show the computations that lead to your answer.

6-5

(h) The function q is defined by $q(x) = 3x \cdot h(x)$. It is known that $f(7) = 14$. Find $q'(7)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -1$, $x = 4$, and $x = 6$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-1.5, 7.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-1.5, 7.5)$. The point is still earned provided the three critical points, $x = -1$, $x = 4$, and $x = 6$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(4, 0)$, $(6, 0)$, read the x -coordinates and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = f(x) = 0$ at $x = -1$, $x = 4$, and $x = 6$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on f , then:

- the response is eligible for all three points if the response presents the connection between $h'(x)$ and $f(x)$ (that is, $h'(x) = f(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or h' as $f(x) = h'(x)$, throughout all parts of this question. m

- the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = f(x)$. Any additional error (for example, not justifying that at $x = -1$, h has a relative minimum)

6-5

is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-1)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

- $h(-1.5) > -46$
- $h(-1) > -46$
- $h(4) = 10$
- $h(6) = 0$
- $h(7) = 8$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = -1$, h has a relative minimum because $h' = f$ is negative for $x < -1$ and $h' = f$ is positive for $x > -1$.
- At $x = 4$, h has a relative maximum because $h' = f$ is positive for $x < 4$ and $h' = f$ is negative for $x > 4$.
- At $x = 6$, h has a relative minimum because $h' = f$ is negative for $x < 6$ and $h' = f$ is positive for $x > 6$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification

6-5

- ☐ Relative maximum with justification
- ☐ Relative minimum with justification

Solution:

At $x = -1$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

At $x = 4$, h has a relative maximum because $h'(x) = f(x)$ changes from positive to negative there.

At $x = 6$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only h' is eligible for all three points.

If a response gives justifications based only on f , then (in parallel with the approach followed in part (b)):

- the response is eligible for all three points if the response presents the connection between $h'(x)$ and $f(x)$ (that is, $h'(x) = f(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of f or h' as $f(x) = h'(x)$, throughout all parts of this question.
- the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = f(x)$.

Justification for the first two points should reference one of the following:

- $h'(x)$ changing from increasing to decreasing, or vice versa
- relative extrema of $h'(x) = f(x)$
- h'' changes sign
- $f'(x)$ changes sign

“ $h'(x) = f(x)$ changes direction” earns neither of the first two points.

Read references to “the graph” as references to f as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn both points.

A response that discusses h' at a particular point (i.e., $h'(-1)$) when the discussion should have been about $h'(x)$ can earn only one of the first two points, with one exception. This communication error should not be

6-5

penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x = 1$ and $x = 5$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ $x = 1$ with justification
- ☐ $x = 5$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = 1$, the graph of h has a point of inflection because $h'(x) = f(x)$ changes from increasing to decreasing there.

At $x = 5$, the graph of h has a point of inflection because $h'(x) = f(x)$ changes from decreasing to increasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 8 is all that is needed to earn the point. No supporting work is required.



0	1
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The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$h(7) = \int_6^7 f(t) dt = 8$$

6-5

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 56 and ± 10 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 56) \pm (\pm 10)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 56 and ± 10 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-56 - (-10)$ earns both points because numerical expressions need not be simplified.
- The expression $(-56) - 10$ earns the first point, but not the second point.
- The expression $-56 - (-10) - 8$ earns neither point.
- The expression $\int_6^{-1} g(t) dt = -46$ earns neither point.
- The expression $-\int_{-1}^4 f(t) dt - \int_4^6 f(t) dt = -46$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 56 and ± 10 , but it does earn the second point.
- $-56, 10$ earns neither point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ $\left| \int_4^{-1} f(t) dt \right| = 56$ and $\left| \int_4^6 f(t) dt \right| = 10$
- ☐ Answer

Solution:

$$\begin{aligned}
 h(-1) &= \int_6^{-1} f(t) dt = - \int_{-1}^6 f(t) dt \\
 &= - \int_{-1}^4 f(t) dt - \int_4^6 f(t) dt = -56 - (-10) = -46
 \end{aligned}$$

6-5

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-1, 4)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $h(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(d)$ equals the average rate of change of $h(x)$ ” earns both points.
- “Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(d)$ equals the average rate of change of $h(x)$ ” does not earn the first point, but does earn second point.
- “Yes because of MVT” does not earn the first point but does earn the second point.
- “Yes, because $h(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $h(x)$ is continuous” earns neither point.
- “Yes, because $h(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$$h' = f \Rightarrow h \text{ is differentiable on } -1 \leq x \leq 4. \Rightarrow h \text{ is continuous on } -1 \leq x \leq 4.$$

6-5

Therefore, the Mean Value Theorem can be applied to h on the interval $-1 \leq x \leq 4$ to guarantee that there exists a value of d , for $-1 < d < 4$, such that $h'(d)$ equals the average rate of change of h over the interval $-1 \leq x \leq 4$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

$$\cdot \frac{h(7)}{28} = \frac{2}{7}$$

$$\cdot \frac{8}{28}$$

Responses that do not earn the point include:

$$\cdot \frac{h(7)}{4 \cdot 7}$$

$$\cdot \frac{h(7)}{28}$$

Note: an incorrect value for $h(7)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that h is continuous. If the response does state that h is continuous, an explanation is not required.

The response does not need to use limit notation.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h' = f \Rightarrow h \text{ is continuous. } \Rightarrow \lim_{x \rightarrow 7} h(x) = h(7)$$

$$\lim_{x \rightarrow 7} \frac{h(x)}{4x} = \frac{h(7)}{4 \cdot (7)} = \frac{8}{28} = \frac{2}{7}$$

Part H

6-5

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of q that contains a sum of two terms, with at least one term written as a product, and one term involving h and the other term involving h' (or f).

Fundamental Theorem of Calculus is evidenced by linking h' and f either directly or indirectly in the presented derivative of $q(x)$ or in $q'(7)$. Since $f(7) = 15$ is given in the question, replacing $h'(7)$ directly with 14 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $h(7)$ that is brought in from part (d) can earn the third point.

$21 \cdot 14 + 3 \cdot 8$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c \cdot d$ where any of a , b , c , or d is incorrect, will earn zero points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$q'(x) = 3x \cdot h'(x) + 3 \cdot h(x)$$

$$= 3x \cdot f(x) + 3 \cdot h(x)$$

$$q'(7) = 3 \cdot 7 \cdot f(7) + 3h(7) = 21 \cdot (14) + 3 \cdot (8) = 318$$

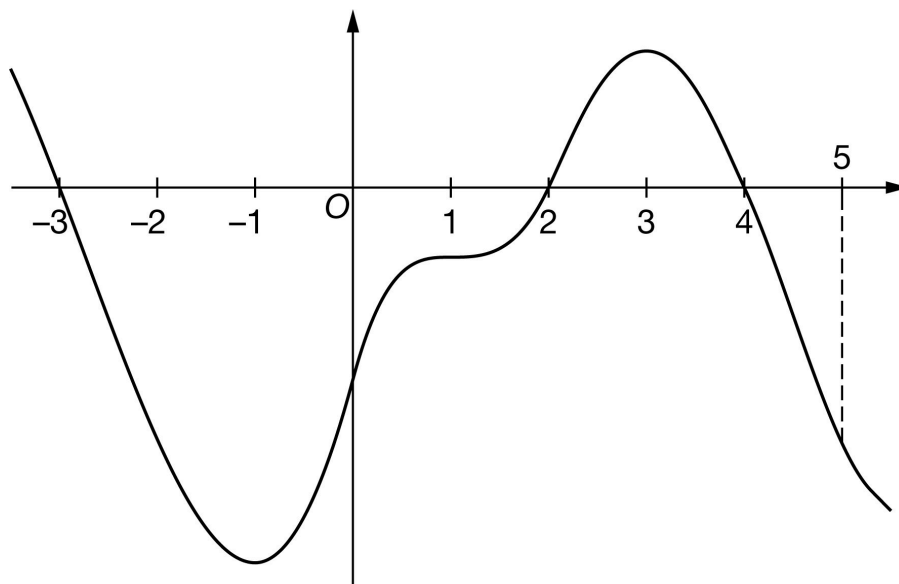
6-5

35. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h

The graph of the differentiable function h with domain $-3.5 \leq x \leq 5.5$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of h on the intervals $[-3, 2]$, $[2, 4]$, and $[4, 5]$ are 58, 11, and 8, respectively. The graph of h has horizontal tangents at $x = -1$, $x = 1$, and

$x = 3$. Let g be the function defined by $g(x) = \int_4^x h(t) \, dt$ for $-3.5 \leq x \leq 5.5$.

- Find the x -coordinate of each critical point of g on the interval $-3.5 < x < 5.5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify each of your answers.
- Find the x -coordinate of each point of inflection for the graph of g on the interval $-3.5 < x < 5.5$. Justify your answer.
- Find the value of $g(5)$.
- Find the value of $g(-3)$. Show the computations that lead to your answer.
- Must there exist a value of b , for $-3 < b < 2$, such that $g'(b)$ is equal to the average rate of change of g over the interval $-3 \leq x \leq 2$? Justify your answer.
- Find $\lim_{x \rightarrow 5} \frac{g(x)}{3x}$. Show the computations that lead to your answer.

6-5

(h) The function a is defined by $a(x) = x^2 \cdot g(x)$. It is known that $h(5) = -16$. Find $a'(5)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3$, $x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3.5, 5.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3.5, 5.5)$. The point is still earned provided the three critical points, $x = -3$, $x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(2, 0)$, $(4, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = h(x) = 0$ at $x = -3$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on h , then:

· the response is eligible for all three points if the response presents the connection between $g'(x)$ and $h(x)$ (that is, $g'(x) = h(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of h or g' as $h(x) = g'(x)$, throughout all parts of this question.

6-5

· the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = h(x)$. Any additional error (for example, not justifying that at $x = 2$, h has a relative minimum) is eligible for at most one of the three points.

Read references to “the graph” as references to $h(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(-3)$ changes from negative to positive”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $g(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

- $g(-3.5) < 47$
- $g(-3) = 47$
- $g(2) = -11$
- $g(4) = 0$
- $g(5) = -8$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = -3$, g has a relative maximum because $g' = h$ is positive for $x < -3$ and $g' = h$ is negative for $x > -3$.
- At $x = 2$, g has a relative minimum because $g' = h$ is negative for $x < 2$ and $g' = h$ is positive for $x > 2$.
- At $x = 4$, g has a relative maximum because $g' = h$ is positive for $x < 4$ and $g' = h$ is negative for $x > 4$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

6-5

- ☐ Relative maximum with justification
- ☐ Relative minimum with justification
- ☐ Relative maximum with justification

Solution:

At $x = -3$, g has a relative maximum because $g'(x) = h(x)$ changes from positive to negative there.

At $x = 2$, g has a relative minimum because $g'(x) = h(x)$ changes from negative to positive there.

At $x = 4$, g has a relative maximum because $g'(x) = h(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only on g is eligible for all three points.

If a response gives justifications based only on h , then (in parallel with the approach followed in part (b)):

- the response is eligible for all three points if the response presents the connection between $g'(x)$ and $h(x)$ (that is, $g'(x) = h(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of h or g' as $h(x) = g'(x)$, throughout all parts of this question.
- the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = h(x)$.

Justification for the first two points should reference one of the following:

- $g'(x)$ changing from increasing to decreasing, or vice versa
- relative extrema of $g'(x)$
- $g''(x)$ changes sign
- $h'(x)$ changes sign

“ $g'(x) = h(x)$ changes direction” earns neither of the first two points.

Read references to “the graph” as references to h as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn both points.

6-5

A response that discusses g' at a particular point (i.e., $g'(-3)$) when the discussion should have been about $g'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x = -1$ and $x = 3$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ $x = -1$ with justification
- ☐ $x = 3$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = -1$, the graph of g has a point of inflection because $g'(x) = h(x)$ changes from decreasing to increasing there.

At $x = 3$, the graph of g has a point of inflection because $g'(x) = h(x)$ changes from increasing to decreasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -8 is all that is needed to earn the point. No supporting work is required.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

6-5

$$g(5) = \int_4^5 h(t) dt = -8$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 58 and ± 11 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 58) \pm (\pm 11)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values. The ± 58 and ± 11 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-(-58) - 11$ earns both points because numerical expressions need not be simplified.
- The expression $(-58) - 11$ earns the first point, but not the second point.
- The expression $-58 - 11 - 8$ earns neither point.
- The expression $\int_4^{-3} h(t) dt = 47$ earns neither point.
- The expression $-\int_{-3}^2 h(t) dt - \int_2^4 h(t) dt = 47$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 58 and ± 11 , but it does earn the second point.
- $-58, 11$ earns neither point



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-3}^2 h(t) dt \right| = 58 \text{ and } \left| \int_4^2 h(t) dt \right| = 11$

☐ Answer

Solution:

$$\begin{aligned}
 g(-3) &= \int_4^{-3} h(t) dt = - \int_{-3}^4 h(t) dt \\
 &= - \int_{-3}^2 h(t) dt - \int_2^4 h(t) dt = -(-58) - 11 = 47
 \end{aligned}$$

6-5

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-3, 2)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $g(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $g(x)$ is continuous and differentiable, therefore there must be a value where $g'(b)$ equals the average rate of change of $g(x)$ ” earns both points.
- “Yes, $g'(x)$ (the derivative of $g(x)$) is continuous and differentiable, therefore there must be a value where $g'(b)$ equals the average rate of change of $g(x)$ ” does not earn the first point, but does earn second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $g(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $g(x)$ is continuous” earns neither point.
- “Yes, because $g(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$$g' = h \Rightarrow g \text{ is differentiable on } -3 \leq x \leq 2. \Rightarrow g \text{ is continuous on } -3 \leq x \leq 2.$$

6-5

Therefore, the Mean Value Theorem can be applied to g on the interval $-3 \leq x \leq 2$ to guarantee that there exists a value of b , for $-3 < b < 2$, such that $g'(b)$ equals the average rate of change of g over the interval $-3 \leq x \leq 2$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

$$\cdot \frac{g(5)}{15} = \frac{-8}{15}$$

$$\cdot \frac{-8}{15}$$

Responses that do not earn the point are:

$$\cdot \frac{g(5)}{15}$$

$$\cdot \frac{g(5)}{3 \cdot 5}$$

Note: an incorrect value for $g(5)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that g is continuous. If the response does state that g is continuous, an explanation is not required.

The response does not need to use limit notation.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g' = h \Rightarrow g \text{ is continuous. } \Rightarrow \lim_{x \rightarrow 5} g(x) = g(5)$$

$$\lim_{x \rightarrow 5} \frac{g(x)}{3x} = \frac{g(5)}{3 \cdot (5)} = \frac{-8}{15}$$

Part H

6-5

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of a that contains a sum of two terms, with at least one term written as a product, and one term involving g and the other term involving g' (or h).

Fundamental Theorem of Calculus is evidenced by linking g' and h either directly or indirectly in the presented derivative of $g(x)$ or in $g'(5)$. Since $h(5) = -16$ is given in the question, replacing $g'(5)$ directly with -16 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $g(5)$ that is brought in from part (d) can earn the third point.

$25 \cdot (-16) + 10 \cdot (-8)$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c \cdot d$ where any of a , b , c , or d is incorrect, will earn zero points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$a'(x) = x^2 \cdot g'(x) + 2x \cdot g(x)$$

$$= x^2 \cdot h(x) + 2x \cdot g(x)$$

$$a'(5) = 25 \cdot h(5) + 2 \cdot 5 \cdot g(5) = 25 \cdot (-16) + 10 \cdot (-8) = -480$$

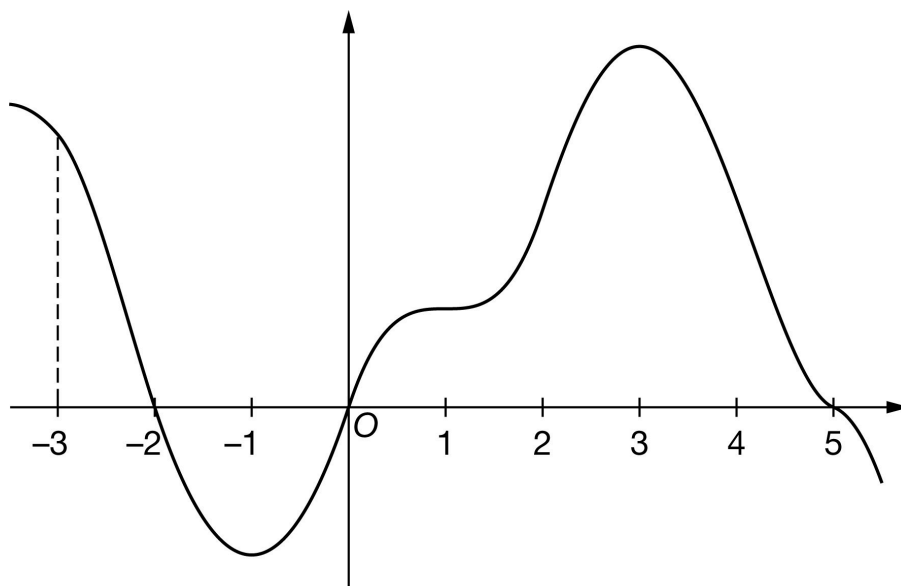
6-5

36. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the differentiable function g with domain $-3.5 \leq x \leq 5.5$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of g on the intervals $[-3, -2]$, $[-2, 0]$, and $[0, 5]$ are 10, 13, and 59, respectively. The graph of g has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. Let h be the function defined by $h(x) = \int_0^x g(t) \, dt$ for $-3.5 \leq x \leq 5.5$.

- Find the x -coordinate of each critical point of h on the interval $-3.5 < x < 5.5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- Find the x -coordinate of each point of inflection for the graph of h on the interval $-3.5 < x < 5.5$. Justify your answer.
- Find the value of $h(5)$.
- Find the value of $h(-3)$. Show the computations that lead to your answer.
- Must there exist a value of p , for $-3 < p < 2$, such that $h'(p)$ is equal to the average rate of change of h over the interval $-3 \leq x \leq 2$? Justify your answer.
- Find $\lim_{x \rightarrow -3} \frac{h(x)}{2x}$. Show the computations that lead to your answer.

6-5

(h) The function f is defined by $f(x) = x \cdot h(x)$. It is known that $g(-3) = 18$. Find $f'(-3)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -2$, $x = 0$, and $x = 5$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3.5, 5.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3.5, 5.5)$. The point is still earned provided the three critical points, $x = -2$, $x = 0$, and $x = 5$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-2, 0)$, $(0, 0)$, $(5, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -2$, $x = 0$, and $x = 5$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g , then:

- the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

- the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$. Any additional error (for example, not justifying that at $x = -2$, h has a relative maximum) is

6-5

eligible for at most one of the three points.

Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-2)$ changes from positive to negative”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

- $h(-3) = 3$
- $h(-2) = 13$
- $h(0) = 0$
- $h(5) = 59$
- $h(5.5) < 59$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = -2$, h has a relative maximum because $h' = g$ is positive for $x < -2$ and $h' = g$ is negative for $x > -2$.
- At $x = 0$, h has a relative minimum because $h' = g$ is negative for $x < 0$ and $h' = g$ is positive for $x > 0$.
- At $x = 5$, h has a relative maximum because $h' = g$ is positive for $x < 5$ and $h' = g$ is negative for $x > 5$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

6-5

- ☐ Relative maximum with justification
- ☐ Relative minimum with justification
- ☐ Relative maximum with justification

Solution:

At $x = -2$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = 0$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = 5$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only h' is eligible for all three points.

If a response gives justifications based only on g , then (in parallel with the approach followed in part (b)):

- the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

- the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$.

Justification for the first two points should reference one of the following:

- $h'(x)$ changing from increasing to decreasing, or vice versa
- relative extrema of $h'(x)$
- $h''(x)$ changes sign
- $g'(x)$ changes sign

“ $h'(x) = g(x)$ changes direction” earns neither of the first two points.

Read references to “the graph” as references to g as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn both points.

6-5

A response that discusses h' at a particular point (i.e., $h'(-1)$) when the discussion should have been about $h'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x = -1$ and $x = 3$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ $x = -1$ with justification
- ☐ $x = 3$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = -1$, the graph of h has a point of inflection because $h'(x) = g(x)$ changes from decreasing to increasing there.

At $x = 3$, the graph of h has a point of inflection because $h'(x) = g(x)$ changes from increasing to decreasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 59 is all that is needed to earn the point. No supporting work is required.



0	1
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The student response accurately includes the criteria below.

- ☐ Answer

Solution:

6-5

$$h(5) = \int_0^5 g(t) dt = 59$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 10 and ± 13 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 10) \pm (\pm 13)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 10 and ± 13 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-10 - (-13)$ earns both points because numerical expressions need not be simplified.
- The expression $(-10) - 13$ earns the first point, but not the second point.
- The expression $-10 - (-13) - 3$ earns neither point.
- The expression $\int_0^{-3} g(t) dt = 3$ earns neither point.
- The expression $-\int_{-3}^{-2} g(t) dt - \int_{-2}^0 g(t) dt = 3$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 10 and ± 13 , but it does earn the second point.
- $-10, 13$ earns neither point



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ $\left| \int_{-2}^{-3} g(t) dt \right| = 10$ and $\left| \int_{-2}^0 g(t) dt \right| = 13$
- ☐ Answer

Solution:

$$\begin{aligned}
 h(-3) &= \int_0^{-3} g(t) dt = - \int_{-3}^0 g(t) dt \\
 &= - \int_{-3}^{-2} g(t) dt - \int_{-2}^0 g(t) dt = -10 - (-13) = 3
 \end{aligned}$$

6-5

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-3, 2)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $h(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(p)$ equals the average rate of change of $h(x)$ ” earns both points.
- “Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(p)$ equals the average rate of change of $h(x)$ ” does not earn the first point, but does the earn second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $h(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $h(x)$ is continuous” earns neither point.
- “Yes, because $h(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$$h' = g \Rightarrow h \text{ is differentiable on } -3 \leq x \leq 2. \Rightarrow h \text{ is continuous on } -3 \leq x \leq 2.$$

6-5

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 2$ to guarantee that there exists a value of p , for $-3 < p < 2$, such that $h'(p)$ equals the average rate of change of h over the interval $-3 \leq x \leq 2$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

$$\cdot \frac{h(-3)}{-6} = \frac{3}{-6}$$

$$\cdot \frac{3}{-6}$$

Responses that do not earn the point are:

$$\cdot \frac{h(-3)}{-6}$$

$$\cdot \frac{h(-3)}{2 \cdot (-3)}$$

Note: an incorrect value for $h(-3)$ can be brought in from part (e) for the numerator of the fraction to earn the point.

The response does not need to state that h is continuous. If the response does state that h is continuous, an explanation is not required.

The response does not need to use limit notation.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h' = g \Rightarrow h \text{ is continuous.} \Rightarrow \lim_{x \rightarrow -3} h(x) = h(-3)$$

$$\lim_{x \rightarrow -3} \frac{h(x)}{2x} = \frac{h(-3)}{2 \cdot (-3)} = \frac{3}{-6} = -\frac{1}{2}$$

Part H

6-5

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of f that contains a sum of two terms, with at least one term written as a product, and one term involving h and the other term involving h' (or g).

Fundamental Theorem of Calculus is evidenced by linking h' and g either directly or indirectly in the presented derivative of $f(x)$ or in $f'(-3)$. Since $g(-3) = 18$ is given in the question, replacing $h'(-3)$ directly with 18 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $h(-3)$ that is brought in from part (e) can earn the third point.

$-3 \cdot 18 + 3$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c$ where any of a , b , or c is incorrect, will earn zero points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$f'(x) = x \cdot h'(x) + 1 \cdot h(x)$$

$$= x \cdot g(x) + h(x)$$

$$f'(-3) = -3 \cdot g(-3) + h(-3) = -3 \cdot (18) + 3 = -51$$

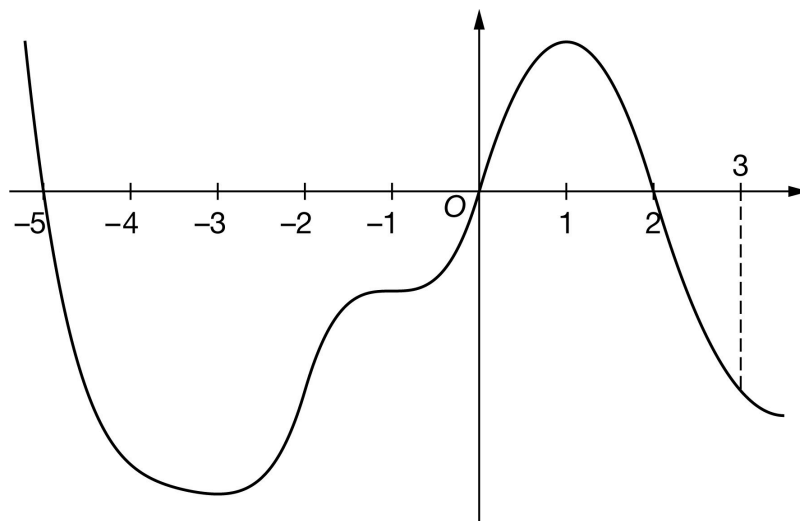
6-5

37. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The graph of the differentiable function f with domain $-5.5 \leq x \leq 3.5$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of f on the intervals $[-5, 0]$, $[0, 2]$, and $[2, 3]$ are 57, 12, and 7, respectively. The graph of f has horizontal tangents at $x = -3$, $x = -1$, and $x = 1$. Let g be the function defined by $g(x) = \int_2^x f(t) dt$ for $-5.5 \leq x \leq 3.5$.

- Find the x -coordinate of each critical point of g on the interval $-5.5 < x < 3.5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify each of your answers.
- Find the x -coordinate of each point of inflection for the graph of g on the interval $-5.5 < x < 3.5$. Justify your answer.
- Find the value of $g(3)$.
- Find the value of $g(-5)$. Show the computations that lead to your answer.
- Must there exist a value of k , for $-5 < k < 0$, such that $g'(k)$ is equal to the average rate of change of g over the interval $-5 \leq x \leq 0$? Justify your answer.
- Find $\lim_{x \rightarrow 3} \frac{5g(x)}{2x}$. Show the computations that lead to your answer.

6-5

(h) The function r is defined by $r(x) = x^2 \cdot g(x)$. It is known that $f(3) = -12$. Find $r'(3)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -5$, $x = 0$, and $x = 2$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-5.5, 3.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-5.5, 3.5)$. The point is still earned provided the three critical points, $x = -5$, $x = 0$, and $x = 2$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-5, 0)$, $(0, 0)$, $(2, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = f(x) = 0$ at $x = -5$, $x = 0$, and $x = 2$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on f , then:

· the response is eligible for all three points if the response presents the connection between $g'(x)$ and $f(x)$ (that is, $g'(x) = f(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or g' as $f(x) = g'(x)$, throughout all parts of this question.

6-5

· the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = f(x)$. Any additional error (for example, not justifying that at $x = -5$, g has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(-5)$ changes from positive to negative”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

- $h(-5.5) < 45$
- $h(-5) = 45$
- $h(0) = -12$
- $h(2) = 0$
- $h(3) = -7$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = -5$, g has a relative maximum because $g' = f$ is positive for $x < -5$ and $g' = f$ is negative for $x > -5$.
- At $x = 0$, g has a relative minimum because $g' = f$ is negative for $x < 0$ and $g'(x) = f(x)$ is positive for $x > 0$.
- At $x = 2$, g has a relative maximum because $g' = f$ is positive for $x < 2$ and $g' = f$ is negative for $x > 2$.



0	1	2	3
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6-5

The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification
- ☐ Relative minimum with justification
- ☐ Relative maximum with justification

Solution:

At $x = -5$, g Alt text: g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

At $x = 0$, g Alt text: g has a relative minimum because $g'(x) = f(x)$ changes from negative to positive there.

At $x = 2$, g Alt text: g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only g' is eligible for all three points.

If a response gives justifications based only on f , then (in parallel with the approach followed in part (b)):

- the response is eligible for all three points if the response presents the connection between $g'(x)$ and $f(x)$ (that is, $g'(x) = f(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of f or g' as $f(x) = g'(x)$, throughout all parts of this question.

- the response is eligible for at most two of the three points if the response does not present the connection $g'(x) = f(x)$.

Justification for the first two points should reference one of the following:

- $g'(x)$ changing from increasing to decreasing, or vice versa
- relative extrema of $g'(x)$
- $g''(x)$ changes sign
- $f'(x)$ changes sign

“ $g'(x) = f(x)$ changes direction” earns neither of the first two points.

Read references to “the graph” as references to f as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of “the function”

6-5

or “it” or “the slope” with g' produces a response that would earn both points.

A response that discusses g' at a particular point (i.e., $g'(-3)$) when the discussion should have been about $g'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x = -3$ and $x = 1$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ $x = -3$ with justification
- ☐ $x = 1$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = -3$, the graph of g has a point of inflection because $g'(x) = f(x)$ changes from decreasing to increasing there.

At $x = 1$, the graph of g has a point of inflection because $g'(x) = f(x)$ changes from increasing to decreasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -7 is all that is needed to earn the point. No supporting work is required.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

6-5

$$g(3) = \int_2^3 f(t) dt = -7$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 57 and ± 12 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 57) \pm (\pm 12)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 57 and ± 12 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-(-57) - (-12)$ earns both points because numerical expressions need not be simplified.
- The expression $(-57) - 12$ earns the first point, but not the second point.
- The expression $-57 - (-12) - 7$ earns neither point.
- The expression $\int_2^{-5} f(t) dt = 45$ earns neither point.
- The expression $-\int_{-5}^0 f(t)dt - \int_0^2 f(t)dt = 45$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 57 and ± 12 , but it does earn the second point.
- $-57, -12$ earns neither point



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ $\left| \int_{-5}^0 f(t) dt \right| = 57$ and $\left| \int_2^0 f(t) dt \right| = 12$
- ☐ Answer

Solution:

6-5

$$\begin{aligned}
 g(-5) &= \int_2^{-5} f(t) \, dt = - \int_{-5}^2 f(t) \, dt \\
 &= - \int_{-5}^0 f(t) \, dt - \int_0^2 f(t) \, dt = -(-57) - 12 = 45
 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-5, 0)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $g(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $g(x)$ is continuous and differentiable, therefore there must be a value where $g'(k)$ equals the average rate of change of $g(x)$ ” earns both points.
- “Yes, $g'(x)$ (the derivative of $g(x)$) is continuous and differentiable, therefore there must be a value where $g'(k)$ equals the average rate of change of $g(x)$ ” does not earn the first point, but does earn second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $g(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $g(x)$ is continuous” earns neither point.
- “Yes, because $g(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

6-5

Solution:

$g' = f \Rightarrow g$ is differentiable on $-5 \leq x \leq 0 \Rightarrow g$ is continuous on $-5 \leq x \leq 0$.

Therefore, the Mean Value Theorem can be applied to g on the interval $-5 \leq x \leq 0$ to guarantee that there exists a value of k , for $-5 < k < 0$, such that $g'(k)$ equals the average rate of change of g over the interval $-5 \leq x \leq 0$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the first point include:

$$\begin{aligned} & \cdot \frac{5(-7)}{2 \cdot 3} \\ & \cdot -\frac{35}{6} \end{aligned}$$

Responses that do not earn the first point include:

$$\begin{aligned} & \cdot \frac{5 \cdot g(3)}{6} \\ & \cdot \frac{5 \cdot g(3)}{2 \cdot 3} \end{aligned}$$

Note: an incorrect value for $g(3)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that g is continuous. If the response does state that g is continuous, an explanation is not required.

The response does not need to use limit notation.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$g' = f \Rightarrow g$ is continuous. $\Rightarrow \lim_{x \rightarrow 3} g(x) = g(3)$

6-5

$$\lim_{x \rightarrow 3} \frac{5g(x)}{2x} = \frac{5g(3)}{2 \cdot (3)} = \frac{5(-7)}{6} = -\frac{35}{6}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of r that contains a sum of two terms, with at least one term written as a product, and one term involving g and the other term involving g' (or f).

Fundamental Theorem of Calculus is evidenced by linking g' and f either directly or indirectly in the presented derivative of $r(x)$ or in $r'(3)$. Since $f(3) = -12$ is given in the question, replacing $g'(3)$ directly with -12 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $g(3)$ that is brought in from part (d) can earn the third point.

$9(-12) + 6(-7)$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c \cdot d$ where any of a , b , c , or d is incorrect, will earn zero points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$r'(x) = x^2 \cdot g'(x) + 2x \cdot g(x)$$

$$= x^2 \cdot f(x) + 2x \cdot g(x)$$

$$r'(3) = 3^2 \cdot f(3) + 2 \cdot 3 \cdot g(3) = 9 \cdot (-12) + 6 \cdot (-7) = -150$$

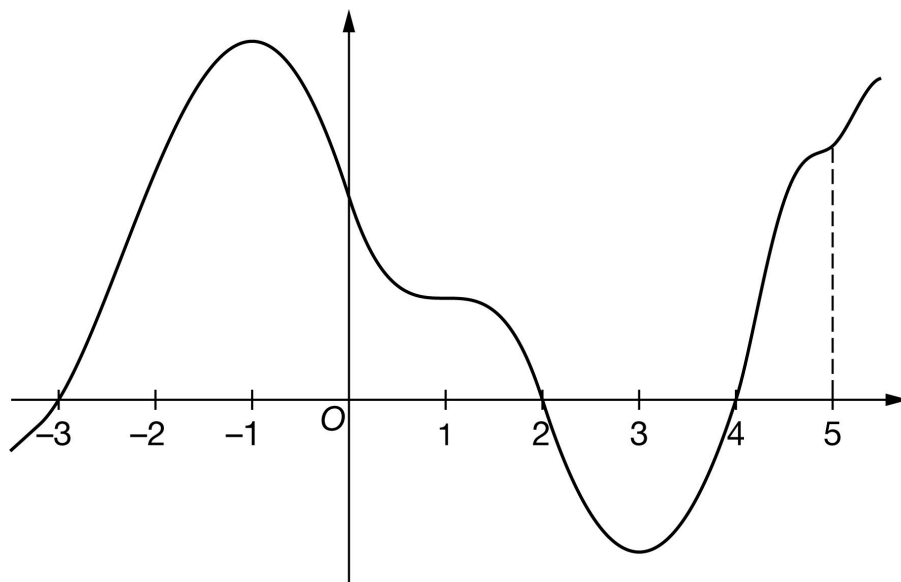
6-5

38. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the differentiable function g with domain $-3.5 \leq x \leq 5.5$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of g on the intervals $[-3, 2]$, $[2, 4]$, and $[4, 5]$ are 55, 12, and 10, respectively. The graph of g has horizontal tangents at $x = -1$, $x = 1$, and $x = 3$. Let h be the function defined by $h(x) = \int_4^x g(t) \, dt$ for $-3.5 \leq x \leq 5.5$.

$h(x) = \int_4^x g(t) \, dt$ for $-3.5 \leq x \leq 5.5$.

- Find the x -coordinate of each critical point of h on the interval $-3.5 < x < 5.5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- Find the x -coordinate of each point of inflection for the graph of h on the interval $-3.5 < x < 5.5$. Justify your answer.
- Find the value of $h(5)$.
- Find the value of $h(-3)$. Show the computations that lead to your answer.
- Must there exist a value of r , for $-3 < r < 2$, such that $h'(r)$ is equal to the average rate of change of h over the interval $-3 \leq x \leq 2$? Justify your answer.
- Find $\lim_{x \rightarrow 5} \frac{h(x)}{2x}$. Show the computations that lead to your answer.

6-5

(h) The function k is defined by $k(x) = x \cdot h(x)$. It is known that $g(5) = 15$. Find $k'(5)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3$, $x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3.5, 5.5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3.5, 5.5)$. The point is still earned provided the three critical points $x = -3$, $x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(2, 0)$, $(4, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -3$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g , then:

- the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.

- the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative minimum) is

6-5

eligible for at most one of the three points.

Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(3)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn these points, all three critical points must be included in the table or list and the relevant endpoint must be considered. The correct values of relevant inputs include:

- $h(-3.5) > -43$
- $h(-3) = -43$
- $h(2) = 12$
- $h(4) = 0$
- $h(5) = 10$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = -3$, h has a relative minimum because $h' = g$ is negative for $x < -3$ and $h' = g$ is positive for $x > -3$.
- At $x = 2$, h has a relative maximum because $h' = g$ h prime equals g is positive for $x < 2$ and $h' = g$ is negative for $x > 2$.
- At $x = 4$, h has a relative minimum because $h' = g$ is negative for $x < 4$ and $h' = g$ is positive for $x > 4$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

6-5

- ☐ Relative minimum with justification
- ☐ Relative maximum with justification
- ☐ Relative minimum with justification

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = 2$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = 4$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each point of inflection must be accompanied by a correct justification to earn the point.

A response giving justification based only h' is eligible for all three points.

If a response gives justifications based only on g , then (in parallel with the approach followed in part (b)):

- the response is eligible for all three points if the response presents the connection between $h'(x)$ and $g(x)$ (that is, $h'(x) = g(x)$). If you do not see this in part (c), look for it in part (a) and part (b). If you find this connection in either part, read all subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.
- the response is eligible for at most two of the three points if the response does not present the connection $h'(x) = g(x)$.

Justification for the first two points should reference one of the following:

- $h'(x)$ changing from increasing to decreasing, or vice versa
- relative extrema of $h'(x) = g(x)$
- $h''(x)$ changes sign
- $g'(x)$ changes sign

“ $h'(x) = g(x)$ changes direction” earns neither of the first two points.

Read references to “the graph” as references to g as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one of the first two points. This type of response earns one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn both points.

6-5

A response that discusses h' at a particular point (i.e., $h'(-3)$) when the discussion should have been about $h'(x)$ can earn only one of the first two points, with one exception. This communication error should not be penalized twice. If a response failed to earn a point for this error in part (b), it will not lose a point in this part.

If $x = -1$ and $x = 3$ are both listed as points of inflection and no other numbers are listed as points of inflection, the third point is earned. A response does not need to say that there are only two inflection points if only two points are listed.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ $x = -1$ with justification
- ☐ $x = 3$ with justification
- ☐ Response includes no other points of inflection

Solution:

At $x = -1$, the graph of h has a point of inflection because $h'(x) = g(x)$ changes from increasing to decreasing there.

At $x = 3$, the graph of h has a point of inflection because $h'(x) = g(x)$ changes from decreasing to increasing there.

There are no other points of inflection.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 10 is all that is needed to earn the point. No supporting work is required.



0	1
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The student response accurately includes the criteria below.

- ☐ Answer

Solution:

6-5

$$h(5) = \int_4^5 g(t) dt = 10$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned when the response identifies ± 55 and ± 12 either linked to their corresponding definite integrals or combined as a sum or difference, often expressed by the sum (or difference) $(\pm 55) \pm (\pm 12)$. The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 55 and ± 12 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-55 - (-12)$ earns both points because numerical expressions need not be simplified.
- The expression $(-55) - 12$ earns the first point, but not the second point.
- The expression $-55 - (-12) - 8$ earns neither point.
- The expression $\int_4^{-3} g(t) dt = -43$ earns neither point.
- The expression $-\int_{-3}^2 g(t) dt - \int_2^4 g(t) dt = -43$ or any other sum or difference of correct symbolic integrals does not earn the first point because there is no representation of ± 55 and ± 12 , but it does earn the second point.
- $-55, 12$ earns neither point



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ $\left| \int_2^{-3} g(t) dt \right| = 55$ and $\left| \int_2^4 g(t) dt \right| = 12$
- ☐ Answer

Solution:

6-5

$$\begin{aligned}
 h(-3) &= \int_4^{-3} g(t) \, dt = - \int_{-3}^4 g(t) \, dt \\
 &= - \int_{-3}^2 g(t) \, dt - \int_2^4 g(t) \, dt = -55 - (-12) = -43
 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must contain the interval $(-3, 2)$ and be a subset of the domain referenced in the stem of the problem with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., by either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $h(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(r)$ equals the average rate of change of $h(x)$ ” earns both points.
- “Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(r)$ equals the average rate of change of $h(x)$ ” does not earn the first point, but does earn second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $h(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $h(x)$ is continuous” earns neither point.
- “Yes, because $h(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

6-5

Solution:

$h' = g \Rightarrow h$ is differentiable on $-3 \leq x \leq 2. \Rightarrow h$ is continuous on $-3 \leq x \leq 2$.

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 2$ to guarantee that there exists a value of r , for $-3 < r < 2$, such that $h'(r)$ equals the average rate of change of h over the interval $-3 \leq x \leq 2$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer supported by some work that must include a fraction. Responses that earn the point include:

$$\cdot \frac{h(5)}{10} = 1$$

$$\cdot \frac{10}{10}$$

Responses that do not earn the point include:

$$\cdot \frac{h(5)}{10}$$

$$\cdot 1$$

Note: an incorrect value for $h(5)$ can be brought in from part (d) for the numerator of the fraction to earn the point.

The response does not need to state that h is continuous. If the response does state that h is continuous, an explanation is not required.

The response does not need to use limit notation.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h' = g \Rightarrow h, \text{ is continuous. } \Rightarrow \lim_{x \rightarrow 5} h(x) = h(5)$$

6-5

$$\lim_{x \rightarrow 5} \frac{h(x)}{2x} = \frac{h(5)}{2 \cdot 5} = \frac{10}{10} = 1$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first (product rule) point is awarded for sufficient evidence of an attempt at the product rule. The evidence required is a derivative of k that contains a sum of two terms, with at least one term written as a product, and one term involving h and the other term involving h' (or g).

Fundamental Theorem of Calculus is evidenced by linking h' and g either directly or indirectly in the presented derivative of $k(x)$ or in $k'(5)$. Since $g(5) = 15$ is given in the question, replacing $h'(5)$ directly with 15 without an intermediate step earns the second point.

It is possible to earn the second point without earning the first point.

A response that contains an incorrect value for $h(5)$ that is brought in from part (d) can earn the third point.

$5 \cdot 15 + 10$ is the minimal answer that earns all three points. If there is no supporting work, an answer of the form $a \cdot b + c$ where any of a , b , or c is incorrect, will earn zero points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$k'(x) = x \cdot h'(x) + 1 \cdot h(x)$$

$$= x \cdot g(x) + h(x)$$

$$k'(5) = 5g(5) + h(5) = 5 \cdot 15 + 10 = 85$$

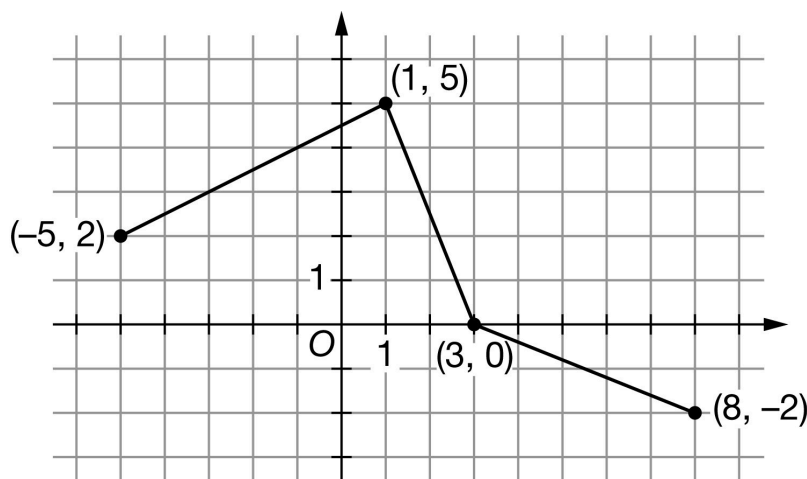
6-5

39. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the function g on the interval $-5 \leq x \leq 8$ consists of three line segments, as shown in the figure above. Let m be the function defined by $m(x) = \int_1^x g(t) \, dt$ for $-5 \leq x \leq 8$.

- For the function m , is $x = 3$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute maximum value of m on the interval $-5 \leq x \leq 8$. Justify your answer.
- On what open intervals contained in $-5 < x < 8$ is the graph of m both increasing and concave down? Give a reason for your answer.
- Find the value of $m''(1)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow 1} \frac{m(x)}{x+1}$. Show the computations that lead to your answer.
- Let k be the function defined by $k(x) = m(x^2)$. Find the value of $k'(1)$ or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-5

The response must include the wording local maximum or relative maximum. We also need to see a connection to g . The obvious is $m' = g$. A response of local maximum with the explanation: " g (or the graph) changes from positive to negative" is an implied connection. A response of local maximum with explanation: " m' changes from positive to negative" without the connection of $m' = g$ is not acceptable.

Simply saying " m changes from increasing to decreasing" is not acceptable.

In this situation, the Second Derivative Test $m''(3) < 0$ is an incorrect statement and the response with this reason does not earn the point.



0	1
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The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 3$ is the location of a relative maximum of m because $m'(x) = g(x)$ changes from positive to negative there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = 3$ is the only critical point. (g is positive on $(-5, 3)$ and negative on $(3, 8)$ is another way to state this.)

Note: Stating that $x = 3$ is the only relative maximum critical point is not enough.

The second point is for arguing that $x = 3$ then produces the absolute maximum. Note that the response does not have to repeat the justification of $x = 3$ being a relative maximum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = 3$. This consideration is more than just the statement $x = 3$. The response must use $x = 3$ and an area calculation in trying to find $m(3)$.

6-5

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as difference of $g(x)$ values.

The second point is for consideration of the endpoints. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $g(x)$ values.

Common errors that show an attempt to find area:

- $m(-5) = -9$ (finding the area of the triangle and forgetting the rectangle)
- $m(8) = 10$ (adding the triangle areas). These can be used to earn the second point.

A correct argument that eliminates $x = -5$ by stating $m(-5)$ is negative will earn the second point as long as $x = 8$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -5$, $x = 3$, and $x = 8$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no calculation mistakes in intermediate work.

NOTE: A coordinate point is not an acceptable answer.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute maximum
- ☐ Answer with justification
- ☐ Considers $x = 3$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

6-5

Because $x = 3$ is the only point on the interval $-5 \leq x \leq 8$ where $m'(x) = g(x) = 0$, $x = 3$ is the only critical point of m on the interval $-5 \leq x \leq 8$.

Because $x = 3$ is also the location of a relative maximum for m , $m(3)$ is the absolute maximum for m in $-5 \leq x \leq 8$.

$$m(3) = \int_1^3 g(t) dt = 5$$

Therefore, the absolute maximum value of m on $-5 \leq x \leq 8$ is $m(3) = 5$.

Alternate solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$x = 3$ is the only critical point in the interval because it is the only point where $m'(x) = g(x) = 0$.

$$m(3) = \int_1^3 g(t) dt = 5$$

$$m(-5) = \int_1^{-5} g(t) dt = -\int_{-5}^1 g(t) dt = -21$$

$$m(8) = \int_1^8 g(t) dt = 0$$

Therefore, the absolute maximum value of m on $-5 \leq x \leq 8$ is $m(3) = 5$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for providing the interval $(1, 3)$ or inequality $1 < x < 3$.

We will allow $[1, 3]$, $(1, 3]$, $[1, 3)$, or an equivalent inequality.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons m' is positive and decreasing or m' is positive and m'' is negative, will earn the second point if the connection $m' = g$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function m , then the second point is not earned.



0	1	2
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The student response accurately includes **both** of the criteria below.

6-5

- ☐ Answer
- ☐ Reason

Solution:

m is increasing where $m' = g$ is positive.

m is concave down where $m' = g$ is decreasing.

The graph of m is increasing and concave down on the open interval $(1, 3)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $m''(x) = g'(x)$.

We are not allowing an import of a previous connection between m'' and g' .

The explicit connection, $m''(x) = g'(x)$, earns the first point.

It is not assumed that a response which only presents $m'(x) = g(x)$ has made that connection.

The implicit connection will be made in the explanation with a connection to g' , the differentiability of g , or the characteristic of the graph of g .

The second point is for the answer with explanation.

This must include "Does not exist" and one of the following:

- limits involving the difference quotient that appears in these scoring guidelines,
- limits involving g' as the argument of the limit,
- or some correct verbal description of the characteristic of the graph of g at $x = 1$.

Note that all of the above descriptions earn the first point for the implicit consideration of $g'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: g' does not exist because g (or "the graph") has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of g . The phrase $m''(1)$ of 1 does not exist because there is a cusp will not earn this point. Whereas, the phrase $m''(1)$ does not exist because there is a cusp on g is acceptable and earns both points.

6-5



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Considers $g'(x)$
- ☐ Answer with explanation

Solution:

$$m''(x) = g'(x)$$

Because $\lim_{x \rightarrow 1^-} \frac{g(x) - g(1)}{x - 1} = \frac{1}{2}$ and $\lim_{x \rightarrow 1^+} \frac{g(x) - g(1)}{x - 1} = -\frac{5}{2}$, $g'(1) = m''(1)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of m does not need to be stated.

An answer of $\frac{m(1)}{2}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{2}$.

Copy/calculation errors in the denominator will not earn the point. Disregard notation errors.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$m \text{ is continuous. } \Rightarrow \lim_{x \rightarrow 1} m(x) = m(1)$$

6-5

$$\lim_{x \rightarrow 1} \frac{m(x)}{x+1} = \frac{m(1)}{1+1} = \frac{0}{2} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $k : m'(x^2) \cdot 2x$.

This point is earned for any of $m'(x^2) \cdot 2x$, $g(x^2) \cdot 2x$, $g(1) \cdot 2$, or $5 \cdot 2$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$$m'(x^2) = g(x^2), m'(1) = g(1), \text{ or } m'(1) = 5.$$

This point would be earned with a response of $g(1) \cdot 2$, or $5 \cdot 2$, even with no preceding work.

The third point is for either $5 \cdot 2$ or 10. Note, the product need not be simplified.

An answer of 10 with no supporting work earns no points in part (f).



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$k'(x) = m'(x^2) \cdot 2x = g(x^2) \cdot 2x$$

$$k'(1) = g(1) \cdot 2 = 5 \cdot 2 = 10$$

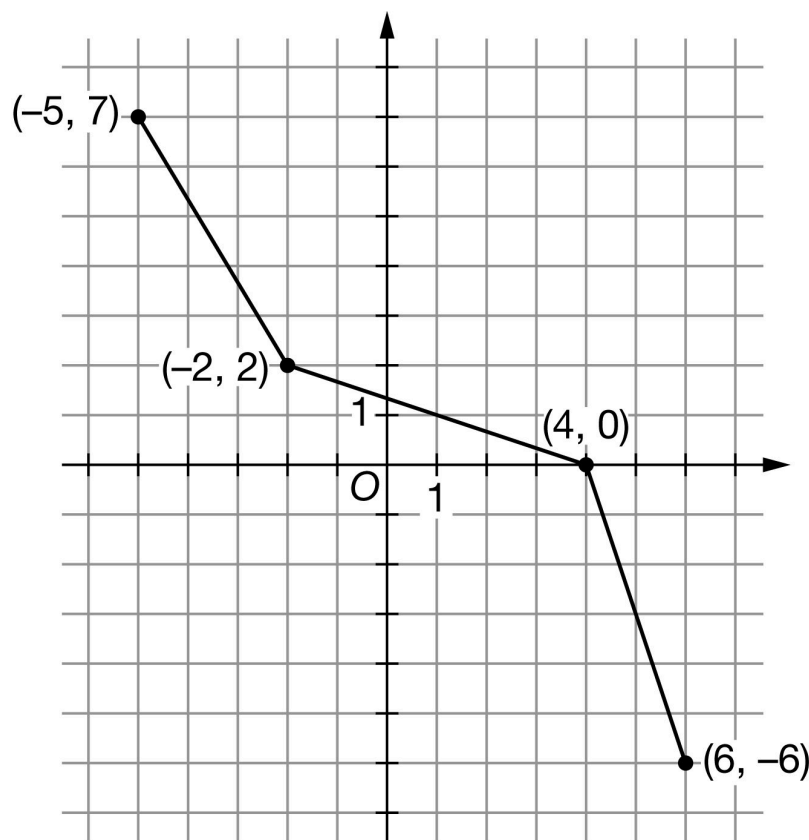
6-5

40. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The graph of the function f on the interval $-5 \leq x \leq 6$ consists of three line segments, as shown in the figure above. Let h be the function defined by $h(x) = \int_{-2}^x f(t) \, dt$ for $-5 \leq x \leq 6$.

- For the function h , is $x = 4$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute maximum value of h on the interval $-5 \leq x \leq 6$. Justify your answer.
- On what open intervals contained in $-5 < x < 6$ is the graph of h both increasing and concave down? Give a reason for your answer.
- Find the value of $h''(-2)$, or explain why it does not exist.

6-5

(e) Find $\lim_{x \rightarrow -2} \frac{h(x)}{x-2}$. Show the computations that lead to your answer.

(f) Let k be the function defined by $k(x) = h(x^2)$. Find the value of $k'(\sqrt{5})$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{5}$ can be keyboarded as sqrt(5))

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local maximum or relative maximum. We also need to see a connection to h . The obvious is $h' = f$. A response of local maximum with the explanation: " f (or the graph) changes from positive to negative" is an implied connection. A response of local maximum with explanation: " h' changes from positive to negative," without the connection of $h' = f$ will not earn the point.

Simply saying " h . changes from increasing to decreasing" is not enough to earn the point.

In this situation, appealing to the Second Derivative Test by claiming that $h''(4) < 0$ is an incorrect statement, so a response with this reason does not earn the point.



0	1
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The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 4$ is the location of a relative maximum of h because $h'(x) = f(x)$ changes from positive to negative at this point.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = 4$ is the only critical point. (f is positive on $(-5, 4)$ and negative on $(4, 6)$ is another way to state this.)

Note: Stating that $x = 4$ is the only relative maximum critical point is not enough.

The second point is for arguing that $x = 4$ then produces the absolute maximum. Note that the response does not have to repeat the justification of $x = 4$ being a relative maximum from part (a). If this is not in part (a), then it needs to be here.

6-5

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct value of the maximum.

The presentation of a coordinate point $(4, 6)$ as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = 4$. This consideration is more than just the statement $x = 4$. The response must use $x = 4$ and an area calculation in trying to find $h(4)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $f(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $f(x)$ values.

The second point is for consideration of the endpoints. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find areas for $h(-5)$ and $h(6)$.

The response cannot simply write the integrals with the correct upper limits. Attempts at both areas must be made.

Using the y -values of the coordinate points from $f(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $f(x)$ values.

Common errors that show an attempt to find area:

· $h(6) = 12$ (adding the triangle areas)

· $Ah(-5) = -\frac{15}{2}$ (finding the area of the triangle and forgetting the rectangle). These can be used to earn the second point.

A correct argument that eliminates $x = -5$ by stating $h(-5)$ is negative will earn the second point as long as $x = 6$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -5$, $x = 4$, and $x = 6$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no numerical calculation mistakes in intermediate work.

A coordinate point $(4, 6)$ is not an acceptable answer.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

6-5

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute maximum
- ☐ Answer with justification
- ☐ Considers $x = 4$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

Because $x = 4$ is the only point on the interval $-5 \leq x \leq 6$ where $h'(x) = f(x) = 0$, $x = 4$ is the only critical point of h on the interval $-5 \leq x \leq 6$.

Because $x = 4$ is also the location of a relative maximum of h , $h(4)$ is the absolute maximum for h on $-5 \leq x \leq 6$.

$$h(4) = \int_{-2}^4 f(t) dt = 6$$

Therefore, the absolute maximum value of h on $-5 \leq x \leq 6$ is $h(4) = 6$.

Alternate solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$x = 4$ is the only critical point in the interval because it is the only point where $h'(x) = f(x) = 0$.

$$h(4) = \int_{-2}^4 f(t) dt = 6$$

$$h(-5) = \int_{-2}^{-5} f(t) dt = -\int_{-5}^{-2} f(t) dt = -13.5$$

$$h(6) = \int_{-2}^6 f(t) dt = 0$$

Therefore, the absolute maximum value of h on $-5 \leq x \leq 6$ is $h(4) = 6$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for providing the interval $(-5, 4)$ or inequality $-5 < x < 4$.

We will allow $[-5, 4]$, $(-5, 4]$, $[-5, 4)$, or an equivalent inequality.

6-5

Presenting $(-5, -2) \cup (-2, 4)$ with correct reasoning earns one out of the two points. Any accepted version (open parentheses or brackets) earns the point. The “ $\cup$ ” is not necessary. Be careful, one bracket on either -2 is actually the entire interval and is eligible for both points.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons “ h' is positive and decreasing” or “ h' is positive and h'' is negative” will earn the second point if the connection $h' = f$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function h , then the second point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

h is increasing where $h' = f$ is positive.

h is concave down where $h' = f$ is decreasing.

The graph of h is both increasing and concave down on the open interval $(-5, 4)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $h''(x) = f'(x)$.

We are not allowing an import of a previous connection between h'' and f' .

The explicit connection, $h''(x) = f'(x)$, earns the first point.

It is not assumed that a response which only presents $h'(x) = f(x)$ has made the connection between h'' and f' .

The second point is for the answer with explanation.

This must include “Does not exist” and one of the following:

- limits involving the difference quotient that appear in these scoring guidelines,
- limits involving h' as the argument of the limit,

6-5

· or some correct verbal description of the characteristic of the graph of f at $x = -2$.

Note that all of the above descriptions earn the first point for the implicit consideration of $h''(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: f' does not exist because f (or "the graph") has a/is a (blank). Fill in the blank with corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of f . The phrase $h''(-2)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $h''(-2)$ does not exist because there is a cusp on f is acceptable and earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $f'(x)$
- ☐ Answer with explanation

Solution:

$$h''(x) = f'(x)$$

Because $\lim_{x \rightarrow -2^-} \frac{f(x) - f(-2)}{x - (-2)} = -\frac{5}{3}$ and $\lim_{x \rightarrow -2^+} \frac{f(x) - f(-2)}{x - (-2)} = -\frac{1}{3}$, $f'(-2) = h''(-2)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of h does not need to be stated.

An answer of $\frac{h(-2)}{-4}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{-4}$.

6-5

Copy/calculation errors in the denominator will not earn the point.

Disregard limit notation errors.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

h is continuous. $\Rightarrow \lim_{x \rightarrow -2} h(x) = h(-2)$

$$\lim_{x \rightarrow -2} \frac{h(x)}{x-2} = \frac{h(-2)}{-2-2} = \frac{0}{-4} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $k : h'(x^2) \cdot 2x$.

This point is earned for any of $h'(x^2) \cdot 2x$, $f(x^2) \cdot 2x$, $f(5) \cdot 2\sqrt{5}$, or $-3 \cdot 2\sqrt{5}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but could still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$h'(x^2) = f(x^2)$, $h'(5) = f(5)$, or $h'(5) = -3$.

This point would be earned with a response of $f(5) \cdot 2\sqrt{5}$ or $-3 \cdot 2\sqrt{5}$ even with no preceding work.

The third point is for either $-3 \cdot 2\sqrt{5}$ or $-6\sqrt{5}$. Note, the product need not be simplified.

An answer of $-6\sqrt{5}$ with no supporting work earns no points in part (f).



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

6-5

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$k'(x) = h'(x^2) \cdot 2x = f(x^2) \cdot 2x$$

$$k'(\sqrt{5}) = f(5) \cdot 2\sqrt{5} = -3 \cdot 2\sqrt{5} = -6\sqrt{5}$$

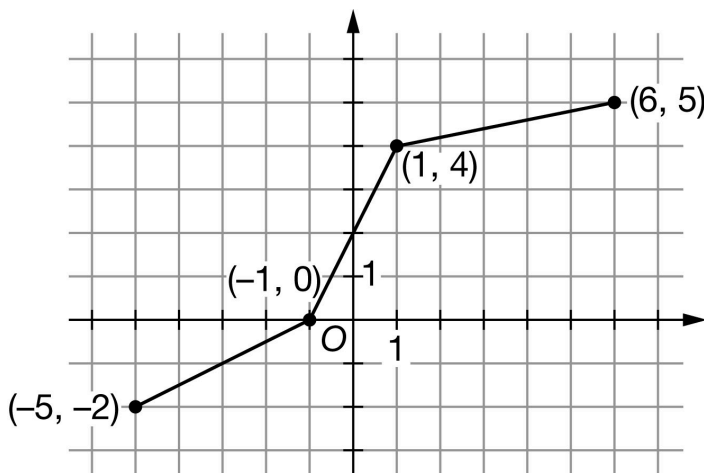
6-5

41. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h

The graph of the function h on the interval $-5 \leq x \leq 6$ consists of three line segments, as shown in the figure above. Let p be the function defined by $p(x) = \int_1^x h(t) \, dt$ for $-5 \leq x \leq 6$.

- For the function p , is $x = -1$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute minimum value of p on the interval $-5 \leq x \leq 6$. Justify your answer.
- On what open intervals contained in $-5 < x < 6$ is the graph of p both decreasing and concave up? Give a reason for your answer.
- Find the value of $p''(1)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow 1} \frac{p(x)}{x+4}$. Show the computations that lead to your answer.
- Let k be the function defined by $k(x) = p(x^2)$. Find the value of $k'(\sqrt{2})$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{2}$ can be keyboarded as sqrt(2))

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-5

The response must include the wording local minimum or relative minimum. We also need to see a connection to h . The obvious is $p' = h$. A response of local minimum with the explanation: " h (or the graph) changes from negative to positive" is an implied connection. A response of local minimum with explanation: " p' changes from negative to positive," without the connection of $p' = h$ is not acceptable.

Simply saying " p changes from decreasing to increasing" is not acceptable.

In this situation, the Second Derivative Test $p''(-1) > 0$ is an incorrect statement and a response with this reason would not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = -1$ is the location of a relative minimum of p because $p'(x) = h(x)$ changes from negative to positive there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = -1$ is the only critical point. (h is negative on $(-5, -1)$ and positive on $(-1, 6)$ is another way to state this.)

Note: Stating that $x = -1$ is the only relative minimum critical point is not enough.

The second point is for arguing that $x = -1$ then produces the absolute minimum. Note that they do not have to repeat the justification of $x = -1$ being a relative minimum from part (a). If this is not in (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = -1$. This consideration is more than just the statement $x = -1$. The response must use $x = -1$ and an area calculation in trying to find $p(-1)$.

6-5

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $h(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $h(x)$ values.

A common error that shows an attempt to find area is $k(-1) = 4$. This earns the first point.

The second point is for consideration of the endpoints. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $h(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as the difference of $h(x)$ values.

Common errors that show an attempt to find area:

· $p(-5) = 8$ (adding the triangle areas)

· $p(6) = \frac{5}{2}$ (finding the area of the triangle and forgetting the rectangle). These can be used to earn the second point.

A correct argument that eliminates $x = 6$ by stating $p(6)$ is positive will earn the second point as long as $x = -5$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -5$, $x = -1$, and $x = 6$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no calculation mistakes in intermediate work.

A coordinate point is not an acceptable answer.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute minimum
- ☐ Answer with justification
- ☐ Considers $x = -1$
- ☐ Considers both endpoints

6-5

☐ Answer with justification
Solution:

Because $x = -1$ is the only point on the interval $-5 \leq x \leq 6$ where $p'(x) = h(x) = 0$, $x = -1$ is the only critical point of p on the interval $-5 \leq x \leq 6$.

Because $x = -1$ is also the location of a relative minimum for p , $p(-1)$ is the absolute minimum for p on $-5 \leq x \leq 6$.

$$p(-1) = \int_1^{-1} h(t) dt = - \int_{-1}^1 h(t) dt = -4$$

Therefore, the absolute minimum value of p on $-3 \leq x \leq 6$ is $p(-1) = -4$.

Alternate solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$x = -1$ is the only critical point in the interval because it is the only point where $p'(x) = h(x) = 0$.

$$p(-1) = \int_1^{-1} h(t) dt = - \int_{-1}^1 h(t) dt = -4$$

$$p(-5) = \int_1^{-5} h(t) dt = - \int_{-5}^1 h(t) dt = 0$$

$$p(6) = \int_1^6 h(t) dt = 22.5$$

Therefore, the absolute minimum value of p on $-3 \leq x \leq 6$ is $p(-1) = -4$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the interval $(-5, -1)$ or inequality $-5 < x < -1$.

We will allow $[-5, -1]$, $(-5, -1]$, $[-5, -1)$, or an equivalent inequality.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons p' is negative and increasing or p' is negative and p'' is positive will earn the second point if the connection $p' = h$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function p , then the second point is not earned.

6-5



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

p is decreasing where $p' = h$ is negative.

p is concave up where $p' = h$ is increasing.

The graph of p is both decreasing and concave up on the open interval $(-5, -1)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $p''(x) = h'(x)$.

We are not allowing an import of a previous connection between p'' and h' .

The explicit connection, $p'' = h'$, earns the first point.

It is not assumed that a response which only presents $p' = h$ has made that connection.

The implicit connection will be made in the explanation with a connection to h' , the differentiability of h , or the characteristic of the graph of h .

The second point is for the answer with explanation

This must include "Does not exist" and one of the following:

- limits involving the difference quotient that appears in these scoring guidelines,
- limits involving h' as the argument of the limit,
- or some correct verbal description of the characteristic of the graph of h at $x = 1$.

Note that all of the above descriptions earn the first point for the implicit consideration of $h'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

6-5

Acceptable verbal descriptions: h' does not exist because h (or "the graph") has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of h . The phrase $p''(1)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $p''(1)$ does not exist because there is a cusp on h is acceptable and earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $h'(x)$
- ☐ Answer with explanation

Solution:

$$p''(x) = h'(x)$$

Because $\lim_{x \rightarrow 1^-} \frac{h(x) - h(1)}{x - 1} = 2$, and $\lim_{x \rightarrow 1^+} \frac{h(x) - h(1)}{x - 1} = \frac{1}{5}$, $h'(1) = p''(1)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of p does not need to be stated.

An answer of $\frac{p(1)}{5}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{5}$.

Copy/calculation errors in the denominator will not earn the point. Disregard notation errors



0	1
---	---

6-5

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$p \text{ is continuous. } \Rightarrow \lim_{x \rightarrow 1} p(x) = p(1) \lim_{x \rightarrow 1} \frac{p(x)}{x+4} = \frac{p(1)}{1+4} = \frac{0}{5} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of k : $p'(x^2) \cdot 2x$.

This point is earned for any of $p'(x^2) \cdot 2x$, $h(x^2) \cdot 2x$, $h(2) \cdot 2\sqrt{2}$, or $\frac{21}{5} \cdot 2\sqrt{2}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$$p'(x^2) = h(x^2), p'(2) = h(2), \text{ or } p'(2) = \frac{21}{5}.$$

This point would be earned with a response of $h(2) \cdot 2\sqrt{2}$, or $\frac{21}{5} \cdot 2\sqrt{2}$, even with no preceding work.

The third point is for either $\frac{21}{5} \cdot 2\sqrt{2}$ or $\frac{42}{5}\sqrt{2}$. Note, the product need not be simplified.

An answer of $\frac{42}{5}\sqrt{2}$ with no supporting work earns no points in part (f).



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$k'(x) = p'(x^2) \cdot 2x = h(x^2) \cdot 2x$$

6-5

$$k'(\sqrt{2}) = h(2) \cdot 2\sqrt{2} = \frac{21}{5} \cdot 2\sqrt{2} = \left(\frac{42}{5}\right)\sqrt{2}$$

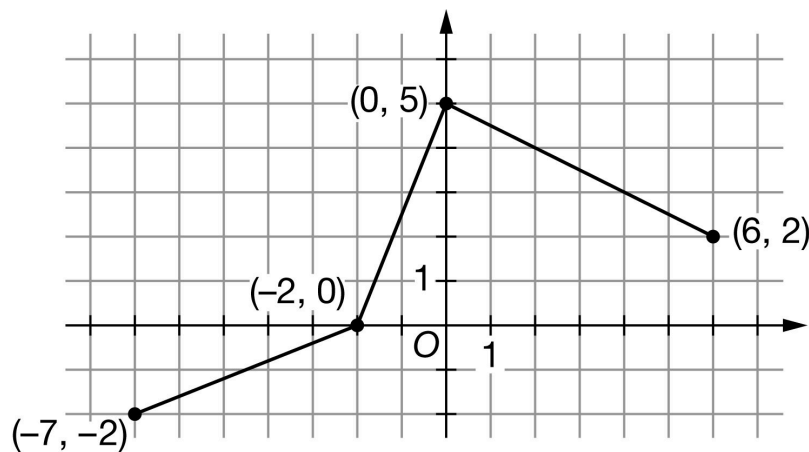
6-5

42. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h

The graph of the function h on the interval $-7 \leq x \leq 6$ consists of three line segments, as shown in the figure above. Let r be the function defined by $r(x) = \int_0^x h(t) \, dt$ for $-7 \leq x \leq 6$.

- For the function r , is $x = -2$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute minimum value of r on the interval $-7 \leq x \leq 6$. Justify your answer.
- On what open intervals contained in $-7 < x < 6$ is the graph of r both increasing and concave down? Give a reason for your answer.
- Find the value of $r''(-2)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow 0} \frac{r(x)}{x-4}$. Show the computations that lead to your answer.
- Let v be the function defined by $v(x) = r(x^2)$. Find the value of $v'(\sqrt{3})$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{3}$ can be keyboarded as sqrt(3))

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-5

The response must include the wording local minimum or relative minimum. We also need to see a connection to h . The obvious is $r' = h$. A response of local minimum with the explanation: " h (or the graph) changes from negative to positive" is an implied connection. A response of local minimum with explanation: " r' changes from negative to positive," without the connection of $r' = h$ is not acceptable.

Simply saying " r changes from decreasing to increasing" is not acceptable.

In this situation, the Second Derivative Test $r''(2) > 0$ is an incorrect statement, and a response with this reason does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = -2$ is the location of a relative minimum of r because $r'(x) = h(x)$ changes from negative to positive there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = -2$ is the only critical point. (h is negative on $(-7, -2)$ and positive on $(-2, 6)$ is another way to state this.)

Note: Stating that $x = -2$ is the only relative minimum critical point is not enough.

The second point is for arguing that $x = -2$ then produces the absolute minimum. Note that the response does not have to repeat the justification of $x = -2$ being a relative minimum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = -2$. This consideration is more than just the statement $x = -2$. The response must use $x = -2$ and an area calculation in trying to find $r(-2)$.

6-5

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $h(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as difference of $h(x)$ values.

A common error that shows an attempt to find area is $r(-2) = 5$. This earns the first point.

The second point is for consideration of the endpoints along with an attempt to find area. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $h(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as the difference of $h(x)$ values.

Common errors that show an attempt to find area:

· $r(-7) = 10$ (adding the triangle areas)

· $r(6) = 9$ (finding the area of the triangle and forgetting the rectangle). These can be used to earn the second point.

A correct argument that eliminates $x = 6$ by stating $r(6)$ is positive will earn the second point as long as $x = -7$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -7$, $x = -2$, and $x = 6$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no numerical calculation mistakes in intermediate work.

Note: A coordinate point is not an acceptable answer.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute minimum
- ☐ Answer with justification
- ☐ Considers $x = -2$
- ☐ Considers both endpoints

6-5

☐ Answer with justification**Solution:**

Because $x = -2$ is the only point on the interval $-7 \leq x \leq 6$ where $r'(x) = h(x) = 0$, $x = -2$ is the only critical point of r on the interval $-7 \leq x \leq 6$.

Because $x = -2$ is also the location of a relative minimum for r , $r(-2)$ is the absolute minimum for r on $-7 \leq x \leq 6$.

$$r(-2) = \int_0^{-2} h(t) dt = - \int_{-2}^0 h(t) dt = -5$$

Therefore, the absolute minimum value of r on $-7 \leq x \leq 6$ is $r(-2) = -5$.

Alternate solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$x = -2$ is the only critical point in the interval because it is the only point where $r'(x) = h(x) = 0$.

$$r(-2) = \int_0^{-2} h(t) dt = - \int_{-2}^0 h(t) dt = -5$$

$$r(-7) = \int_0^{-7} h(t) dt = - \int_{-7}^0 h(t) dt = 0$$

$$r(6) = \int_0^6 h(t) dt = 21$$

Therefore, the absolute minimum value of r on $-7 \leq x \leq 6$ is $r(-2) = -5$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for providing the interval $(0, 6)$ or inequality $0 < x < 6$.

We will allow $[0, 6]$, $(0, 6]$, $[0, 6)$, or an equivalent inequality.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons r' is positive and decreasing or r' is positive and r'' is negative will earn the second point if the connection $r' = h$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function r , then the second point is not earned.

6-5



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

r is increasing where $r' = h$ is positive.

r is concave down where $r' = h$ is decreasing.

The graph of r is both increasing and concave down on the open interval $(0, 6)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a consideration of $r''(x) = h'(x)$.

We are not allowing an import of a previous connection between r'' and h' .

The explicit connection $r''(x) = h'(x)$ earns the first point.

It is not assumed that a response which presents $r'(x) = h(x)$ has made that connection.

The implicit connection will be made in the explanation with a connection to h' , the differentiability of h , or the characteristic of the graph of h .

The second point is for the answer with explanation

This must include "Does not exist" and one of the following:

- limits involving the difference quotient that appears in these scoring guidelines,
- limits involving h' as the argument of the limit, or some correct verbal description of the characteristic of the graph of h at $x = -2$.

Note that all of the above descriptions earn the first point for the implicit consideration of $h'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

6-5

Acceptable verbal descriptions: h' does not exist because h (or "the graph") has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of h . The phrase $r''(1)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $r''(1)$ does not exist because there is a cusp on h is acceptable and earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $h'(x)$
- ☐ Answer with explanation

Solution:

Because $\lim_{x \rightarrow -2^-} \frac{h(x) - h(-2)}{x - (-2)} = \frac{2}{5}$ and $\lim_{x \rightarrow -2^+} \frac{h(x) - h(-2)}{x - (-2)} = \frac{5}{2}$, $h'(-2) = r''(-2)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of r does not need to be stated.

An answer of $\frac{r(0)}{-4}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{-4}$.

Copy/calculation errors in the denominator will not earn the point.

Disregard notation errors



0	1
---	---

6-5

The student response accurately includes the criteria below.

☐ Answer

Solution:

r is continuous. $\Rightarrow \lim_{x \rightarrow 0} r(x) = r(0)$

$$\lim_{x \rightarrow 0} \frac{r(x)}{x-4} = \frac{r(0)}{0-4} = \frac{0}{-4} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $v : r'(x^2) \cdot 2x$.

This point is earned for any of $r'(x^2) \cdot 2x$, $h(x^2) \cdot 2x$, $h(3) \cdot 2\sqrt{3}$, or $\frac{7}{2} \cdot 2\sqrt{3}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$$r'(x^2) = h(x^2), r'(3) = h(3), \text{ or } r'(3) = \frac{7}{2}.$$

This point would be earned with a response of $h(3) \cdot 2\sqrt{3}$, or $\frac{7}{2} \cdot 2\sqrt{3}$, even with no preceding work.

The third point is for either $\frac{7}{2} \cdot 2\sqrt{3}$ or $7\sqrt{3}$. Note, the product need not be simplified.

An answer of $7\sqrt{3}$ with no supporting work earns no points in part (f).



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$v'(x) = r'(x^2) \cdot 2x = h(x^2) \cdot 2x$$

6-5

$$v'(\sqrt{3}) = h(3) \cdot 2\sqrt{3} = \frac{7}{2} \cdot 2\sqrt{3} = 7\sqrt{3}$$

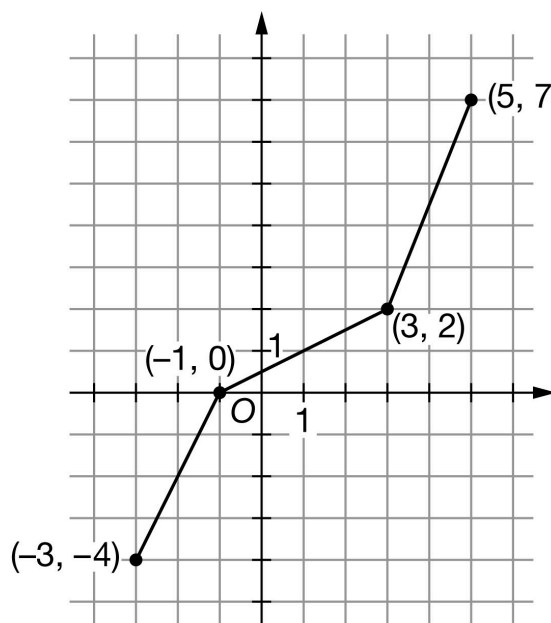
6-5

43. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the function g on the interval $-3 \leq x \leq 5$ consists of three line segments, as shown in the figure above. Let k be the function defined by $k(x) = \int_3^x g(t) \, dt$ for $-3 \leq x \leq 5$.

- For the function k , is $x = -1$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute minimum value of k on the interval $-3 \leq x \leq 5$. Justify your answer.
- On what open intervals contained in $-3 < x < 5$ is the graph of k both increasing and concave up? Give a reason for your answer.
- Find the value of $k''(-1)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow 3} \frac{k(x)}{x+3}$. Show the computations that lead to your answer.
- Let h be the function defined by $h(x) = k(x^2)$. Find the value of $h'(\sqrt{3})$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{3}$ can be keyboarded as sqrt(3))

6-5

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local minimum or relative minimum. We also need to see a connection to g . The obvious is $k' = g$. A response of local minimum with the explanation: “ g (or the graph) changes from negative to positive” is an implied connection. A response of local minimum with explanation: “ k' changes from negative to positive,” without the connection of $k' = g$ is not acceptable.

Simply saying “ k changes from decreasing to increasing” is not acceptable.

In this situation, the Second Derivative Test $k''(-1) > 0$ is an incorrect statement and a response with this reason does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = -1$ is the location of a relative minimum of k because $k'(x) = g(x)$ changes from negative to positive there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = -1$ is the only critical point. (g is negative on $(-3, -1)$ and positive on $(-1, 5)$ is another way to state this.)

Note: Stating that $x = -1$ is the only relative minimum critical point is not enough.

The second point is for arguing that $x = -1$ then produces the absolute minimum. Note that the response does not have to repeat the justification of $x = -1$ being a relative minimum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

6-5

The Candidates Test method:

The first point is for the consideration of $x = -1$. This consideration is more than just the statement $x = -1$. The response must use $x = -1$ AND an area calculation in trying to find $k(-1)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $g(x)$ values.

A common error that shows an attempt to find area is $k(-1) = 4$. This earns the first point.

The second point is for consideration of the endpoints. The words “must check the endpoints” is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $g(x)$ values.

Common errors that show an attempt to find area:

- $k(-3) = 8$ (adding the triangle areas)
- $k(5) = 5$ (finding the area of the triangle and forgetting the rectangle)
- These can be used to earn the second point.

A correct argument that eliminates $x = 5$ by stating $k(5)$ is positive will earn the second point as long as $x = -3$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -3$, $x = -1$, and $x = 5$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no numerical calculation mistakes in intermediate work.

A coordinate point is not an acceptable answer.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute minimum

6-5

- ☐ Answer with justification
- ☐ Considers $x = -1$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

Because $x = -1$ is the only point on the interval $-3 \leq x \leq 5$ where $k'(x) = g(x) = 0$, $x = -1$ is the only critical point of k on the interval $-3 \leq x \leq 5$.

Because $x = -1$ is also the location of a relative minimum for k , $k(-1)$ is the absolute minimum for k on $-3 \leq x \leq 5$.

$$k(-1) = \int_3^{-1} g(t)dt = - \int_{-1}^3 g(t)dt = -4$$

Therefore, the absolute minimum value of k on $-3 \leq x \leq 5$ is $k(-1) = -4$.

Alternate solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$x = -1$ is the only critical point in the interval because it is the only point where $k'(x) = g(x) = 0$.

$$k(-1) = \int_3^{-1} g(t)dt = - \int_{-1}^3 g(t)dt = -4$$

$$k(-3) = \int_3^{-3} g(t)dt = - \int_{-3}^3 g(t)dt = 0$$

$$k(5) = \int_3^5 g(t)dt = 9$$

Therefore, the absolute minimum value of k on $-3 \leq x \leq 5$ is $k(-1) = -4$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the interval $(-1, 5)$ or inequality $-1 < x < 5$.

We will allow $[-1, 5]$, $(-1, 5]$, $[-1, 5)$, or an equivalent inequality.

Presenting $(-1, 3) \cup (3, 5)$ with correct reasoning earns one out of the two points. Any accepted version (open parentheses or brackets) earns the point. The “ $\cup$ ” is not necessary. Be careful, one bracket on either 3 is actually the entire interval and is eligible for both points.

6-5

The **second point** is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons “ k' is positive and increasing” or “ k' is positive and k'' is positive” will earn the second point if the connection $k' = g$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function k , then the second point is not earned.

The interval $(-1, 7)$ with the correct justification will earn one of the two points. (The response took the y -coordinate of the right-hand point of the interval $(5, 7)$). This is considered a copy error. This is the only copy error that is allowed.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

k is increasing where $k' = g$ is positive.

k is concave up where $k' = g$ is increasing.

The graph of k is both increasing and concave up on the open interval $(-1, 5)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $k''(x) = g'(x)$.

We are not allowing an import of a previous connection between k'' and g' .

The explicit connection, $k''(x) = g'(x)$ earns the first point.

It is not assumed that a response which only presents $k'(x) = g(x)$ has made that connection.

The implicit connection will be made in the explanation with a connection to g' , the differentiability of g , or the characteristic of the graph of g .

The second point is for the answer with explanation.

This must include “Does not exist” and one of the following:

6-5

- limits involving the difference quotient that appears in these scoring guidelines,
- limits involving g' as the argument of the limit,
- or some correct verbal description of the characteristic of the graph of g at $x = -1$.

Note that all of the above descriptions earn the first point for the implicit consideration of $g'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: g' does not exist because g (or “the graph”) has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will NOT accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of g . The phrase $k''(1)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $k''(1)$ does not exist because there is a cusp on g is acceptable and earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $g'(x)$
- ☐ Answer with explanation

Solution:

Because $\lim_{x \rightarrow -1^-} \frac{g(x) - g(-1)}{x - (-1)} = 2$ and $\lim_{x \rightarrow -1^+} \frac{g(x) - g(-1)}{x - (-1)} = \frac{1}{2}$, $g''(-1) = k''(-1)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of k does not need to be stated.

An answer of $\frac{k(3)}{6}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

6-5

The minimal answer is $\frac{0}{6}$.

Copy/calculation errors in the denominator will not earn the point.

Disregard notation errors.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

k is continuous. $\Rightarrow \lim_{x \rightarrow 3} k(x) = k(3)$

$$\lim_{x \rightarrow 3} \frac{k(x)}{x+3} = \frac{k(3)}{3+3} = \frac{0}{6} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $h : k'(x^2) \cdot 2x$.

This point is earned for any of $k'(x^2) \cdot 2x$, $g(x^2) \cdot 2x$, $g(3) \cdot 2\sqrt{3}$, or $2 \cdot 2\sqrt{3}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$$k'(x^2) = g(x^2), k'(3) = g(3), \text{ or } k'(3) = 2.$$

This point would be earned with a response of $g(3) \cdot 2\sqrt{3}$, or $2 \cdot 2\sqrt{3}$, even with no preceding work.

The third point is for either $2 \cdot 2\sqrt{3}$, or $4\sqrt{3}$. Note, the product need not be simplified.

An answer of $4\sqrt{3}$ with no supporting work earns no points in part (f).



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

6-5

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$h'(x) = k'(x^2) \cdot 2x = g(x^2) \cdot 2x$$

$$h'(\sqrt{3}) = g(3) \cdot 2\sqrt{3} = 2 \cdot 2\sqrt{3} = 4\sqrt{3}$$

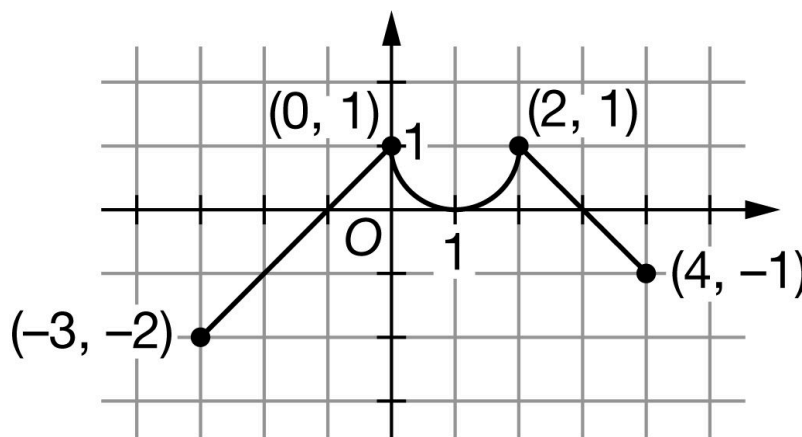
6-5

44. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The continuous function g has domain $-3 \leq x \leq 4$. The graph of g , consisting of two line segments and a semicircle, is shown in the figure above. The graph of g has a horizontal tangent at $x = 1$. Let h be the function defined by $h(x) = \int_0^x g(t) \, dt$ for $-3 \leq x \leq 4$.

- Find the x -coordinate of each critical point of h on the interval $-3 < x < 4$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- For $-3 < x < 4$, on what open intervals is h decreasing and concave down? Give a reason for your answer.
- Find the value of $h(2)$. Show the computations that lead to your answer.
- Find the value of $h(-3)$. Show the computations that lead to your answer.
- Find the value of $h''(-1)$, or explain why it does not exist.
- Must there exist a value of r , for $-3 < r < 1$, such that $h'(r)$ is equal to the average rate of change of h over the interval $-3 \leq x \leq 1$? Justify your answer.
- Find $\lim_{x \rightarrow 2} \frac{h'(x)}{5x}$. Show the computations that lead to your answer.
- The function k is defined by $k(x) = h(x^2)$. Find $k'(-2)$. Show the computations that lead to your answer.

6-5

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -1$, $x = 1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 4)$. The point is still earned provided the three critical points $x = -1$, $x = 1$, and $x = 3$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(1, 0)$, and $(3, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -1$, $x = 1$, and $x = 3$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g , then:

- the response is eligible for all three points, if the response presents the connection between $h'(x)$ and $g(x)$, (that is, $h' = g$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $h' = g$, throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection $h'(x) = g(x)$. Any additional error (for example, not justifying that at $x = -1$, h has a relative minimum) is eligible for at most one of the three points.

6-5

Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(3)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $h(x)$ values for $x = -1, 1, 3$ must be included in the table or list of values. The correct values of relevant inputs include:

$$\cdot h(-3) = \frac{3}{2}$$

$$\cdot h(-1) = -\frac{1}{2}$$

$$\cdot h(1) = 1 - \frac{\pi}{4}$$

$$\cdot h(3) = \frac{5}{2} - \frac{\pi}{2}$$

$$\cdot h(4) = 2 - \frac{\pi}{2}$$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = -1$, h has a relative minimum because $h' = g$ is negative for $x < -1$ and $h' = g$ is positive for $x > -1$.
- At $x = 1$, h has neither because $h' = g$ is positive for $x < 1$ and $h' = g$ is positive for $x > 1$.
- At $x = 3$, h has a relative maximum because $h' = g$ is positive for $x < 3$ and $h' = g$ is negative for $x > 3$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification

6-5

☐ Relative maximum with justification
Solution:

At $x = -1$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = 1$, h has neither because $h'(x) = g(x)$ does not change sign there.

At $x = 3$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[3, 4]$, $(3, 4]$, $[3, 4)$, and $(3, 4)$ all earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only h' is eligible to earn the second point.

A response referencing only g is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason “ h' is decreasing” can also be expressed by either “ h'' is negative” or “the slope of h' is negative.”

A response referencing just “the function” or “it” or “the slope” is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason
Solution:

h is decreasing where $h' = g$ is negative.

h is concave down where $h' = g$ is decreasing.

6-5

h is decreasing and concave down on the open interval $(3, 4)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $h(2) = 0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(2) = 2 \cdot 1 - \frac{\pi}{2} \cdot 1^2 = 2 - \frac{\pi}{2}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $-\frac{1}{2} + 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $h(-3) = \frac{3}{2}$ with no supporting work earns the second point only.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\int_{-3}^{-1} g(t) dt = -2$ – OR – $\int_{-1}^0 g(t) dt = \frac{1}{2}$

☐ Answer

Solution:

6-5

$$\begin{aligned}
 h(-3) &= \int_0^{-3} g(t) \, dt = - \int_{-3}^0 g(t) \, dt \\
 &= - \int_{-3}^{-1} g(t) \, dt - \int_{-1}^0 g(t) \, dt = -(-2) - \frac{1}{2} = \frac{3}{2}
 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $h''(-1) = 1$ or $g'(-1) = 1$ or even simply stating a numerical value of 1 clearly marked as the answer earns the point.

No supporting work is required because the answer of 1 can easily be read from the graph.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h''(x) = g'(x)$$

$$h''(-1) = g'(-1) = \frac{1 - (-2)}{0 - (-3)} = 1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $h(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(r)$ equals the average rate of change of $h'(x)$ ” earns both points.

6-5

- “Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(r)$ equals the average rate of change of $h(x)$ ” does not earn the first point, but does earn the second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $h(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $h(x)$ is continuous” earns neither point.
- “Yes, because $h(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims “ h is not differentiable, so therefore the MVT does not apply” earns one of the two points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$$h' = g \Rightarrow h \text{ is differentiable on } -3 \leq x \leq 1 \Rightarrow h \text{ is continuous on } -3 \leq x \leq 1.$$

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 1$ to guarantee that there exists a value of r , for $-3 < r < 1$, such that $h'(r)$ equals the average rate of change of h over the interval $-3 \leq x \leq 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim_{x \rightarrow 2} \frac{h'(x)}{5x} = \frac{1}{10}$ earns the point, as does just stating the answer of $\frac{1}{10}$.

No additional supporting work is required.

The response does not need to state that h' is continuous. If the response does state that h' is continuous, an explanation is not required.

6-5



0

1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h' = g \Rightarrow h' \text{ is continuous.} \Rightarrow \lim_{x \rightarrow 2} h'(x) = h'(2)$$

$$\lim_{x \rightarrow 2} \frac{h'(x)}{5x} = \frac{h'(2)}{5 \cdot 2} = \frac{g(2)}{10} = \frac{1}{10}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -2$ is substituted for x , then $x = -2$ must appear unsimplified in at least one factor in order to earn the first point. Responses that earn the point include:

$$\cdot h'(x^2) \cdot 2x$$

$$\cdot g(x^2) \cdot 2x$$

$$\cdot h'((-2)^2) \cdot 2 \cdot (-2)$$

$$\cdot g((-2)^2) \cdot 2 \cdot (-2)$$

$$\cdot h'(4) \cdot 2 \cdot (-2)$$

$$\cdot g(4) \cdot 2 \cdot (-2)$$

$$\cdot h'((-2)^2) \cdot (-4)$$

$$\cdot g((-2)^2) \cdot (-4)$$

Responses that do not earn the point include:

$$\cdot h'(x^2) \cdot 2$$

$$\cdot g(x^2) \cdot 2$$

$$\cdot h'(x) \cdot 2x$$

6-5

$$\cdot g(x) \cdot 2x$$

$$\cdot h'(4) \cdot (-4)$$

$$\cdot g(4) \cdot (-4)$$

$$\cdot h(x^2) \cdot 2x$$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $k(x)$ or in $k'(-2)$. For example, the second point is earned by $g(x^2)$ or by $g((-2)^2)$.

The second point may be earned without earning the first point. For example, $h'(x^2) \cdot 2 = g(x^2) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and Ag are linked.

The second point may be earned by going directly from $h'(4)$ to -1 . The function g does not have to be stated explicitly. Responses that earn the first two points include:

$$\cdot h'((-2)^2) \cdot 2 \cdot (-2) = -1 \cdot 2 \cdot (-2)$$

$$\cdot h'((-2)^2) \cdot (-4) = -1 \cdot (-4)$$

Recall, if $x = -2$ is substituted for x , then $x = -2$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $g(x^2) \cdot 2x$ and $g((-2)^2) \cdot 2 \cdot (-2)$ earn the first two points.

The third point is earned for the correct answer of 4 simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (i). Responses of the form $h'(x)^2 \cdot 2x$ or $(h'(x))^2 \cdot 2x$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

6-5**Solution:**

$$k'(x) = h'(x^2) \cdot 2x$$

$$= g(x^2) \cdot 2x$$

$$k'(-2) = g((-2)^2) \cdot 2 \cdot (-2) = g(4) \cdot (-4) = -1 \cdot (-4) = 4$$

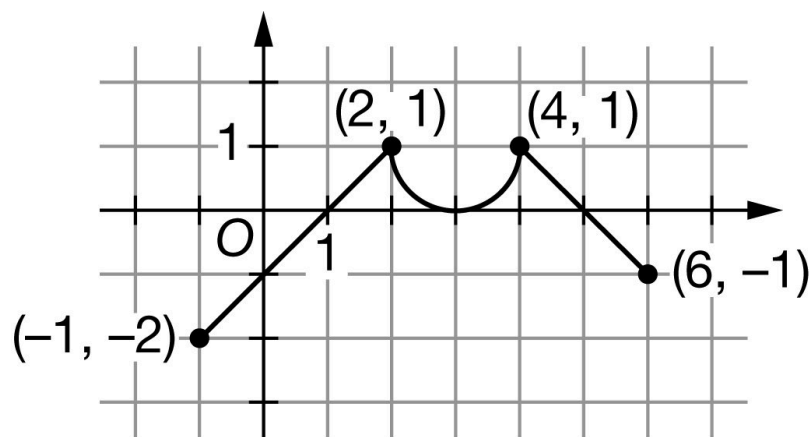
6-5

45. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h

The continuous function h has domain $-1 \leq x \leq 6$. The graph of h , consisting of two line segments and a semicircle, is shown in the figure above. The graph of h has a horizontal tangent at $x = 3$. Let g be the function defined by $g(x) = \int_2^x h(t) \, dt$ for $-1 \leq x \leq 6$.

- Find the x -coordinate of each critical point of g on the interval $-1 \leq x \leq 6$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify each of your answers.
- For $-1 \leq x \leq 6$, on what open intervals is g decreasing and concave up? Give a reason for your answer.
- Find the value of $g(4)$. Show the computations that lead to your answer.
- Find the value of $g(-1)$. Show the computations that lead to your answer.
- Find the value of $g''(0)$, or explain why it does not exist.
- Must there exist a value of d , for $-1 < d < 3$, such that $g'(d)$ is equal to the average rate of change of g over the interval $-1 \leq x \leq 3$? Justify your answer.
- Find $\lim_{x \rightarrow 4} \frac{g'(x)}{3x}$. Show the computations that lead to your answer.
- The function a is defined by $a(x) = g(2x)$. Find $a'(0)$. Show the computations that lead to your answer.

6-5

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = 1$, $x = 3$, and $x = 5$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-1, 6)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-1, 6)$. The point is still earned provided the three critical points $x = 1$, $x = 3$, and $x = 5$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(1, 0)$, $(3, 0)$, and $(5, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = h(x) = 0$ at $x = 1$, $x = 3$, and $x = 5$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on h , then:

- the response is eligible for all three points, if the response presents the connection between $g'(x)$ and $h(x)$, (that is, $g' = h$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of h or g' as $g' = h$, throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection $g'(x) = h(x)$. Any additional error (for example, not justifying that at $x = 1$, g has a relative minimum) is eligible for at most one of the three points.

6-5

Read references to “the graph” as references to $h(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(1)$ changes from negative to positive”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $g(x)$ value for $x = 1, 3, 5$ must be included in the table or list of values. The correct values of relevant inputs include:

$$\cdot g(-1) = \frac{3}{2}$$

$$\cdot g(1) = -\frac{1}{2}$$

$$\cdot g(3) = 1 - \frac{\pi}{4}$$

$$\cdot g(5) = \frac{5}{2} - \frac{\pi}{4}$$

$$\cdot g(6) = 2 - \frac{\pi}{4}$$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = 1$, g has a relative minimum because $g' = h$ is negative for $x < 1$ and $g' = h$ is positive for $x > 1$.
- At $x = 3$, g has neither because $g' = h$ is positive for $x < 3$ and $g' = h$ is positive for $x > 3$.
- At $x = 5$, g has a relative maximum because $g' = h$ is positive for $x < 5$ and $g' = h$ is negative for $x > 5$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification

6-5

☐ Relative maximum with justification
Solution:

At $x = 1$, g has a relative minimum because $g'(x) = h(x)$ changes from negative to positive there.

At $x = 3$, g has neither because $g'(x) = h(x)$ does not change sign there.

At $x = 5$, g has a relative maximum because $g'(x) = h(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[-1, 1]$, $(-1, 1]$, $[-1, 1)$, and $(-1, 1)$ all earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only g' is eligible to earn the second point.

A response referencing only h is not eligible to earn the second point, unless the response has established the link between $h(x)$ and $g'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason “ g' is increasing” can also be expressed by either “ g'' is positive” or “the slope of g' is positive.”

A response referencing just “the function” or “it” or “the slope” is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason
Solution:

g is decreasing where $g' = h$ is negative.

g is concave up where $g' = h$ is increasing.

6-5

g is decreasing and concave up on the open interval $(-1, 1)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $g(4) = 0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(4) = 2 \cdot 1 - \frac{\pi}{2} \cdot 1^2 = 2 - \frac{\pi}{2}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $-\frac{1}{2} + 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $g(-1) = \frac{3}{2}$ with no supporting work earns the second point only.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\int_{-1}^1 h(t) \, dt = -2$ – OR – $\int_1^2 h(t) \, dt = \frac{1}{2}$

☐ Answer

Solution:

6-5

$$\begin{aligned}
 g(-1) &= \int_2^{-1} h(t) \, dt = - \int_{-1}^2 h(t) \, dt \\
 &= - \int_{-1}^1 h(t) \, dt - \int_1^2 h(t) \, dt = -(-2) - \frac{1}{2} = \frac{3}{2}
 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $g''(0) = 1$ or $h'(0) = 1$ or even simply stating a numerical value of 1 clearly marked as the answer earns the point.

No supporting work is required because the answer of 1 can easily be read from the graph.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g''(x) = h'(x)$$

$$g''(0) = h'(0) = \frac{1}{2} = \frac{1 - (-2)}{2 - (-1)}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $g(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $g(x)$ is continuous and differentiable, therefore there must be a value where $g'(d)$ equals the average rate of change of $g(x)$ ” earns both points.

6-5

- “Yes, $g'(x)$ (the derivative of $g(x)$) is continuous and differentiable, therefore there must be a value where $g'(d)$ equals the average rate of change of $g(x)$ ” does not earn the first point, but does earn the second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $g(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $g(x)$ is continuous” earns neither point.
- “Yes, because $g(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims “ g is not differentiable, so therefore the MVT does not apply” earns one of the two points.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$g' = h \Rightarrow g$ is differentiable on $-1 \leq x \leq 3$. $\Rightarrow g$ is continuous on $-1 \leq x \leq 3$.

Therefore, the Mean Value Theorem can be applied to g on the interval $-1 \leq x \leq 3$ to guarantee that there exists a value of d , for $-1 < d < 3$, such that $g'(d)$ equals the average rate of change of g over the interval $-1 \leq x \leq 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim_{x \rightarrow 4} \frac{g'(x)}{3x} = \frac{1}{12}$ earns the point, as does just stating the answer of $\frac{1}{12}$.

No additional supporting work is required.

The response does not need to state that g' is continuous. If the response does state that g' is continuous, an explanation is not required.

6-5



0

1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g' = h \Rightarrow g' \text{ is continuous.} \Rightarrow \lim_{x \rightarrow 4} g'(x) = g'(4)$$

$$\lim_{x \rightarrow 4} \frac{g'(x)}{3x} = \frac{g'(4)}{3 \cdot 4} = \frac{h(4)}{12} = \frac{1}{12}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 0$ is substituted for x , then $x = 0$ must appear unsimplified to earn the first point. Responses that earn the point include:

$$\cdot g'(2x) \cdot 2$$

$$\cdot h(2x) \cdot 2$$

$$\cdot g'(2 \cdot 0) \cdot 2$$

$$\cdot h(2 \cdot 0) \cdot 2$$

Responses that do not earn the point include:

$$\cdot g'(x) \cdot 2$$

$$\cdot h(x) \cdot 2$$

$$\cdot g'(0) \cdot 2$$

$$\cdot h(0) \cdot 2$$

$$\cdot g(2x) \cdot 2$$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking g' and h either directly or indirectly in the presented derivative of $a(x)$ or in $a'(0)$. For example, the second point is earned by $h(2x)$ or by $h(2 \cdot 0)$.

The second point may be earned without earning the first point. For example, $g'(2x) = h(2x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because g' and h are

6-5

linked.

The second point may be earned by going directly from $g'(0)$ to -1 . The function h does not have to be stated explicitly. For example, $g'(2 \cdot 0) \cdot 2 = -1 \cdot 2$ earns the first two points. Recall, if $x = 0$ is substituted for x , then $x = 0$ must appear unsimplified to earn the first point.

The first two points may be earned by a single expression. For example, both $h(2x) \cdot 2$ and $h(2 \cdot 0) \cdot 2$ earn the first two points.

The third point is earned for the correct answer of -2 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$a'(x) = g'(2x) \cdot 2$$

$$= h(2x) \cdot 2$$

$$a'(0) = g'(2 \cdot 0) \cdot 2 = g'(0) \cdot 2 = h(0) \cdot 2 = -1 \cdot 2 = -2$$

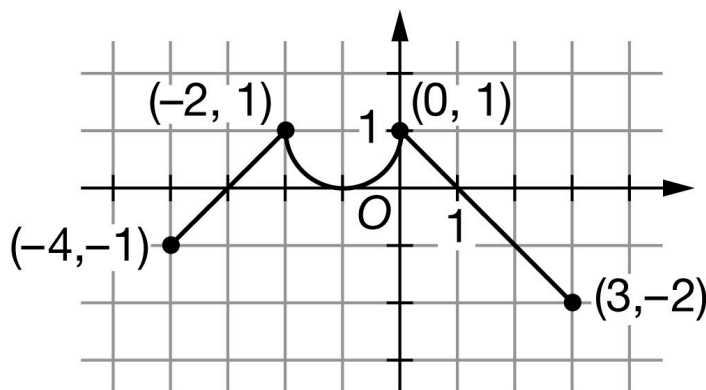
6-5

46. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The continuous function f has domain $-4 \leq x \leq 3$. The graph of f , consisting of two line segments and a semicircle, is shown in the figure above. The graph of f has a horizontal tangent at $x = -1$. Let h be

the function defined by $h(x) = \int_0^x f(t) \, dt$ for $-4 \leq x \leq 3$.

- Find the x -coordinate of each critical point of h on the interval $-4 < x < 3$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- For $-4 < x < 3$, on what open intervals is h increasing and concave down? Give a reason for your answer.
- Find the value of $h(-2)$. Show the computations that lead to your answer.
- Find the value of $h(3)$. Show the computations that lead to your answer.
- Find the value of $h''(2)$, or explain why it does not exist.
- Must there exist a value of k , for $-4 < k < 0$, such that $h'(k)$ is equal to the average rate of change of h over the interval $-4 \leq x \leq 0$? Justify your answer.
- Find $\lim_{x \rightarrow -2} \frac{h'(x)}{3x}$. Show the computations that lead to your answer.
- The function r is defined by $r(x) = h(-3x)$. Find $r'(-1)$. Show the computations that lead to your answer.

Part A

6-5

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3$, $x = -1$, and $x = 1$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4, 3)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4, 3)$. The point is still earned provided the three critical points $x = -3$, $x = -1$, and $x = 1$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(-1, 0)$, and $(1, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = f(x) = 0$ at $x = -3$, $x = -1$, and $x = 1$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on f , then:

- the response is eligible for all three points, if the response presents the connection between $h'(x)$ and $f(x)$, (that is, $h' = f$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or h' as $h' = f$, throughout all parts of this question.
- the response is eligible for at most two of the three points, if the response does not present the connection $h'(x) = f(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative minimum) is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

6-5

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-3)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $h(x)$ values for $x = -3, -1, 1$ must be included in the table or list of values. The correct values of relevant inputs include:

$$\cdot h(-4) = \frac{\pi}{2} - 2$$

$$\cdot h(-3) = \frac{\pi}{2} - \frac{5}{2}$$

$$\cdot h(-1) = \frac{\pi}{4} - 1$$

$$\cdot h(1) = \frac{1}{2}$$

$$\cdot h(3) = -\frac{3}{2}$$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -3$, h has a relative minimum because $h' = f$ is negative for $x < -3$ and $h' = f$ is positive for $x > -3$.

· At $x = -1$, h has neither because $h' = f$ is positive for $x < -1$ and $h' = f$ is positive for $x > -1$.

· At $x = 1$, h has a relative maximum because $h' = f$ is positive for $x < 1$ and $h' = f$ is negative for $x > 1$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

6-5

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

At $x = -1$, h has neither because $h'(x) = f(x)$ does not change sign there.

At $x = 1$, h has a relative maximum because $h'(x) = f(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[-2, -1]$ and $(0, 1]$ or $[-2, -1)$ and $(0, 1)$ both earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only h' is eligible to earn the second point.

A response referencing only f is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $h'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason “ h' is decreasing” can also be expressed by either “ h'' is negative” or “the slope of h' is negative.”

A response referencing just “the function” or “it” or “the slope” is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

Responses giving a completely correct justification for only one of the two subintervals earn one of the two points.



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

h is increasing where $h' = f$ is positive.

h is concave down where $h' = f$ is decreasing.

h is increasing and concave down on the open intervals $(-2, -1)$ and $(0, 1)$.

6-5

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $h(-2) = -0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(-2) = -\left(2 \cdot 1 - \frac{\pi}{2} \cdot 1^2\right) = -\left(2 - \frac{\pi}{2}\right) = \frac{\pi}{2} - 2$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\frac{1}{2} - 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $h(3) = -\frac{3}{2}$ with no supporting work earns the second point only.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\int_0^1 f(t) dt = \frac{1}{2}$ – OR – $\int_1^3 f(t) dt = -2$

☐ Answer

Solution:

6-5

$$h(3) = \int_0^3 f(t) dt$$

$$= \int_0^1 f(t) dt + \int_1^3 f(t) dt = \frac{1}{2} + (-2) = -\frac{3}{2}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $h''(2) = -1$ or $f'(2) = -1$ or even simply stating a numerical value of -1 clearly marked as the answer earns the point.

No supporting work is required because the answer of -1 can easily be read from the graph.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h''(x) = f'(x)$$

$$h''(2) = f'(2) = \frac{1 - (-2)}{0 - 3} = -1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $h(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(k)$ equals the average rate of change of $h(x)$ ” earns both points.

6-5

- “Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(k)$ equals the average rate of change of $h(x)$ ” does not earn the first point, but does earn the second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $h(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $h(x)$ is continuous” earns neither point.
- “Yes, because $h(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims “ h is not differentiable, so therefore the MVT does not apply” earns one of the two points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$h' = f \Rightarrow h$ is differentiable on $-4 \leq x \leq 0$. $\Rightarrow h$ is continuous on $-4 \leq x \leq 0$.

Therefore, the Mean Value Theorem can be applied to h on the interval $-4 \leq x \leq 0$ to guarantee that there exists a value of k , for $-4 < k < 0$, such that $h'(k)$ equals the average rate of change of h over the interval $-4 \leq x \leq 0$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim_{x \rightarrow -2} \frac{h'(x)}{3x} = \frac{-1}{6}$ earns the point, as does just stating the answer of $\frac{-1}{6}$.

No additional supporting work is required.

The response does not need to state that h' is continuous. If the response does state that h' is continuous, an explanation is not required.

6-5



0

1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h' = f \Rightarrow h' \text{ is continuous.} \Rightarrow \lim_{x \rightarrow -2} h'(x) = h'(-2)$$

$$\lim_{x \rightarrow -2} \frac{h'(x)}{3x} = \frac{h'(-2)}{3 \cdot (-2)} = \frac{f(-2)}{-6} = \frac{1}{-6}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -1$ is substituted for x , then $x = -1$ must appear unsimplified to earn the first point. Responses that earn the point include:

$$\cdot h'(-3x) \cdot (-3)$$

$$\cdot f(-3x) \cdot (-3)$$

$$\cdot h'(-3 \cdot -1) \cdot (-3)$$

$$\cdot f(-3 \cdot -1) \cdot (-3)$$

Responses that do not earn the point include:

$$\cdot h'(x) \cdot (-3)$$

$$\cdot f(x) \cdot (-3)$$

$$\cdot h'(3) \cdot (-3)$$

$$\cdot f(3) \cdot (-3)$$

$$\cdot h(-3x) \cdot (-3)$$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking h' and f either directly or indirectly in the presented derivative of $r(x)$ or in $r'(-1)$. For example, the second point is earned by $f(-3x)$ or by $f(-3 \cdot -1)$.

6-5

The second point may be earned without earning the first point. For example, $h'(-3x) = f(-3x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and f are linked.

The second point may be earned by going directly from $h'(3)$ to -2 . The function f does not have to be stated explicitly. For example, $h'(-3 \cdot -1) \cdot (-3) = -2 \cdot (-3)$ earns the first two points. Recall, if $x = -1$ is substituted for x , then $x = -1$ must appear unsimplified to earn the first point.

The first two points may be earned by a single expression. For example, both $f(-3x) \cdot (-3)$ and $f(-3 \cdot -1) \cdot (-3)$ earn the first two points.

The third point is earned for the correct answer of 6, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$r'(x) = h'(-3x) \cdot (-3)$$

$$= f(-3x) \cdot (-3)$$

$$r'(-1) = f((-3) \cdot (-1)) \cdot (-3) = f(3) \cdot (-3) = -2 \cdot (-3) = 6$$

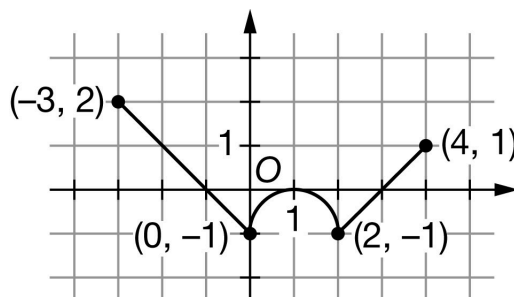
6-5

47. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The continuous function f has domain $-3 \leq x \leq 4$. The graph of f , consisting of two line segments and a semicircle, is shown in the figure above. The graph of f has a horizontal tangent at $x = 1$. Let g be the function defined by $g(x) = \int_0^x f(t) \, dt$ for $-3 \leq x \leq 4$.

- Find the x -coordinate of each critical point of g on the interval $-3 < x < 4$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify each of your answers.
- For $-3 < x < 4$, on what open intervals is g decreasing and concave up? Give a reason for your answer.
- Find the value of $g(2)$. Show the computations that lead to your answer.
- Find the value of $g(-3)$. Show the computations that lead to your answer.
- Find the value of $g''(-2)$, or explain why it does not exist.
- Must there exist a value of b , for $-3 < b < 1$, such that $g'(b)$ is equal to the average rate of change of g over the interval $-3 \leq x \leq 1$? Justify your answer.
- Find $\lim_{x \rightarrow 2} \frac{g'(x)}{4x}$. Show the computations that lead to your answer.
- The function z is defined by $z(x) = g(x^2)$. Find $z'(-2)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-5

To earn this point, the response must include all three critical values: $x = -1$, $x = 1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 4)$. The point is still earned provided the three critical points $x = -1$, $x = 1$, and $x = 3$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(1, 0)$, and $(3, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = f(x) = 0$ at $x = -1$, $x = 1$, and $x = 3$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on f , then:

- the response is eligible for all three points, if the response presents the connection between $g'(x)$ and $f(x)$, (that is, $g' = f$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or g' as $g' = f$, throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection $g'(x) = f(x)$. Any additional error (for example, not justifying that at $x = -1$, f has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

6-5

A response that discusses g' at a particular point (e.g., “ $g'(3)$ changes from negative to positive”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $g(x)$ values for $x = -1, 1, 3$ must be included in the table or list of values. The correct values of relevant inputs include:

$$\cdot g(-3) = -\frac{3}{2}$$

$$\cdot g(-1) = \frac{1}{2}$$

$$\cdot g(1) = \frac{\pi}{4} - 1$$

$$\cdot g(3) = \frac{\pi}{2} - \frac{5}{2}$$

$$\cdot g(4) = \frac{\pi}{2} - 2$$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = -1$, g has a relative maximum because $g' = f$ is positive for $x < -1$ and $g' = f$ is negative for $x > -1$.
- At $x = 1$, g has neither because $g' = f$ is negative for $x < 1$ and $g' = f$ is negative for $x > 1$.
- At $x = 3$, g has a relative minimum because $g' = f$ is negative for $x < 3$ and $g' = f$ is positive for $x > 3$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification
- ☐ Neither with justification
- ☐ Relative minimum with justification

Solution:

At $x = -1$, g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

6-5

At $x = 1$, g has neither because $g'(x) = f(x)$ does not change sign there.

At $x = 3$, g has a relative minimum because $g'(x) = f(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[0, 1]$ and $(2, 3]$ or $[0, 1]$ and $(2, 3)$ both earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only g' is eligible to earn the second point.

A response referencing only f is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $g'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

A response referencing just “the function” or “it” or “the slope” is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

The reason “ g' is increasing” can also be expressed by either “ g'' is positive” or “the slope of g' is positive.”

Responses giving a completely correct justification for only one of the two subintervals earn one of the two points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

g is decreasing where $g' = f$ is negative.

g is concave up where $g' = f$ is increasing.

g is decreasing and concave up on the open intervals $(0, 1)$ and $(2, 3)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

6-5

Correct supporting work is needed to earn this point.

The correct decimal value is $g(2) = -0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

$$g(2) = -\left(2 \cdot 1 - \frac{\pi}{2} \cdot 1^2\right) = -\left(2 - \frac{\pi}{2}\right) = \frac{\pi}{2} - 2$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\frac{1}{2} - 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, 2 or $-\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $g(-3) = -\frac{3}{2}$ with no supporting work earns the second point only.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\int_{-3}^{-1} f(t) dt = 2$ – OR – $\int_{-1}^0 f(t) dt = -\frac{1}{2}$

☐ Answer

Solution:

$$\begin{aligned} g(-3) &= \int_0^{-3} f(t) dt = -\int_{-3}^0 f(t) dt \\ &= -\int_{-3}^{-1} f(t) dt - \int_{-1}^0 f(t) dt = -2 - \left(-\frac{1}{2}\right) = -\frac{3}{2} \end{aligned}$$

Part F

6-5

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $g''(-2) = -1$ or $f'(-2) = -1$ or even simply stating a numerical value of -1 clearly marked as the answer earns the point.

No supporting work is required because the answer of -1 can easily be read from the graph.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g''(x) = f'(x)$$

$$g''(-2) = f'(-2) = \frac{-1-2}{0-(-3)} = -1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $g(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $g(x)$ is continuous and differentiable, therefore there must be a value where $g'(b)$ equals the average rate of change of $g(x)$ ” earns both points.
- “Yes, $g'(x)$ (the derivative of $g(x)$) is continuous and differentiable, therefore there must be a value where $g'(b)$ equals the average rate of change of $g(x)$ ” does not earn the first point, but does earn the second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $g(x)$ is continuous and differentiable” earns the first point, but not the second point.

6-5

- “Yes, because $g(x)$ is continuous” earns neither point.
- “Yes, because $g(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims “ g is not differentiable, so therefore the MVT does not apply” earns one of the two points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$g' = f \Rightarrow g$ is differentiable on $-3 \leq x \leq 1$. $\Rightarrow g$ is continuous on $-3 \leq x \leq 1$.

Therefore, the Mean Value Theorem can be applied to g on the interval $-3 \leq x \leq 1$ to guarantee that there exists a value of b , for $-3 < b < 1$, such that $g'(b)$ equals the average rate of change of g over the interval $-3 \leq x \leq 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim_{x \rightarrow 2} \frac{g'(x)}{4x} = \frac{-1}{8}$ earns the point, as does just stating the answer of $\frac{-1}{8}$.

No additional supporting work is required.

The response does not need to state that g' is continuous. If the response does state that g' is continuous, an explanation is not required.



0	1
---	---

The student response accurately includes the criteria below.

Answer

6-5

**Solution:**

$$g' = f \Rightarrow g' \text{ is continuous. } \Rightarrow \lim_{x \rightarrow 2} g'(x) = g'(2)$$

$$\lim_{x \rightarrow 2} \frac{g'(x)}{4x} = \frac{g'(2)}{4 \cdot 2} = \frac{f(2)}{8} = \frac{-1}{8}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -2$ is substituted for x , then $x = -2$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

$$\cdot g'(x^2) \cdot 2x$$

$$\cdot f(x^2) \cdot 2x$$

$$\cdot g'((-2)^2) \cdot 2 \cdot (-2)$$

$$\cdot f((-2)^2) \cdot 2 \cdot (-2)$$

$$\cdot g'(4) \cdot 2 \cdot (-2)$$

$$\cdot f(4) \cdot 2 \cdot (-2)$$

$$\cdot g'((-2)^2) \cdot (-4)$$

$$\cdot f((-2)^2) \cdot (-4)$$

Responses that do not earn the point include:

$$\cdot g'(x^2) \cdot 2$$

$$\cdot f(x^2) \cdot 2$$

$$\cdot g'(x) \cdot 2x$$

$$\cdot f(x) \cdot 2x$$

$$\cdot g'(4) \cdot (-4)$$

$$\cdot f(4) \cdot (-4)$$

6-5

$$\cdot g(x^2) \cdot 2x$$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking g' and f either directly or indirectly in the presented derivative of $z(x)$ or in $z'(-2)$. For example, the second point is earned by $f(x^2)$ or by $f((-2)^2)$.

The second point may be earned without earning the first point. For example, $g'(x^2) \cdot 2 = f(x^2) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because g' and f are linked.

The second point may be earned by going directly from $g'(4)$ to 1. The function f does not have to be stated explicitly. Responses that earn the first two points include:

$$\cdot g'((-2)^2) \cdot 2 \cdot (-2) = 1 \cdot 2 \cdot (-2)$$

$$\cdot g'((-2)^2) \cdot (-4) = 1 \cdot (-4)$$

Recall, if $x = -2$ is substituted for x , then $x = -2$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $f(x^2) \cdot 2x$ and $f((-2)^2) \cdot 2 \cdot (-2)$ earn the first two points.

The third point is earned for the correct answer of -4 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (i). Responses of the form $g'(x)^2 \cdot 2x$ or $(g'(x))^2 \cdot 2x$ are not uncommon. These type of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$z'(x) = g'(x^2) \cdot 2x$$

6-5

$$= f(x^2) \cdot 2x$$

$$z'(-2) = f((-2)^2) \cdot 2 \cdot (-2) = f(4) \cdot (-4) = 1 \cdot (-4) = -4$$

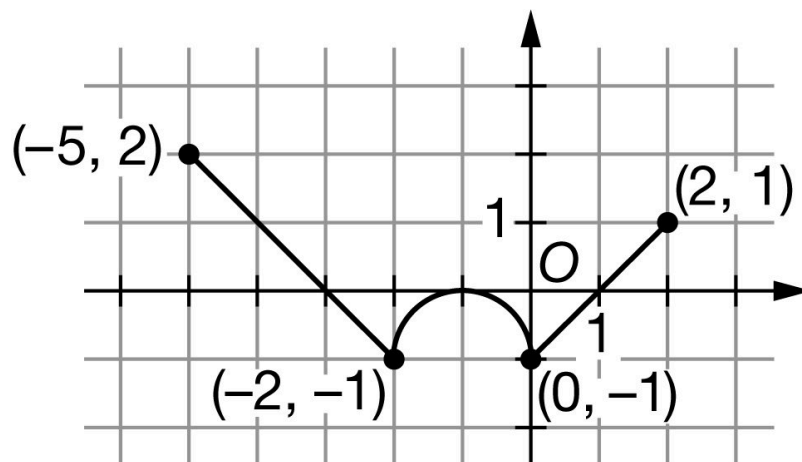
6-5

48. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The continuous function g has domain $-5 \leq x \leq 2$. The graph of g , consisting of two line segments and a semicircle, is shown in the figure above. The graph of g has a horizontal tangent at $x = -1$. Let h be the function defined by $h(x) = \int_{-2}^x g(t) \, dt$ for $-5 \leq x \leq 2$.

- Find the x -coordinate of each critical point of h on the interval $-5 < x < 2$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- For $-5 < x < 2$, on what open intervals is h increasing and concave up? Give a reason for your answer.
- Find the value of $h(0)$. Show the computations that lead to your answer.
- Find the value of $h(-5)$. Show the computations that lead to your answer.
- Find the value of $h''(-3)$, or explain why it does not exist.
- Must there exist a value of p , for $-3 < p < 1$, such that $h'(p)$ is equal to the average rate of change of h over the interval $-3 \leq x \leq 1$? Justify your answer.
- Find $\lim_{x \rightarrow -2} \frac{h'(x)}{4x}$. Show the computations that lead to your answer.

6-5

(i) The function f is defined by $f(x) = h(-2x)$. Find $f'(1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3$, $x = -1$, and $x = 1$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-5, 2)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-5, 2)$. The point is still earned provided the three critical points $x = -3$, $x = -1$ and $x = 1$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(-1, 0)$ and $(1, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -3$, $x = -1$, and $x = 1$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g , then:

- the response is eligible for all three points, if the response presents the connection between $h'(x)$ and $g(x)$, (that is, $h' = g$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $h' = g$, throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection $h'(x) = g(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative maximum) is

6-5

eligible for at most one of the three points.

Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(1)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $h(x)$ values for $x = -3, -1, 1$ must be included in the table or list of values. The correct values of relevant inputs include:

$$\cdot h(-5) = -\frac{3}{2}$$

$$\cdot h(-3) = \frac{1}{2}$$

$$\cdot h(-1) = \frac{\pi}{4} - 1$$

$$\cdot h(1) = \frac{\pi}{2} - \frac{5}{2}$$

$$\cdot h(2) = \frac{\pi}{2} - 2$$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -3$, h has a relative maximum because $h' = g$ is positive for $x < -3$ and $h' = g$ is negative for $x > -3$.

· At $x = -1$, h has neither because $h' = g$ is negative for $x < -1$ and $h' = g$ is negative for $x > -1$.

· At $x = 1$, h has a relative minimum because $h' = g$ is negative for $x < 1$ and $h' = g$ is positive for $x > 1$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

6-5

- ☐ Relative maximum with justification
- ☐ Neither with justification
- ☐ Relative minimum with justification

Solution:

At $x = -3$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = -1$, h has neither because $h'(x) = g(x)$ does not change sign there.

At $x = 1$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[1, 2]$, $(1, 2]$, $[1, 2)$, and $(1, 2)$ all earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only h' is eligible to earn the second point.

A response referencing only g is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason “ h' is increasing” can also be expressed by either “ h'' is positive” or “the slope of h' is positive.”

A response referencing just “the function” or “it” or “the slope” is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

h is increasing where $h' = g$ is positive.

6-5

h is concave up where $h' = g$ is increasing.

h is increasing and concave up on the open interval $(1, 2)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $h(0) = -0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(0) = -\left(2 \cdot 1 - \frac{\pi}{2} \cdot 1^2\right) = -\left(2 - \frac{\pi}{2}\right) = \frac{\pi}{2} - 2$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\frac{1}{2} - 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $-\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $h(-5) = -\frac{3}{2}$ with no supporting work earns the second point only.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\int_{-5}^{-3} g(t) dt = 2$ – OR – $\int_{-3}^{-2} g(t) dt = -\frac{1}{2}$

☐ Answer

6-5

Solution:

$$\begin{aligned}
 h(-5) &= \int_{-2}^{-5} g(t) \, dt = - \int_{-5}^{-2} g(t) \, dt \\
 &= - \int_{-5}^{-3} g(t) \, dt - \int_{-3}^{-2} g(t) \, dt = -(2) - \left(-\frac{1}{2}\right) = -\frac{3}{2}
 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $h''(-3) = -1$ or $g'(-3) = -1$ or even simply stating a numerical value of -1 clearly marked as the answer earns the point.

No supporting work is required because the answer of -1 can easily be read from the graph.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h''(x) = g'(x)$$

$$h''(-3) = g'(-3) = \frac{2 - (-1)}{-5 - (-2)} = -1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

• “Yes, $h(x)$ is continuous and differentiable so MVT applies” earns both points.

6-5

· “Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(p)$ equals the average rate of change of $h(x)$ ” earns both points.

· “Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(p)$ equals the average rate of change of $h(x)$ ” does not earn the first point, but does earn the second point.

· “Yes because of MVT” does not earn the first point, but does earn the second point.

· “Yes, because $h(x)$ is continuous and differentiable” earns the first point, but not the second point.

· “Yes, because $h(x)$ is continuous” earns neither point.

· “Yes, because $h(x)$ is differentiable” earns neither point.

· “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims “ h is not differentiable, so therefore the MVT does not apply” earns one of the two points.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$h' = g \Rightarrow h$ is differentiable on $-3 \leq x \leq 1$. $\Rightarrow h$ is continuous on $-3 \leq x \leq 1$.

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 1$ to guarantee that there exists a value of p , for $-3 < p < 1$, such that $h'(p)$ equals the average rate of change of h over the interval $-3 \leq x \leq 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim_{x \rightarrow 2} \frac{h'(x)}{4x} = \frac{1}{8}$ earns the point, as does just stating the answer of $\frac{1}{8}$.

No additional supporting work is required.

6-5

The response does not need to state that h' is continuous. If the response does state that h' is continuous, an explanation is not required.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h' = g \Rightarrow h' \text{ is continuous.} \Rightarrow \lim_{x \rightarrow -2} h'(x) = h'(-2)$$

$$\lim_{x \rightarrow -2} \frac{h'(x)}{4x} = \frac{h'(-2)}{4 \cdot (-2)} = \frac{g(-2)}{-8} = \frac{-1}{-8} = \frac{1}{8}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 1$ is substituted for x , then $x = 1$ must appear unsimplified to earn the first point. Responses that earn the point include:

$$\cdot h'(-2x) \cdot (-2)$$

$$\cdot g(-2x) \cdot (-2)$$

$$\cdot h'(-2 \cdot 1) \cdot (-2)$$

$$\cdot g(-2 \cdot 1) \cdot (-2)$$

Responses that do not earn the point include:

$$\cdot h'(x) \cdot (-2)$$

$$\cdot g(x) \cdot (-2)$$

$$\cdot h'(-2) \cdot (-2)$$

$$\cdot g(-2) \cdot (-2)$$

$$\cdot h(-2x) \cdot (-2)$$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $f(x)$ or in $f'(1)$. For example, the second point is earned by $g(-2x)$ or by $g(-2 \cdot 1)$.

6-5

The second point may be earned without earning the first point. For example, $h'(-2x) = g(-2x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and g are linked.

The second point may be earned by going directly from $h'(-2)$ to -1 . The function g does not have to be stated explicitly. For example, $h'(-2 \cdot 1) \cdot (-2) = -1 \cdot (-2)$ earns the first two points. Recall, if $x = 1$ is substituted for x , then $x = 1$ must appear unsimplified to earn the first point.

The first two points may be earned by a single expression. For example, both $g(-2x) \cdot (-2)$ and $g(-2 \cdot 1) \cdot (-2)$ earn the first two points.

The third point is earned for the correct answer of 2, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$f'(x) = h'(-2x) \cdot (-2)$$

$$= g(-2x) \cdot (-2)$$

$$f'(1) = g(-2 \cdot 1) \cdot (-2) = g(-2) \cdot (-2) = (-1) \cdot (-2) = 2$$

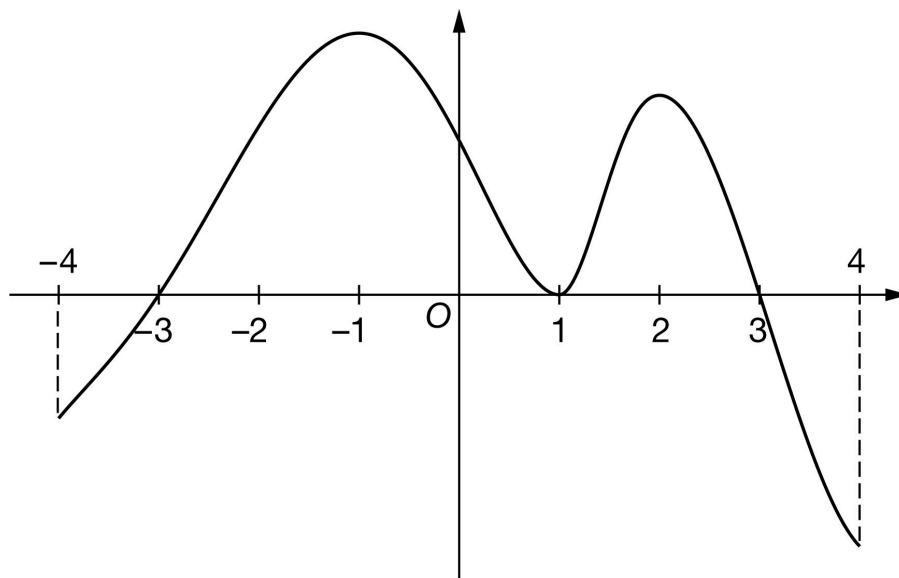
6-5

49. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The graph of the differentiable function f with domain $-4 \leq x \leq 4$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of f on the intervals $[-4, -3]$, $[-3, 1]$, $[1, 3]$, and $[3, 4]$ are 5, 46, 18, and 11, respectively. The graph of f has horizontal tangents at $x = -1$, $x = 1$, and $x = 2$. Let h be the function defined by $h(x) = \int_1^x f(t) \, dt$ for $-4 \leq x \leq 4$.

- Find the value of $h(3)$.
- Find the value of $h(-4)$. Show the computations that lead to your answer.
- Find the x -coordinate of each critical point of h on the interval $-4 < x < 4$.
- Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- For $-4 < x < 4$, find the x -coordinate of each point of inflection of the graph of h . Justify your answers.
- Explain why there must be a value of c , for $1 < c < 3$, such that $h(c) = 13$.
- Find $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 6x}$. Give a reason for your answer.

6-5

(h) The function m is defined by $m(x) = h(x^2)$. It is known that $f(4) = -19.3$. Find $m'(2)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 18 is all that is needed to earn the point. No supporting work is required.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(3) = \int_1^3 f(t) dt = 18$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 5 and ± 46 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 5 and ± 46 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-(-5) - 46$ earns both points because numerical expressions need not be simplified.
- The expression $-5 - 46$ earns the first point, but not the second point.
- The expression $-(-5) - 46 + 8$ earns neither point.

Note: Just the list $-5, 46$ earns neither point. However, if the values -5 and 46 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals that equals -41 earns just the second point because not enough work has been shown.

6-5



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-4}^{-3} f(t) \, dt \right| = 5$ and $\left| \int_{-3}^1 f(t) \, dt \right| = 46$

☐ Answer

Solution:

$$\begin{aligned} h(-4) &= \int_1^{-4} f(t) \, dt = - \int_{-4}^1 f(t) \, dt \\ &= - \int_{-4}^{-3} f(t) \, dt - \int_{-3}^1 f(t) \, dt = -(-5) - 46 = -41 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = 1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4, 4)$. The point is still earned provided the three critical points, $x = -3$, $x = 1$, and $x = 3$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(1, 0)$, and $(3, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = f(x) = 0 \text{ at } x = -3, x = 1, \text{ and } x = 3.$$

6-5

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on f , then:

- the response is eligible for all three points, if the response presents the connection between $f(x)$ and $h'(x)$ (that is, $f(x) = h'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of f or h' as $f(x) = h'(x)$, throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection $f(x) = h'(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-3)$ changes from positive to negative”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -3$, $x = 1$, $x = 3$) must be included in the table or list. The correct values of relevant inputs are:

- $h(-4) = -41$

- $h(-3) = -46$

- $h(1) = 0$

- $h(3) = 18$

- $h(4) = 7$

6-5

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

- At $x = -3$, h has a relative minimum because $h' = f$ is negative for $x < -3$ and $h' = f$ is positive for $x > -3$.
- At $x = 1$, h has neither because $h' = f$ is positive for $x < 1$ and $h' = f$ is positive for $x > 1$.
- At $x = -3$, h has a relative maximum because $h' = f$ is positive for $x < 3$ and $h' = f$ is negative for $x > 3$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

At $x = 1$, h has neither a relative minimum nor a relative maximum because $h'(x) = f(x)$ does not change sign there.

At $x = 3$, h has a relative maximum because $h'(x) = f(x)$ changes from positive to negative there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list $-1, 1, 2$ and only this list, earns the first point.

A response referring only to $h'(x)$ is eligible to earn the second point.

A response referring only to $f(x)$ is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $h'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

- h' changes from increasing to decreasing or vice versa
- h'' changes from positive to negative or vice versa

6-5

h' has a local extrema

Any incorrect statement (such as h' changes from decreasing to increasing at $x = -1$) fails to earn the second point.

If the response addresses at least one correct value of x it is eligible to earn the second point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of h occur where $h' = f$ changes from increasing to decreasing or vice versa. Therefore, the points of inflection occur at $x = -1$, $x = 1$, and $x = 2$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that h is continuous in order to be eligible to earn the point. The response is not required to state that h is differentiable $\Rightarrow$ that h is continuous.

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $1 < x < 3$, or the point is not earned.

A response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, in the presence of the required function evaluation, the statement “since h is continuous, and $0 < 13 < 18$, such a point exists” would earn the point.

The value of 13 must be bounded by 0 and 18 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $h(1)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. $h(1) = 0$ must be present.

A response may reference a value of $h(3)$ determined in part (a). If the value of $h(3)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is greater than 13.

6-5



0	1
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The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

$h' = f$ is differentiable on $1 \leq x \leq 3$. $\Rightarrow h$ is continuous on $1 \leq x \leq 3$.

$$h(1) = 0 < 13 < 18 = h(3)$$

By the Intermediate Value Theorem, there must be a value of c , for $1 < c < 3$, such that $h(c) = 13$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that h and h' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

- The first point may be earned by stating $3^2 - 9 = 0$ and $h(3) - 18 = 0$.
 - All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3^2 = 9$ and $h(3) = 18$ are not enough to earn the first point.
 - The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.
 - Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 6x} = \frac{0}{0}$ does not earn the first point. The statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 6x} \rightarrow \frac{0}{0}$ does earn the first point.
- The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.
- If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.
 - Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 6x} = \frac{2x}{h'(x) - 6}$, will earn the second point, but not the third point.

6-5

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x \rightarrow 3} \frac{x^2-9}{h(x)-6x} = \frac{0}{0}$ and $\lim_{x \rightarrow 3} \frac{x^2-9}{h(x)-6x} = \frac{2x}{h'(x)-6}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of -1 , or its numeric equivalent.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

$h' = f$ is differentiable. $\Rightarrow h' = f$ is continuous.
 $\Rightarrow h$ is continuous. $\Rightarrow \lim_{x \rightarrow 3} h(x) = h(3)$ and $\lim_{x \rightarrow 3} h'(x) = h'(3)$.

$$\lim_{x \rightarrow 3} (x^2 - 9) = 3^2 - 9 = 0$$

$$\lim_{x \rightarrow 3} (h(x) - 6x) = h(3) - 6 \cdot 3 = 18 - 18 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{x^2-9}{h(x)-6x} &= \lim_{x \rightarrow 3} \frac{2x}{h'(x)-6} \\ &= \frac{2 \cdot 3}{h'(3)-6} = \frac{6}{f(3)-6} = \frac{6}{0-6} = -1 \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 2$ is substituted for x , then $x = 2$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

- $\cdot h'(x^2) \cdot 2x$
- $\cdot f(x^2) \cdot 2x$
- $\cdot h'(2^2) \cdot 2 \cdot 2$

6-5

$$\cdot f(2^2) \cdot 2 \cdot 2$$

$$\cdot h'(4) \cdot 2 \cdot 2$$

$$\cdot f(4) \cdot 2 \cdot 2$$

$$\cdot h'(2^2) \cdot 4$$

$$\cdot f(2^2) \cdot 4$$

Responses that do not earn the point include:

$$\cdot h'(x^2) \cdot 2$$

$$\cdot f(x^2) \cdot 2$$

$$\cdot h'(x) \cdot 2x$$

$$\cdot f(x) \cdot 2x$$

$$\cdot h'(4) \cdot 4$$

$$\cdot f(4) \cdot 4$$

$$\cdot h(x^2) \cdot 2x$$

The second point is earned for the correct application of the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $m(x)$ or in $m'(2)$. For example, the second point is earned by $f(x^2)$ or by $f(2^2)$.

The second point may be earned without earning the first point. For example, $h'(x^2) \cdot 2 = f(x^2) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and f are linked.

The second point may be earned by going directly from $h'(1)$ to -19.3 . The function f does not have to be stated explicitly. Responses that earn the first two points include:

$$\cdot h'(2^2) \cdot 2 \cdot 2 = -19.3 \cdot 2 \cdot 2$$

$$\cdot h'(2^2) \cdot 4 = -19.3 \cdot 4$$

Recall, if $x = 2$ is substituted for x , then $x = 2$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $f(x^2) \cdot 2x$ and $f(2^2) \cdot 2 \cdot 2$ earn the first two points.

The third point is earned for the correct answer of -77.2 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

6-5

Be careful of parenthesis errors in part (h). Responses of the form $h'(x^2) \cdot 2x$ or $(h'(x))^2 \cdot 2x$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$m'(x) = h'(x^2) \cdot 2x = f(x^2) \cdot 2x$$

$$m'(2) = f(2^2) \cdot 2 \cdot 2 = f(4) \cdot 4 = -19.3 \cdot 4 = -77.2$$

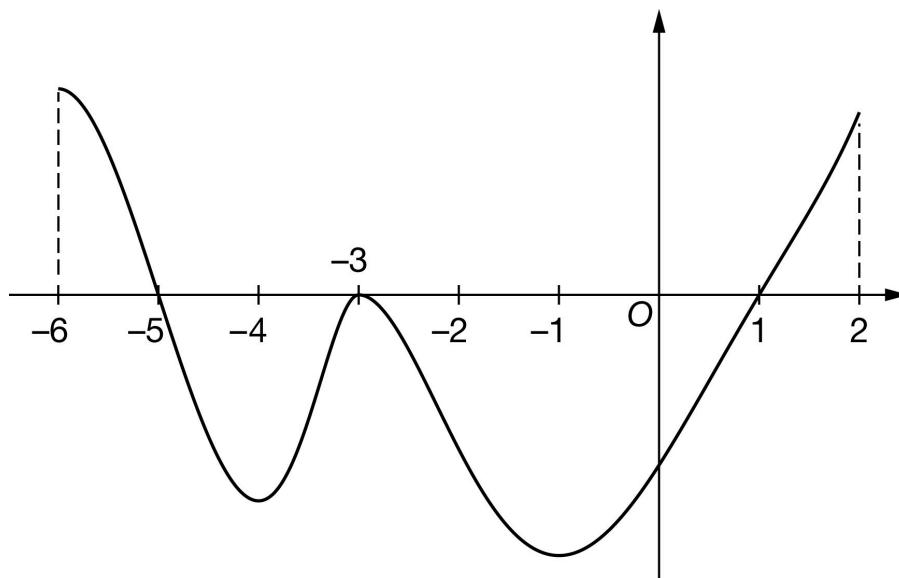
6-5

50. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the differentiable function g with domain $-6 \leq x \leq 2$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of g on the intervals $[-6, -5]$, $[-5, -3]$, $[-3, 1]$, and $[1, 2]$ are 9, 17, 42, and 6, respectively. The graph of g has horizontal tangents at $x = -4$, $x = -3$, and $x = -1$. Let h be the function defined by $h(x) = \int_{-3}^x g(t) dt$ for $-6 \leq x \leq 2$.

- Find the value of $h(1)$.
- Find the value of $h(-6)$. Show the computations that lead to your answer.
- Find the x -coordinate of each critical point of h on the interval $-6 \leq x \leq 2$.
- Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- For $-6 \leq x \leq 2$, find the x -coordinate of each point of inflection of the graph of h . Justify your answers.
- Explain why there must be a value of k , for $-3 < k < 1$, such that $h(k) = -33$.
- Find $\lim_{x \rightarrow 1} \frac{3x^2 - 3}{h(x) + 42x}$. Give a reason for your answer.

6-5

(h) The function p is defined by $p(x) = h(2x)$. It is known that $g(-2) = -10.8$. Find $p'(-1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -42 is all that is needed to earn the point. No supporting work is required.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(1) = \int_{-3}^1 g(t) dt = -42$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 9 and ± 17 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 9 and ± 17 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-9 - (-17)$ earns both points because numerical expressions need not be simplified.
- The expression $-9 - 17$ earns the first point, but not the second point.
- The expression $-9 - (-17) + 42$ earns neither point.

Note: Just the list $-9, -17$ earns neither point. However, if the values -9 and -17 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals equal to 8 earns just the second point because not enough work has been shown.

6-5



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-6}^{-5} f(t) \, dt \right| = 9$ and $\left| \int_{-5}^{-3} f(t) \, dt \right| = 17$

☐ Answer

Solution:

$$\begin{aligned} h(-6) &= \int_{-3}^{-6} g(t) \, dt = - \int_{-6}^{-3} g(t) \, dt \\ &= - \int_{-6}^{-5} g(t) \, dt - \int_{-5}^{-3} g(t) \, dt = -9 - (-17) = 8 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -5$, $x = -3$, and $x = 1$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-6, 2)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-6, 2)$. The point is still earned provided the three critical points, $x = -5$, $x = -3$, and $x = 1$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-5, 0)$, $(-3, 0)$, and $(1, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = g(x) = 0 \text{ at } x = -5, x = -3, \text{ and } x = 1.$$

6-5

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g , then:

- the response is eligible for all three points, if the response presents the connection between $g(x)$ and $h'(x)$ (that is, $g(x) = h'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.
- the response is eligible for at most two of the three points, if the response does not present the connection $g(x) = h'(x)$. Any additional error (for example, not justifying that at $x = -5$, h has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-5)$ changes from positive to negative”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -5$, $x = -3$, $x = 1$) must be included in the table or list. The correct values of relevant inputs are:

- $h(-6) = 8$
- $h(-5) = 17$
- $h(-3) = 0$
- $h(1) = -42$
- $h(2) = -36$

6-5

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

- At $x = -5$, h has a relative maximum because $h' = g$ is positive for $x < -5$ and $h' = g$ is negative for $x > -5$.
- At $x = -3$, h has neither because $h' = g$ is negative for $x < -3$ and $h' = g$ is negative for $x > -3$.
- At $x = 1$, h has a relative minimum because $h' = g$ is negative for $x < 1$ and $h' = g$ is positive for $x > 1$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification
- ☐ Neither with justification
- ☐ Relative minimum with justification

Solution:

At $x = -5$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = -3$, h has neither a relative minimum nor a relative maximum, because $h'(x) = g(x)$ does not change sign there.

At $x = 1$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list -4 , -3 , -1 , and only this list, earns the first point.

A response referring only to $h'(x)$ is eligible to earn the second point.

A response referring only to $g(x)$ is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

- h' changes from increasing to decreasing or vice versa
- h'' changes from positive to negative or vice versa

6-5

· h' has a local extrema

Any incorrect statement (such as h' changes from increasing to decreasing at $x = -4$) fails to earn the second point.

If the response addresses at least one correct value of x , it is eligible to earn the second point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of h occur where $h' = g$ changes from increasing to decreasing, or vice versa. Therefore, the points of inflection occur at $x = -4$, $x = -3$, and $x = -1$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that h is continuous in order to be eligible to earn the point. The response is not required to state that h is differentiable $\Rightarrow h$ is continuous.

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $-3 < x < 1$, or the point is not earned.

A response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, in the presence of the required function evaluation, the statement “since h is continuous, and $-42 < -33 < 0$, such a point exists” would earn the point.

The value of -33 must be bounded by -42 and 0 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $h(-3)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. We need to see $Ah(-3) = 0$.

A response may reference a value of $h(1)$ determined in part (a). If the value of $h(1)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is less than -33 .

6-5



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

$h' = g$ is differentiable on $-3 \leq x \leq 1$. $\Rightarrow h$ is continuous on $-3 \leq x \leq 1$.

$$h(1) = -42 < -33 < 0 = h(-3)$$

By the Intermediate Value Theorem, there must be a value of k , for $-3 < k < 1$, such that $h(k) = -33$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that h and h' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

· The first point may be earned by stating $3 \cdot 1^2 - 3 = 0$ and $h(1) + 42 = 0$.

· All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3 \cdot 1^2 = 3$ and $h(1) = -42$ are not enough to earn the first point.

· The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

· Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim_{x \rightarrow 1} \frac{3x^2 - 3}{h(x) + 42x} = \frac{0}{0}$ does not earn the first point. The statement

$\lim_{x \rightarrow 1} \frac{3x^2 - 3}{h(x) + 42x} \rightarrow \frac{0}{0}$ does earn the first point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

· If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

· Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 1} \frac{3x^2 - 3}{h(x) + 42x} = \frac{6x}{h'(x) + 42}$, will earn the second point, but not the third point.

6-5

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x \rightarrow 1} \frac{3x^2-3}{h(x)+42x} = \frac{0}{0}$ and $\lim_{x \rightarrow 1} \frac{3x^2-3}{h(x)+42x} = \frac{6x}{h'(x)+42}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{1}{7}$, or its numeric equivalent.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$h' = g$ is differentiable. $\Rightarrow h' = g$ is continuous.
 $\Rightarrow h$ is continuous. $\Rightarrow \lim_{x \rightarrow 1} h(x) = h(1)$ and $\lim_{x \rightarrow 1} h'(x) = h'(1)$.

$$\lim_{x \rightarrow 1} (3x^2 - 3) = 3 \cdot 1^2 - 3 = 0$$

$$\lim_{x \rightarrow 1} (h(x) + 42x) = h(1) + 42 \cdot 1 = -42 + 42 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{3x^2-3}{h(x)+42x} &= \lim_{x \rightarrow 1} \frac{6x}{h'(x)+42} \\ &= \frac{6 \cdot 1}{h'(1)+42} = \frac{6}{g(1)+42} = \frac{6}{0+42} = \frac{1}{7} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -1$ is substituted for x , then $x = -1$ must appear unsimplified to earn the first point. Responses that earn the point include:

$$\cdot h'(2x) \cdot 2$$

$$\cdot g(2x) \cdot 2$$

6-5

$$\cdot h'(2 \cdot (-1)) \cdot 2$$

$$\cdot g(2 \cdot (-1)) \cdot 2$$

Responses that do not earn the point include:

$$\cdot h'(x) \cdot 2$$

$$\cdot g(x) \cdot 2$$

$$\cdot h'(-2) \cdot 2$$

$$\cdot g(-2) \cdot 2$$

$$\cdot h(2x) \cdot 2$$

The second point is earned for the application of the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $p(x)$ or in $p'(-1)$. For example, the second point is demonstrated by $g(2x)$ or by $g(2 \cdot 2)$.

The second point may be earned without earning the first point. For example, $h'(2x) = g(2x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and g are linked.

The second point may be earned by going directly from $h'(-2)$ to -10.8 . The function g does not have to be stated explicitly. For example, $h'(2 \cdot (-1)) \cdot 2 = -10.8 \cdot 2$ earns the first two points. Recall, if $x = -1$ is substituted for x , then $x = -1$ must appear unsimplified to earn the first point.

The first two points can be earned by a single expression. For example, both $g(2x) \cdot 2$ and $g(2 \cdot (-1)) \cdot 2$ earn the first two points.

The third point is earned for the correct answer of -21.6 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$p'(x) = h'(2x) \cdot 2$$

6-5

$$= g(2x) \cdot 2$$

$$p'(-1) = g(2 \cdot (-1)) \cdot 2 = g(-2) \cdot 2 = (-10.8) \cdot 2 = -21.6$$

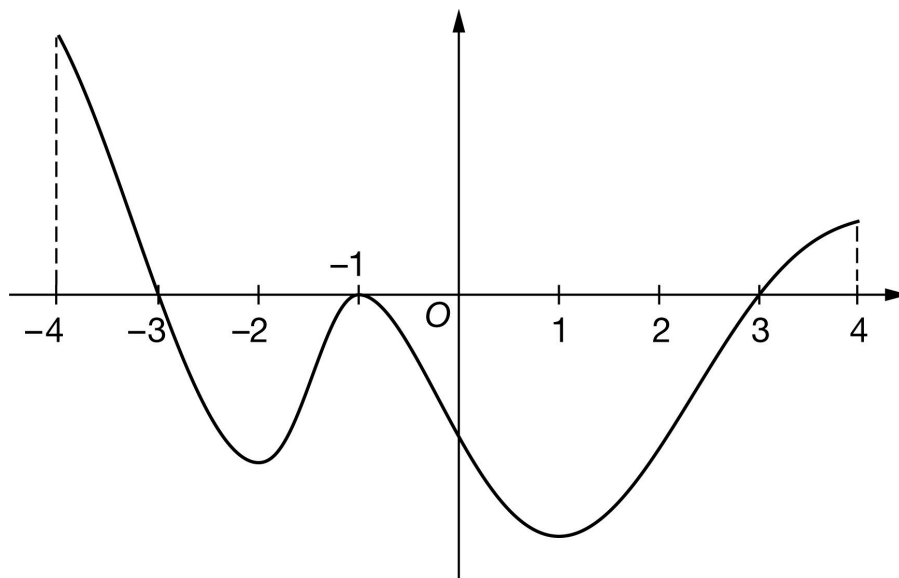
6-5

51. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The graph of the differentiable function f with domain $-4 \leq x \leq 4$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of f on the intervals $[-4, -3]$, $[-3, -1]$, $[-1, 3]$, and $[3, 4]$ are 12, 17, 48, and 4, respectively. The graph of f has horizontal tangents at $x = -2$, $x = -1$, and $x = 1$. Let g be the function defined by $g(x) = \int_{-1}^x f(t) dt$ for $-4 \leq x \leq 4$.

- Find the value of $g(3)$.
- Find the value of $g(-4)$. Show the computations that lead to your answer.
- Find the x -coordinate of each critical point of g on the interval $-4 < x < 4$.
- Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for g . Justify each of your answers.
- For $-4 < x < 4$, find the x -coordinate of each point of inflection of the graph of g . Justify your answers.
- Explain why there must be a value of d , for $-1 < d < 3$, such that $g(d) = -25$.
- Find $\lim_{x \rightarrow 3} \frac{x^2 - 9}{g(x) + 16x}$. Give a reason for your answer.

6-5

(h) The function a is defined by $a(x) = g(x^3)$. It is known that $f(1) = -21.1$. Find $a'(1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of -48 is all that is needed to earn the point. No supporting work is required.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(3) = \int_{-1}^3 f(t) dt = -48$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 12 and ± 17 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 12 and ± 17 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-12 - (-17)$ earns both points because numerical expressions need not be simplified.
- The expression $-12 - (-17)$ earns the first point, but not the second point.
- The expression $-12 - (-17) + 48$ earns neither point.

Note: Just the list $-12, -17$ earns neither point. However, if the values -12 and -17 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals that equals 5 earns just the second point because not enough work has been shown.

6-5



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-3}^{-4} f(t) dt \right| = 12$ and $\left| \int_{-3}^{-1} f(t) dt \right| = 17$

☐ Answer

Solution:

$$\begin{aligned} g(-4) &= \int_{-1}^{-4} f(t) dt = - \int_{-4}^{-1} f(t) dt \\ &= - \int_{-4}^{-3} f(t) dt - \int_{-3}^{-1} f(t) dt = -12 - (-17) = 5 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = -1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4, 4)$. The point is still earned provided the three critical points, $x = -3$, $x = -1$, and $x = 3$, are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(-1, 0)$, and $(3, 0)$ read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g'(x) = f(x) = 0 \text{ at } x = -3, x = -1, \text{ and } x = 3.$$

6-5

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on f , then:

- the response is eligible for all three points, if the response presents the connection between $f(x)$ and $g'(x)$ (that is, $f(x) = g'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of f or g' as $f(x) = g'(x)$, throughout all parts of this question.
- the response is eligible for at most two of the three points, if the response does not present the connection $f(x) = g'(x)$. Any additional error (for example, not justifying that at $x = -3$, g has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(-3)$ changes from positive to negative”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $g(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -3$, $x = -1$, $x = 3$) must be included in the table or list. The correct values of relevant inputs are:

- $g(-4) = 5$
- $g(-3) = 17$
- $g(-1) = 0$
- $g(3) = -48$
- $g(4) = -44$

6-5

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

- At $x = -3$, g has a relative maximum because $g' = f$ is positive for $x < -3$ and $g' = f$ is negative for $x > -3$.
- At $x = -1$, g has neither because $g' = f$ is negative for $x < -1$ and $g' = f$ is negative for $x > -1$.
- At $x = 3$, g has a relative minimum because $g' = f$ is negative for $x < 3$ and $g' = f$ is positive for $x > 3$.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ Relative maximum with justification
- ☐ Neither with justification
- ☐ Relative minimum with justification

Solution:

At $x = -3$, g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

At $x = -1$, g has neither a relative minimum nor a relative maximum because $g'(x) = f(x)$ does not change sign there.

At $x = 3$, g has a relative minimum because $g'(x) = f(x)$ changes from negative to positive there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list -2 , -1 , 1 , and only this list, earns the first point.

A response referring only to $g'(x)$ is eligible to earn the second point.

A response referring only to $f(x)$ is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $g'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

- g' changes from increasing to decreasing or vice versa
- g'' changes from positive to negative or vice versa

6-5

g' has a local extrema

Any incorrect statement (such as g' changes from increasing to decreasing at $x = -2$) fails to earn the second point.

If the response addresses at least one correct value of x , it is eligible to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of g occur where $g' = f$ changes from increasing to decreasing or vice versa. Therefore, the points of inflection occur at $x = -2$, $x = -1$, and $x = 1$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that g is continuous in order to be eligible to earn the point. The response is not required to state that g is differentiable $\Rightarrow g$ is continuous.

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $-1 < x < 3$, or the point is not earned.

A response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, in the presence of the required function evaluation, the statement “since g is continuous, and $-48 < -25 < 0$, such a point exists” would earn the point.

The value of -25 must be bounded by -48 and 0 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $g(-1)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. $g(-1) = 0$ must be present.

A response may reference a value of $g(3)$ determined in part (a). If the value of $g(3)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is less than -25 .

6-5



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

$g' = f$ is differentiable on $-1 \leq x \leq 3$. $\Rightarrow g$ is continuous on $-1 \leq x \leq 3$.

$$g(3) = -48 < -25 < 0 = g(-1)$$

By the Intermediate Value Theorem, there must be a value of d , for $-1 < d < 3$, such that $g(d) = -25$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that g and g' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

- The first point may be earned by stating $3^2 - 9 = 0$ and $g(3) + 48 = 0$.
 - All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3^2 = 9$ and $g(3) = -48$ are not enough to earn the first point.
 - The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.
 - Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{g(x) + 16x} = \frac{0}{0}$ does not earn the first point. The statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{g(x) + 16x} \rightarrow \frac{0}{0}$ does earn the first point.
- The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.
- If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.
 - Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 3} \frac{x^2 - 9}{g(x) + 16x} = \frac{2x}{g'(x) + 16}$, will earn the second point, but not the third point.

6-5

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x \rightarrow 3} \frac{x^2 - 9}{g(x) + 16x} = \frac{0}{0}$ and $\lim_{x \rightarrow 3} \frac{x^2 - 9}{g(x) + 16x} = \frac{2x}{g'(x) + 16}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{3}{8}$, or its numeric equivalent.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$g' = f$ is differentiable. $\Rightarrow g' = f$ is continuous.
 $\Rightarrow g$ is continuous. $\Rightarrow \lim_{x \rightarrow 3} g(x) = g(3)$ and $\lim_{x \rightarrow 3} g'(x) = g'(3)$.

$$\lim_{x \rightarrow 3} (x^2 - 9) = 3^2 - 9 = 0$$

$$\lim_{x \rightarrow 3} (g(x) + 16x) = g(3) + 16 \cdot 3 = -48 + 48 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{x^2 - 9}{g(x) + 16x} &= \lim_{x \rightarrow 3} \frac{2x}{g'(x) + 16} \\ &= \frac{2 \cdot 3}{g'(3) + 16} = \frac{6}{f(3) + 16} = \frac{6}{0 + 16} = \frac{3}{8} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 1$ is substituted for x , then $x = 1$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

$$\cdot g'(x^3) \cdot 3x^2$$

$$\cdot f(x^3) \cdot 3x^2$$

6-5

$$\cdot g'(1^3) \cdot 3 \cdot 1^2$$

$$\cdot f(1^3) \cdot 3 \cdot 1^2$$

$$\cdot g'(1) \cdot 3 \cdot 1^2$$

$$\cdot f(1^3) \cdot 3 \cdot 1$$

Responses that do not earn the point include:

$$\cdot g'(x^3) \cdot 3x$$

$$\cdot f(x^3) \cdot 3x$$

$$\cdot g'(x) \cdot 3x^2$$

$$\cdot f(x) \cdot 3x^2$$

$$\cdot g'(1) \cdot 3$$

$$\cdot f(1) \cdot 3$$

The second point is earned for the correct application of the Fundamental Theorem of Calculus as evidenced by linking g' and f either directly or indirectly in the presented derivative of $a(x)$ or in $a'(1)$. For example, the second point is earned by $f(x^3)$ or by $f(1^3)$.

The second point may be earned without earning the first point. For example, $g'(x^3) \cdot 3x = f(x^3) \cdot 3x$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because g' and f are linked.

The second point may be earned by going directly from $g'(1)$ to -21.1 . The function f does not have to be stated explicitly. For example, $g'(1^3) \cdot 3 \cdot 1^2 = -21.1 \cdot 3 \cdot 1$ earns the first two points. Recall, if $x = 1$ is substituted for x , then $x = 1$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $f(x^3) \cdot 3x^2$ and $f(1^3) \cdot 3 \cdot 1^2$ earn the first two points.

The third point is earned for the correct answer of -63.3 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (h). Responses of the form $g'(x)^3 \cdot 3x^2$ or $(g'(x))^3 \cdot 3x^2$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.



0	1	2	3
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6-5

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$a'(x) = g'(x^3) \cdot 3x^2$$

$$= f(x^3) \cdot 3x^2$$

$$a'(1) = g'(1^3) \cdot 3 \cdot 1^2 = g'(1) \cdot 3 = f(1) \cdot 3 = (-21.1) \cdot 3 = -63.3$$

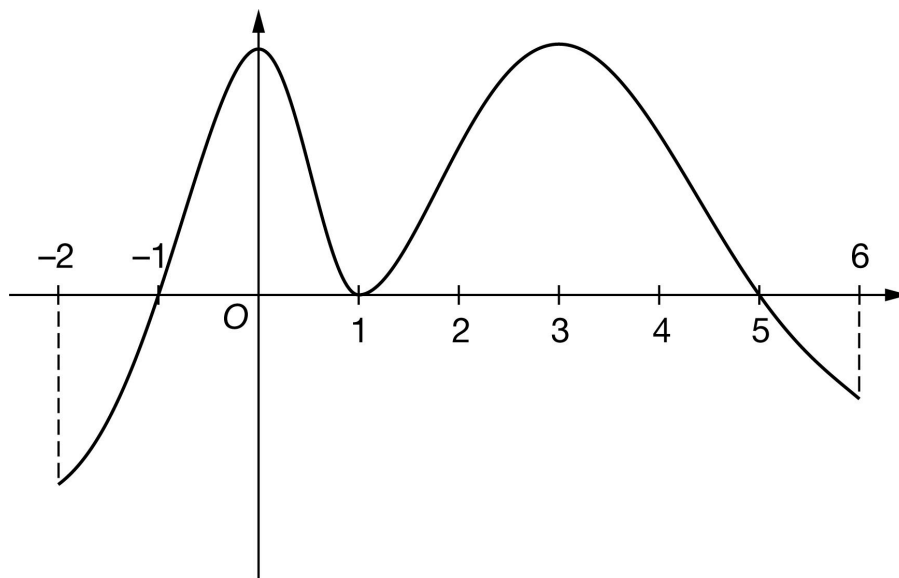
6-5

52. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h

The graph of the differentiable function h with domain $-2 \leq x \leq 6$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of h on the intervals $[-2, -1]$, $[-1, 1]$, $[1, 5]$, and $[5, 6]$ are 8, 19, 40, and 4, respectively. The graph of h has horizontal tangents at $x = 0$, $x = 1$, and $x = 3$. Let g be the function defined by $g(x) = \int_1^x h(t) \, dt$ for $-2 \leq x \leq 6$.

- (a) Find the value of $g(5)$.
- (b) Find the value of $g(-2)$. Show the computations that lead to your answer.
- (c) Find the x -coordinate of each critical point of g on the interval $-2 < x < 6$.
- (d) Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for g . Justify each of your answers.
- (e) For $-2 < x < 6$, find the x -coordinate of each point of inflection of the graph of g . Justify your answers.
- (f) Explain why there must be a value of w , for $1 < w < 5$, such that $g(w) = 5$.
- (g) Find $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x}$. Give a reason for your answer.
- (h) The function z is defined by $z(x) = g(2x)$. It is known that $h(4) = 11.25$. Find $z'(2)$. Show the computations that lead to your answer.

6-5

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 40 is all that is needed to earn the point. No supporting work is required.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(5) = \int_1^5 h(t) dt = 40$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 8 and ± 19 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 8 and ± 19 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-(-8) - 19$ earns both points because numerical expressions need not be simplified.
- The expression $(-8) - 19$ earns the first point, but not the second point.
- The expression $-(-8) - 19 - 40$ earns neither point.

Note: Just the list $-8, 19$ earns neither point. However, if the values -8 and 19 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals that equals -11 earns just the second point because not enough work has been shown.

6-5



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-2}^{-1} h(t) dt \right| = 8$ and $\left| \int_1^{-1} h(t) dt \right| = 19$

☐ Answer

Solution:

$$\begin{aligned} g(-2) &= \int_1^{-2} h(t) dt = - \int_{-2}^1 h(t) dt \\ &= - \int_{-2}^{-1} h(t) dt - \int_{-1}^1 h(t) dt = -(-8) - 19 = -11 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -1$, $x = 1$, and $x = 5$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-2, 6)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-2, 6)$. The point is still earned provided the three critical points, $x = -1$, $x = 1$, and $x = 5$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(1, 0)$, and $(5, 0)$ read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g'(x) = h(x) = 0 \text{ at } x = -1, x = 1, \text{ and } x = 5.$$

6-5

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justification based on g' is eligible for all three points.

If a response gives justifications based only on h , then:

- the response is eligible for all three points, if the response presents the connection between $h(x)$ and $g'(x)$ (that is, $h(x) = g'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of h or g' as $h(x) = g'(x)$, throughout all parts of this question.
- the response is eligible for at most two of the three points, if the response does not present the connection $h(x) = g'(x)$. Any additional error (for example, not justifying that at $x = -1$, g has a relative minimum) is eligible for at most one of the three points.

Read references to “the graph” as references to $h(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(-1)$ changes from negative to positive”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $g(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -1$, $x = 1$, $x = 5$) must be included in the table or list. The correct values of relevant inputs are:

- $g(-2) = -11$
- $g(-1) = -19$
- $g(1) = 0$
- $g(5) = 40$
- $g(6) = 36$

6-5

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

- At $x = -1$, g has a relative minimum because $g' = h$ is negative for $x < -1$ and $g' = h$ is positive for $x > -1$.
- At $x = 1$, g has neither because $g' = h$ is positive for $x < 1$ and $g' = h$ is positive for $x > 1$.
- At $x = 5$, g has a relative maximum because $g' = h$ is positive for $x < 5$ and $g' = h$ is negative for $x > 5$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

Solution:

At $x = -1$, g has a relative minimum because $g'(x) = h(x)$ changes from negative to positive there.

At $x = 1$, g has neither a relative minimum nor a relative maximum because $g'(x) = h(x)$ does not change sign there.

At $x = 5$, g has a relative maximum because $g'(x) = h(x)$ changes from positive to negative there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list 0, 1, 3, and only this list, earns the first point.

A response referring only to $g'(x)$ is eligible to earn the second point.

A response referring only to $h(x)$ is not eligible to earn the second point, unless the response has established the link between $h(x)$ and $g'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

- g' changes from increasing to decreasing or vice versa
- g'' changes from positive to negative or vice versa

6-5

g' has a local extrema

Any incorrect statement (such as g' changes from decreasing to increasing at $x = 0$) fails to earn the second point.

If the response addresses at least one correct value of x , it is eligible to earn the second point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of g occur where $g' = h$ changes from increasing to decreasing, or vice versa. Therefore, the points of inflection occur at $x = 0$, $x = 1$, and $x = 3$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that g is continuous in order to be eligible to earn the point. The response is not required to state that “ g is differentiable $\Rightarrow g$ is continuous.”

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval $1 < x < 5$, or the point is not earned.

A response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, in the presence of the required function evaluation, the statement “since g is continuous, and $0 < 5 < 40$, such a point exists” would earn the point.

The value of 5 must be bounded by 0 and 40 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $g(1)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. $g(1) = 0$ must be present.

A response may reference a value of $g(5)$ determined in part (a). If the value of $g(5)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is greater than 5.

6-5



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

$g' = h$ is differentiable on $1 \leq x \leq 5$. $\Rightarrow g$ is continuous on $1 \leq x \leq 5$.

$$g(1) = 0 < 5 < 40 = g(5)$$

By the Intermediate Value Theorem, there must be a value of w , for $1 < w < 5$, such that $g(w) = 5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that g and g' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

- The first point may be earned by stating $5^2 - 25 = 0$ and $g(5) - 40 = 0$.
 - All arithmetic/algebraic expressions must be completely simplified to zero. For example, $5^2 = 25$ and $g(5) = 40$ are not enough to earn the first point.
 - The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.
 - Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example, the statement $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} = \frac{0}{0}$ does not earn the first point. The statement $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} \rightarrow \frac{0}{0}$ does earn the first point.
- The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.
- If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.
 - Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} = \frac{2x}{g'(x) - 8}$, will earn the second point, but not the third point.

6-5

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} = \frac{0}{0}$ and $\lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} = \frac{2x}{g'(x) - 8}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $-5/4$, or its numeric equivalent.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$g' = h$ is differentiable. $\Rightarrow g' = h$ is continuous.
 $\Rightarrow g$ is continuous. $\Rightarrow \lim_{x \rightarrow 5} g(x) = g(5)$ and $\lim_{x \rightarrow 5} g'(x) = g'(5)$.

$$\lim_{x \rightarrow 5} (x^2 - 25) = 5^2 - 25 = 0$$

$$\lim_{x \rightarrow 5} (g(x) - 8x) = g(5) - 8 \cdot 5 = 40 - 40 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 5} \frac{x^2 - 25}{g(x) - 8x} &= \lim_{x \rightarrow 5} \frac{2x}{g'(x) - 8} \\ &= \frac{2 \cdot 5}{g'(5) - 8} = \frac{10}{h(5) - 8} = \frac{10}{0 - 8} = -\frac{5}{4} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 2$ is substituted for x , then $x = 2$ must appear unsimplified to earn the first point. Responses that earn the point include:

$$\cdot g'(2x) \cdot 2$$

$$\cdot h(2x) \cdot 2$$

6-5

$$\cdot g'(2 \cdot 2) \cdot 2$$

$$\cdot h(2 \cdot 2) \cdot 2$$

Responses that do not earn the point include:

$$\cdot g'(x) \cdot 2$$

$$\cdot h(x) \cdot 2$$

$$\cdot g'(2) \cdot 2$$

$$\cdot h(2) \cdot 2$$

$$\cdot g(2x) \cdot 2$$

The second point is earned for the application of the Fundamental Theorem of Calculus as evidenced by linking g' prime and h either directly or indirectly in the presented derivative of $z(x)$ or in $z'(2)$. For example, the second point is demonstrated by $h(2x)$ or by $h(2 \cdot 2)$.

The second point may be earned without earning the first point. For example, $g'(2x) = h(2x)$ does not earn the first point because the Chain rule is applied incorrectly, but does earn the second point because g' and h are linked.

The second point may be earned by going directly from $g'(4)$ to 11.25. The function h does not have to be stated explicitly. For example, $g'(2 \cdot 2) \cdot 2 = 11.25 \cdot 2$ earns the first two points. Recall, if $x = 2$ is substituted for x , then $x = 2$ must appear unsimplified to earn the first point.

The first two points can be earned by a single expression. For example, both $h(2x) \cdot 2$ and $h(2 \cdot 2) \cdot 2$ earn the first two points.

The third point is earned for the correct answer of 22.5, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$z'(x) = g'(2x) \cdot 2$$

6-5

$$= h(2x) \cdot 2$$

$$z'(2) = h(2 \cdot 2) \cdot 2 = h(4) \cdot 2 = 11.25 \cdot 2 = 22.5$$

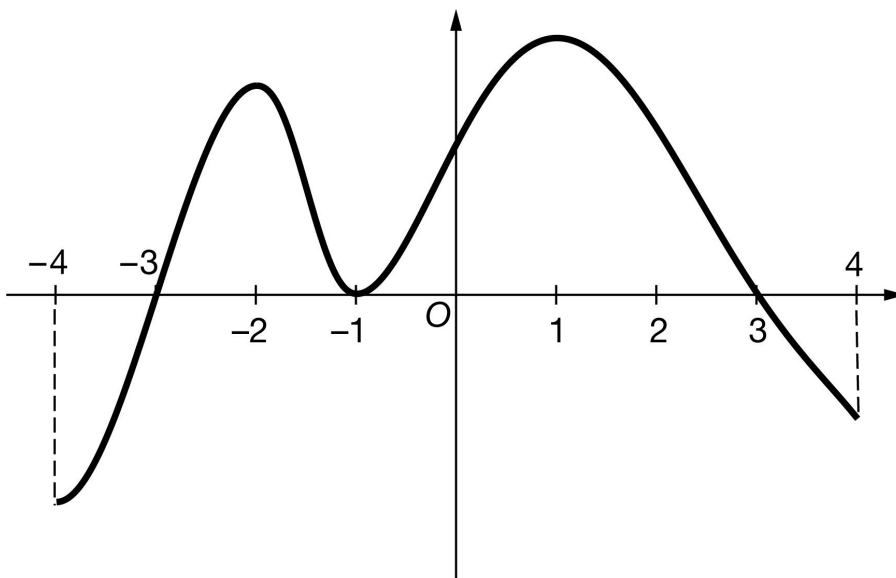
6-5

53. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the differentiable function g with domain $-4 \leq x \leq 4$ is shown in the figure above. The areas of the regions bounded by the x -axis and the graph of g on the intervals $[-4, -3]$, $[-3, -1]$, $[-1, 3]$, and $[3, 4]$ are 10, 18, 45, and 5, respectively. The graph of g has horizontal tangents at $x = -2$, $x = -1$, and $x = 1$. Let h be the function defined by $h(x) = \int_{-1}^x g(t) dt$ for $-4 \leq x \leq 4$.

- Find the value of $h(3)$.
- Find the value of $h(-4)$. Show the computations that lead to your answer.
- Find the x -coordinate of each critical point of h on the interval $-4 < x < 4$.
- Classify each critical point from part (c) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- For $-4 < x < 4$, find the x -coordinate of each point of inflection of the graph of h . Justify your answers.
- Explain why there must be a value of r , for $-1 < r < 3$, such that $h(r) = 20$.
- Find $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x}$. Give a reason for your answer.

6-5

(h) The function k is defined by $k(x) = h(x^2)$. It is known that $g(1) = 19.7$. Find $k'(-1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct value of 45 is all that is needed to earn the point. No supporting work is required.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(3) = \int_{-1}^3 g(t) dt = 45$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned when the response identifies ± 10 and ± 18 in context, often expressed by the sum (or difference). The values do not need to be combined, but if they are, they cannot be combined with any additional values.

The ± 10 and ± 18 terms need not be linked to a definite integral but if they are, this link must be correct.

The second point is earned for a correct answer in the presence of correct supporting work. Some examples and scores include:

- The expression $-(-10) - 18$ earns both points because numerical expressions need not be simplified.
- The $(-10) - 18$ expression earns the first point, but not the second point.
- The expression $-(-10) - 18 - 45$ earns neither point.

Note: Just the list $-10, 18$ earns neither point. However, if the values -10 and 18 are correctly linked with the corresponding definite integrals and never combined, the response earns the first point, but not the second point.

Note: A sum or difference of correct symbolic integrals $= -8$ earns just the second point because not enough work has been shown.

6-5



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ $\left| \int_{-4}^{-3} g(t) \, dt \right| = 10$ and $\left| \int_{-3}^{-1} g(t) \, dt \right| = 18$

☐ Answer

Solution:

$$\begin{aligned} h(-4) &= \int_{-1}^{-4} g(t) \, dt = - \int_{-4}^{-1} g(t) \, dt \\ &= - \int_{-4}^{-3} g(t) \, dt - \int_{-3}^{-1} g(t) \, dt = -(-10) - 18 = -8 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = -1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4, 4)$. The point is still earned provided the three critical points, $x = -3$, $x = -1$, and $x = 3$, are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(-1, 0)$, and $(3, 0)$, read the x -coordinates and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = g(x) = 0 \text{ at } x = -3, x = -1, \text{ and } x = 3.$$

6-5

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (c) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g , then:

- the response is eligible for all three points, if the response presents the connection between $g(x)$ and $h'(x)$ (that is, $g(x) = h'(x)$). If you do not see this in part (d), look for it in part (c). If you find this connection in either part, read ALL subsequent occurrences of g or h' as $g(x) = h'(x)$, throughout all parts of this question.
- the response is eligible for at most two of the three points, if the response does not present the connection $g(x) = h'(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-3)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no point.

A sign chart by itself is not sufficient to earn the points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. Every $h(x)$ value given in the table or list of values must be correct to earn any points. In order to earn the second point, all three critical points (namely, $x = -3$, $x = -1$, $x = 3$) must be included in the table or list. The correct values of relevant inputs are:

- $h(-4) = -8$
- $h(-3) = -18$
- $h(-1) = 0$
- $h(3) = 45$
- $h(4) = 40$

6-5

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (d) earns two points:

- At $x = -3$, h has a relative minimum because $h' = g$ is negative for $x < -3$ and $h' = g$ is positive for $x > -3$.
- At $x = -1$, h has neither because $h' = g$ is positive for $x < -1$ and $h' = g$ is positive for $x > -1$.
- At $x = 3$, h has a relative maximum because $h' = g$ is positive for $x < 3$ and $h' = g$ is negative for $x > 3$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = -1$, h has neither a relative minimum nor a relative maximum, because $h'(x) = g(x)$ does not change sign there.

At $x = 3$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with the list -2 , -1 , 1 , and only this list, earns the first point.

A response referring only to $h'(x)$ is eligible to earn the second point.

A response referring only to $g(x)$ is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously (most likely in part (c) or part (d)), in which case it is eligible to earn the second point.

The second point can be earned by such responses as:

- “ h' changes from increasing to decreasing or vice versa”
- “ h'' changes from positive to negative or vice versa”

6-5

· “ h' has a local extrema”

Any incorrect statement (such as h' changes from decreasing to increasing at $x = -2$) fails to earn the second point.

If the response addresses at least one correct value of x , it is eligible to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Location of inflection points
- ☐ Justification

Solution:

The points of inflection for the graph of h occur where $h' = g$ changes from increasing to decreasing, or vice versa. Therefore, the points of inflection occur at $x = -2$, $x = -1$, and $x = 1$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response must state that h is continuous in order to be eligible to earn the point. The response is not required to state that “ h is differentiable $\Rightarrow h$ is continuous.”

A response need not mention the interval of continuity. If it does, the interval stated must contain the interval stated must contain the interval $-1 < x < 3$, or the point is not earned.

A response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, in the presence of the required function evaluation, the statement “since h is continuous, and $0 < 20 < 45$, such a point exists” would earn the point.

The value of 20 must be bounded by 0 and 45 either symbolically or verbally in order to earn the point.

A response will not earn the point by invoking a non-applicable theorem (such as the Mean Value Theorem).

The connection between $h(-1)$ and 0 must be made explicit. Simply mentioning one or the other is not sufficient. We need to see $h(-1) = 0$.

A response may reference a value of $h(3)$ determined in part (a). If the value of $h(3)$ presented in part (a) is incorrect, that value can be used to earn this point provided the presented value is greater than 20.

6-5



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

$h' = g$ is differentiable on $-1 \leq x \leq 3$. $\Rightarrow h$ is continuous on $-1 \leq x \leq 3$.

$$h(-1) = 0 < 20 < 45 = h(3)$$

By the Intermediate Value Theorem, there must be a value of r , for $-1 < r < 3$, such that $h(r) = 20$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response need not state that h and h' are continuous.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

- The first point may be earned by stating $3^2 - 9 = 0$ and $h(3) - 45 = 0$.
- All arithmetic/algebraic expressions must be completely simplified to zero. For example, $3^2 = 9$ and $h(3) = 45$ are not enough to earn the first point.
- The statement that the limit gives the indeterminate form $0/0$ earns the first point.

· Indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $0/0$. For example, the statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x} = \frac{0}{0}$ does not earn the first point, but the statement $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x} \rightarrow \frac{0}{0}$ earns the first point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

- If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.
- Any incorrect statement of equality here, e.g., $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x} = \frac{2x}{h'(x) - 15}$, will earn the second point, but not the third point.

6-5

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with the statements $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x} = \frac{0}{0}$ and $\lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x} = \frac{2x}{h'(x) - 15}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $-\frac{2}{5}$, or its numeric equivalent.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$h' = g$ is differentiable. $\Rightarrow h' = g$ is continuous.
 $\Rightarrow h$ is continuous $\Rightarrow \lim_{x \rightarrow 3} h(x) = h(3)$ and $\lim_{x \rightarrow 3} h'(x) = h'(3)$.

$$\lim_{x \rightarrow 3} (x^2 - 9) = 3^2 - 9 = 0$$

$$\lim_{x \rightarrow 3} (h(x) - 15x) = h(3) - 15 \cdot 3 = 45 - 45 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{x^2 - 9}{h(x) - 15x} &= \lim_{x \rightarrow 3} \frac{2x}{h'(x) - 15} \\ &= \frac{2 \cdot 3}{h'(3) - 15} = \frac{6}{g(3) - 15} = \frac{6}{0 - 15} = -\frac{2}{5} \end{aligned}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -1$ is substituted for x , then $x = -1$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

$$\cdot h'(x^2) \cdot 2x$$

6-5

$$\cdot g(x^2) \cdot 2x$$

$$\cdot h'((-1)^2) \cdot 2 \cdot (-1)$$

$$\cdot g((-1)^2) \cdot 2 \cdot (-1)$$

$$\cdot h'(1) \cdot 2 \cdot (-1)$$

$$\cdot g(1) \cdot 2 \cdot (-1)$$

$$\cdot h'((-1)^2) \cdot (-2)$$

$$\cdot g((-1)^2) \cdot (-2)$$

Responses that do not earn the point include:

$$\cdot h'(x^2) \cdot 2$$

$$\cdot g(x^2) \cdot 2$$

$$\cdot h'(x) \cdot 2x$$

$$\cdot g(x) \cdot 2x$$

$$\cdot h'(1) \cdot (-2)$$

$$\cdot g(1) \cdot (-2)$$

$$\cdot h(x^2) \cdot 2x$$

The second point is earned for the correct application of the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $k(x)$ or in $k'(-1)$. For example, the second point is earned by $g(x^2)$ or by $g((-1)^2)$.

The second point may be earned without earning the first point. For example, $h'(x^2) \cdot 2 = g(x^2) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and g are linked.

The second point may be earned by going directly from $h'(1)$ to 19.7. The function g does not have to be stated explicitly. Responses that earn the first two points include:

$$\cdot h'((-1)^2) \cdot 2 \cdot (-1) = 19.7 \cdot 2 \cdot (-1)$$

$$\cdot h'((-1)^2) \cdot (-2) = 19.7 \cdot (-2)$$

6-5

Recall, if $x = -1$ is substituted for x , $x = -1$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $g(x^2) \cdot 2x$ and $g((-1)^2) \cdot 2 \cdot (-1)$ earn the first two points.

The third point is earned for the correct answer of -39.4 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (h). Responses of the form $h'(x)^2 \cdot 2x$ or $(h'(x))^2 \cdot 2x$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

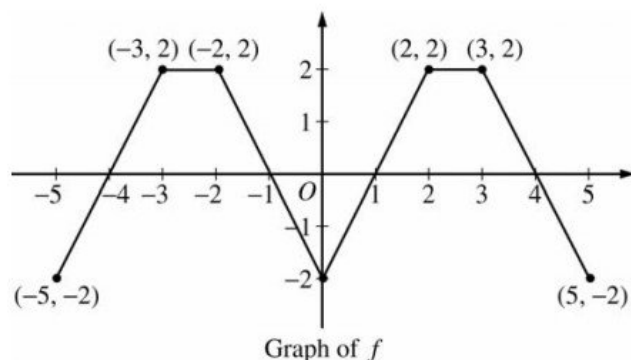
$$k'(x) = h'(x^2) \cdot 2x$$

$$= g(x^2) \cdot 2x$$

$$k'(-1) = g((-1)^2) \cdot 2 \cdot (-1) = g(1) \cdot (-2) = 19.7 \cdot (-2) = -39.4$$

6-5

54.



The graph of the function f shown above consists of six line segments. Let g be the function given by

$$g(x) = \int_0^x f(t) dt.$$

(a) Find $g(4)$, $g'(4)$, and $g''(4)$.

(b) Does g have a relative minimum, a relative maximum, or neither at $x = 1$? Justify your answer.

(c) Suppose that f is defined for all real numbers x and is periodic with a period of length 5. The graph above shows two periods of f . Given that $g(5) = 2$, find $g(10)$ and write an equation for the line tangent to the graph of g at $x = 108$.

Part A

1 point is earned for: $g(4)$

1 point is earned for: $g'(4)$

1 point is earned for: $g''(4)$

$$g(4) = \int_0^4 f(t) dt = 3$$

$$g'(4) = f(4) = 0$$

$$g''(4) = f'(4) = -2$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $g(4)$

1 point is earned for: $g'(4)$

1 point is earned for: $g''(4)$

6-5

$$g(4) = \int_0^4 f(t) dt = 3$$

$$g'(4) = f(4) = 0$$

$$g''(4) = f'(4) = -2$$

Part B

1 point is earned for the answer

1 point is earned for the reason

g has a relative minimum at $x = 1$

because $g' = f$ changes from negative to positive at $x = 1$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the reason

g has a relative minimum at $x = 1$

because $g' = f$ changes from negative to positive at $x = 1$.

Part C

1 point is earned for: $g(10)$

1 point is earned for: $g(108)$

1 point is earned for: $g'(108)$

1 point is earned for the equation of tangent line

$g(0) = 0$ and the function values of g increase by 2 for every increase of 5 in x .

$$g(10) = 2g(5) = 4$$

6-5

$$\begin{aligned}
 g(108) &= \int_0^{105} f(t)dt + \int_{105}^{108} f(t)dt \\
 &= 21g(5) + g(3) = 44
 \end{aligned}$$

$$g'(108) = f(108) = f(3) = 2$$

An equation for the line tangent to the graph of g at $x = 108$ is $y - 44 = 2(x - 108)$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for: $g(10)$

1 point is earned for: $g(108)$

1 point is earned for: $g'(108)$

1 point is earned for the equation of tangent line

$g(0) = 0$ and the function values of g increase by 2 for every increase of 5 in x .

$$g(10) = 2g(5) = 4$$

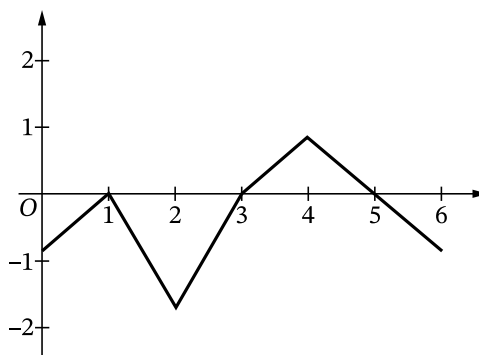
$$\begin{aligned}
 g(108) &= \int_0^{105} f(t)dt + \int_{105}^{108} f(t)dt \\
 &= 21g(5) + g(3) = 44
 \end{aligned}$$

$$g'(108) = f(108) = f(3) = 2$$

An equation for the line tangent to the graph of g at $x = 108$ is $y - 44 = 2(x - 108)$.

6-5

55. The graph of the piecewise linear function f is shown.

Graph of f

Let g be the function given by $g(x) = \int_0^x f(t)dt$. For $0 \leq x \leq 6$, what is the value of x at which the absolute maximum of g occurs?

- (A) 0
(B) 4
(C) 5
(D) 6



Answer A

Correct. Since $g'(x) = f(x)$, g has critical points at $x = 1$, $x = 3$, and $x = 5$. But $x = 1$ is neither a local minimum nor a local maximum since g' does not change signs there, and $x = 3$ is a local minimum since g' changes from negative to positive at $x = 3$. There is a local maximum at $x = 5$ since g' changes from negative to positive at $x = 5$. Therefore the candidates for the absolute maximum are at $x = 5$ and the endpoints $x = 0$ and $x = 6$.

$$g(0) = 0$$

$$g(5) = \int_0^5 f(t)dt = \int_0^3 f(t)dt + \int_3^5 f(t)dt < 0 \text{ since the area above the axis from 3 to 5 is smaller than the area below the axis from 0 to 3.}$$

$$g(6) = \int_0^6 f(t)dt = \int_0^5 f(t)dt + \int_5^6 f(t)dt < \int_0^5 f(t)dt = g(5) < 0 \text{ since } \int_5^6 f(t)dt < 0.$$

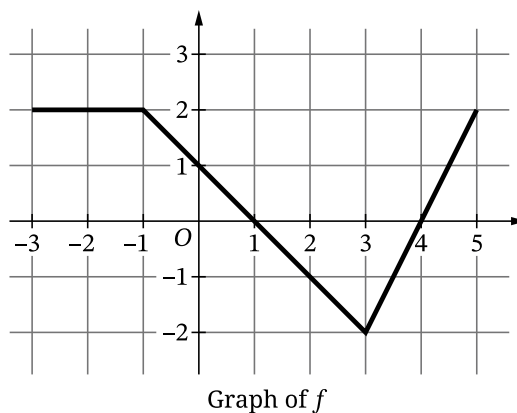
Thus the absolute maximum of g occurs at $x = 0$.

OR

Since $g'(x) = f(x)$, g is decreasing from 0 to 3, then increasing from 3 to 5. But the area above the axis from 3 to 5 is smaller than the area below the axis from 0 to 3, so g does not increase enough from 3 to 5 to become larger than it was at $x = 0$. After that, g decreases further. So the largest value of g must be at $x = 0$.

6-5

56. The graph of the function f on the closed interval $[-3, 5]$ is shown.



Let g be the function given by $g(x) = \int_2^x f(t)dt$. At what value of x in the closed interval $[-3, 5]$ does g have an absolute minimum?

(A) -3

(B) 1

(C) 3

(D) 4



Answer A

Correct. Since $g'(x) = f(x)$, the function g has critical points at $x = 1$ and $x = 4$. The critical point at $x = 1$ is a local maximum since $g' = f$ changes from positive to negative at $x = 1$. The graph of g is increasing for $4 < x < 5$, so $x = 5$ cannot be a minimum. Therefore the only candidates for an absolute minimum are at the critical point $x = 4$ and the endpoint $x = -3$.

$$g(-3) = \int_2^{-3} f(t)dt = -\int_{-3}^2 f(t)dt = -\left((2)(2) + \frac{1}{2}(2)(2) - \frac{1}{2}(1)(1)\right) = -\frac{11}{2}$$

$$g(4) = \int_2^4 f(t)dt = -\left(\frac{1}{2}(1)(1+2) + \frac{1}{2}(1)(2)\right) = -\frac{5}{2}$$

Therefore the absolute minimum is at $x = -3$.

Alternatively, since $g'(x) = f(x)$, the graph of f indicates that g increases on the interval $-3 \leq x \leq 1$ by the amount $\int_{-3}^1 g'(x)dx = \int_{-3}^1 f(x)dx = 4 + 2 = 6$, then decreases on the interval $1 \leq x \leq 4$ by the amount $\left|\int_1^4 g'(x)dx\right| = \left|\int_1^4 f(x)dx\right| = 3$, then increases again on the interval $4 \leq x \leq 5$. Thus g never decreases enough on the interval $1 \leq x \leq 4$ to go below the initial value at $x = -3$, so the absolute minimum of g must occur at $x = -3$.

6-5

57. Let f be the function defined by $f(x) = -2x + \int_2^x \sqrt{1+t^2} \, dt$. Which of the following statements is true on the interval $0 < x < 1$?

- (A) f is decreasing, and the graph of f is concave up. ✓
- (B) f is decreasing, and the graph of f is concave down.
- (C) f is increasing, and the graph of f is concave up.
- (D) f is increasing, and the graph of f is concave down.

Answer A

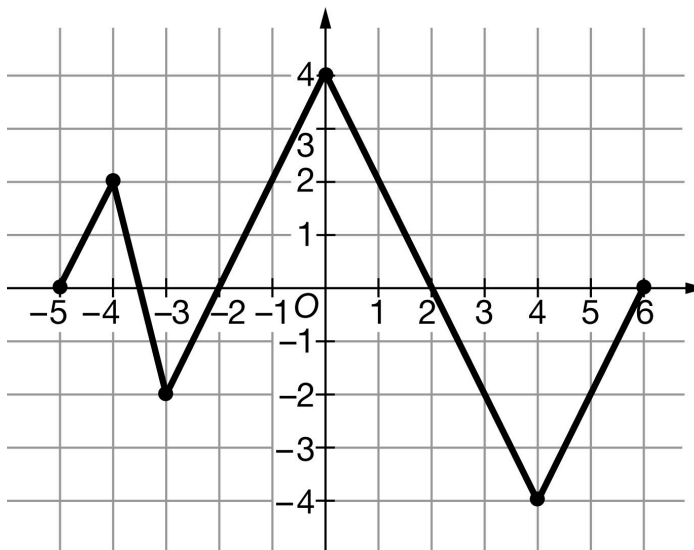
Correct. This statement is true.

$f'(x) = -2 + \sqrt{1+x^2} < 0$ for $0 < x < 1$ since $\sqrt{1+x^2} < 2$ for $0 < x < 1$. Therefore, f is decreasing for $0 < x < 1$.

$f''(x) = \frac{x}{\sqrt{1+x^2}} > 0$ for $0 < x < 1$. Therefore, the graph of f is concave up for $0 < x < 1$.

6-5

58.

Graph of f

The graph of the piecewise linear function f is shown in the figure above. If $g(x) = \int_{-5}^x f(t) \, dt$, how many zeros does g have on the closed interval $[-5, 6]$?

- (A) One
 (B) Three
 (C) Four
 (D) Five

**Answer B**

Correct. The function g has a zero at $x = -5$ because $g(-5) = \int_{-5}^{-5} f(t) \, dt = 0$.

When accumulating values for $\int_{-5}^x f(t) \, dt$ as x increases starting at $x = -5$, the next zero of g will occur when the area of the region bounded by the graph of f above the t -axis is equal to the area of the region bounded by the graph of f below the t -axis. So the next zero of g occurs at $x = -2$. This can be verified by noting that

$$g(-2) = \int_{-5}^{-2} f(t) \, dt = \int_{-5}^{-3.5} f(t) \, dt + \int_{-3.5}^{-2} f(t) \, dt = \frac{1}{2}(1.5)(2) - \frac{1}{2}(1.5)(2) = 0.$$

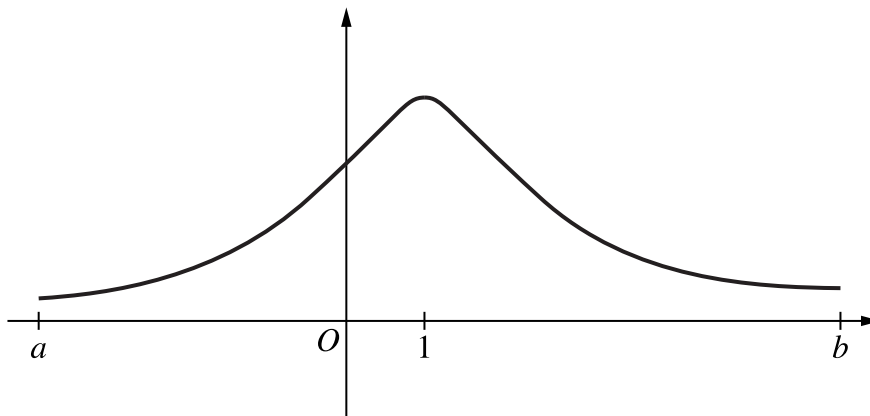
Continuing to accumulate values for $\int_{-5}^x f(t) \, dt$ as x increases from $x = -2$, the next zero of g will occur at $x = 6$. This can be verified by noting that

$$g(6) = \int_{-5}^{-2} f(t) \, dt + \int_{-2}^2 f(t) \, dt + \int_2^6 f(t) \, dt = 0 + \frac{1}{2}(4)(4) - \frac{1}{2}(4)(4) = 0.$$

Therefore, g has three zeros on the interval $[-5, 6]$.

6-5

59.

Graph of f

The figure shows the graph of the function f on the open interval (a, b) . Let g be the function given by $g(x) = \int_1^x f(t) dt$. Which of the following statements must be true?

- I. g is increasing on (a, b) .
 - II. g has a relative maximum on (a, b) .
 - III. g has exactly one zero on (a, b) .
- (A) I only
- (B) II only
- (C) I and II only
- (D) I and III only



Answer D

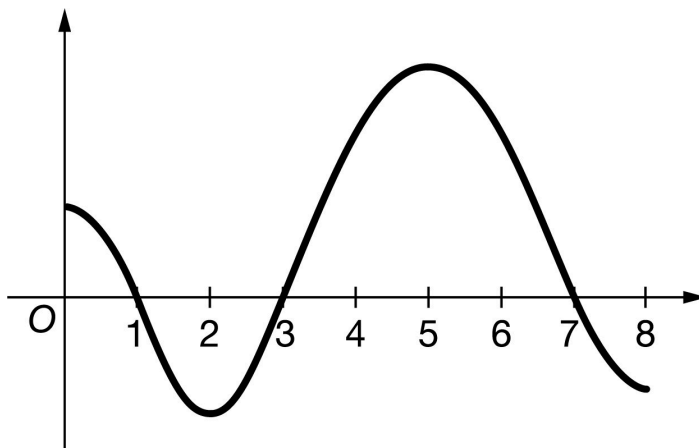
Correct. Statement I is true because $g'(x) = f(x) > 0$ for $a < x < b$.

Statement II is false because $g'(x) = f(x) \neq 0$ for any x in (a, b) , so g has no critical point in (a, b) and therefore cannot have a relative extremum on (a, b) .

Statement III is true because $g(1) = \int_1^1 f(t) dt = 0$, and since g is increasing on (a, b) , $x = 1$ can be the only zero on (a, b) .

6-5

60.

Graph of g

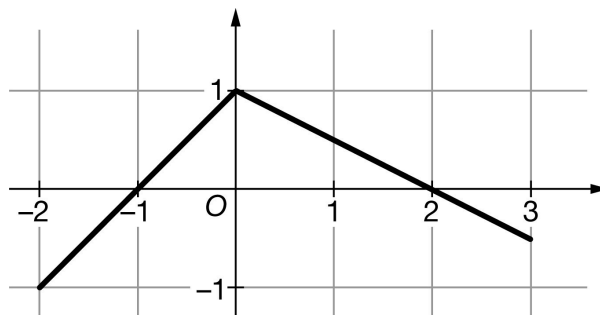
The figure above shows the graph of the continuous function g on the interval $[0, 8]$. Let h be the function defined by $h(x) = \int_3^x g(t) \, dt$. On what intervals is h increasing?

- (A) $[2, 5]$ only
(B) $[1, 7]$
(C) $[0, 1]$ and $[3, 7]$ ✓
(D) $[1, 3]$ and $[7, 8]$

Answer C

6-5

61.

Graph of f

The graph of the function f consists of two line segments, as shown in the figure above. Let g be the function defined by $g(x) = \int_{-2}^x f(t) \, dt$. At what value of x does g attain its global maximum on the closed interval $[-2, 3]$?

- (A) -2
- (B) -1
- (C) 0
- (D) 2

**Answer D**

Correct. The global maximum occurs at a critical point of g or at an endpoint. Since $g'(x) = f(x)$, the critical points on g in the interval $[-2, 3]$ are at $x = -1$ and $x = 2$ where $g'(x) = f(x) = 0$. The critical point at $x = -1$ is a local minimum of g , however, since the graph of $g' = f$ changes from negative to positive at $x = -1$. Therefore, the three candidates are only at $x = -2$, $x = 2$, and $x = 3$. Using the areas of triangles to evaluate the definite integrals shows the following.

$$g(-2) = \int_{-2}^{-2} f(t) \, dt = 0$$

$$g(2) = \int_{-2}^2 f(t) \, dt = -\frac{1}{2} + \frac{3}{2} = 1$$

$$g(3) = \int_{-2}^3 f(t) \, dt = -\frac{1}{2} + \frac{3}{2} - \frac{1}{4} = \frac{3}{4}$$

Therefore, g attains its global maximum at $x = 2$.

6-5

62. Let f be the function given by $f(x) = \int_6^x (-t^2 - t + 6) \, dt$. On which of the following intervals is f increasing?
- (A) $(-\infty, -\frac{1}{2}]$
- (B) $[-3, 2]$ ✓
- (C) $[-\frac{1}{2}, \infty)$
- (D) $(-\infty, -3]$ and $[2, \infty)$

Answer B

Correct. By the Fundamental Theorem of Calculus, $f'(x) = -x^2 - x + 6 = -(x+3)(x-2)$. The zeros of f' are at $x = -3$ and $x = 2$. Since the graph of f' is an inverted parabola, f' is positive on the interval $-3 < x < 2$. The function f is therefore increasing on the interval $[-3, 2]$.

63.  The function h is given by $h(x) = \int_{1.1}^x (2 + 2 \ln(\ln t) - (\ln t)^2) \, dt$ for $1.1 \leq x \leq 7$. On what intervals, if any, is h increasing?
- (A) $1.1 \leq x \leq 2.718$
- (B) $1.489 \leq x \leq 5.893$ ✓
- (C) $1.1 \leq x \leq 1.489$ and $5.893 \leq x \leq 7$
- (D) There are no intervals on which h is increasing.

Answer B

Correct. The function h will be increasing on intervals where $h'(x) \geq 0$. The second formulation of the Fundamental Theorem of Calculus says that

$$h'(x) = \frac{d}{dx} \left(\int_{1.1}^x (2 + 2 \ln(\ln t) - (\ln t)^2) \, dt \right) = 2 + 2 \ln(\ln x) - (\ln x)^2.$$

The calculator is used to graph this function over the interval $1.1 \leq x \leq 7$ and find the two zeros at 1.489 and 5.893. The graph of h' is above the x -axis between these two zeros.

6-5

64. Let f be the function given by $f(x) = \int_{10}^x (-t^2 + 2t + 3) dt$. On what intervals is f increasing?

(A) $(-\infty, 1]$

(B) $[-1, 3]$ ✓

(C) $[1, \infty)$

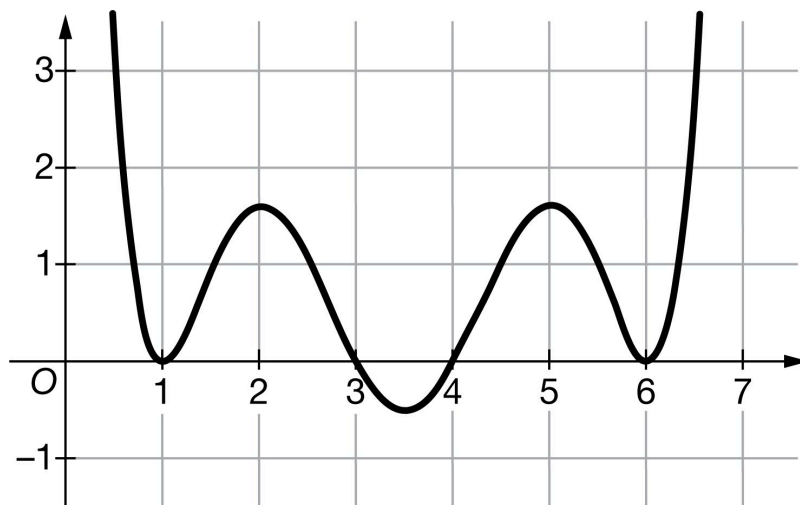
(D) $(-\infty, -1]$ and $[3, \infty)$

Answer B

Correct. By the Fundamental Theorem of Calculus, $f'(x) = -x^2 + 2x + 3 = -(x - 3)(x + 1)$. The zeros of f' are at $x = -1$ and $x = 3$. Since the graph of f' is an inverted parabola, f' is positive on the interval $-1 < x < 3$. The function f is therefore increasing on the interval $[-1, 3]$.

6-5

65.

Graph of f

The graph of the function f is shown above. Let g be the function defined by $g(x) = \int_1^x f(t) \, dt$. At what values of x in the interval $0.5 < x < 6.5$ does g have a relative maximum?

- (A) 3 only
(B) 4 only
(C) 2 and 5
(D) 1, 3, 4, and 6

**Answer A**

Correct. By the Fundamental Theorem of Calculus, $g'(x) = f(x)$. Since $g'(x) = f(x)$ changes sign from positive to negative at $x = 3$, this is the location of a relative maximum of g .

6-5

66.  The function h is given by $h(x) = \int_1^x \ln(t \sin t + 5) dt$ for $1 \leq x \leq 7$. On what intervals, if any, is h decreasing?
- (A) $2.029 \leq x \leq 4.413$
- (B) $4.323 \leq x \leq 5.461$ ✓
- (C) $1 \leq x \leq 4.322$ and $5.461 \leq x \leq 7$
- (D) There are no intervals on which h is decreasing.

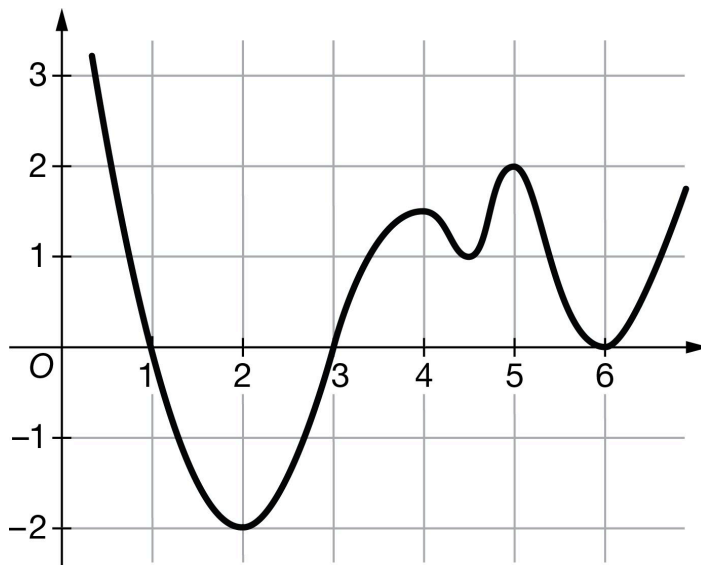
Answer B

Correct. The function h will be decreasing on intervals where $h'(x) \leq 0$. The second formulation of the Fundamental Theorem of Calculus says that $h'(x) = \frac{d}{dx} \left(\int_1^x \ln(t \sin t + 5) dt \right) = \ln(x \sin x + 5)$.

The calculator is used to graph this function over the interval $1 \leq x \leq 7$ and find the two zeros at $x = 4.323$ and $x = 5.461$. The graph of h' is below the x -axis between these two zeros.

6-5

67.

Graph of f

The graph of the function f is shown above. Let g be the function defined by $g(x) = \int_1^x f(t) \, dt$. At what values of x in the interval $0.5 < x < 6.5$ does g have a relative maximum?

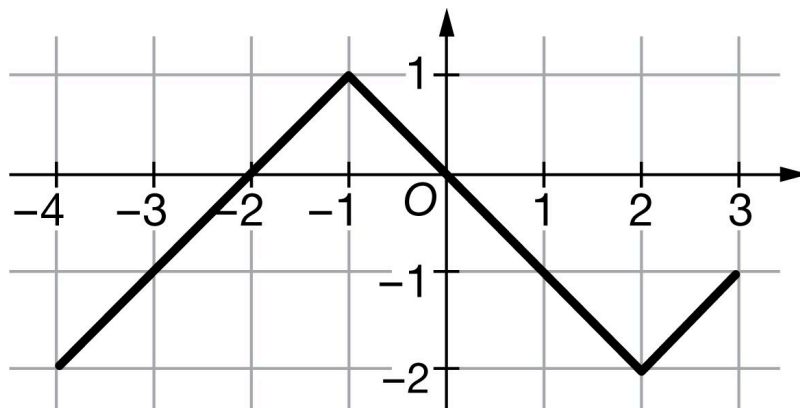
- (A) 1 only
(B) 3 only
(C) 4 and 5
(D) 1, 3, and 6

**Answer A**

Correct. By the Fundamental Theorem of Calculus, $g'(x) = f(x)$. Since $g'(x) = f(x)$ changes sign from positive to negative at $x = 1$, this is the location of a relative maximum of g .

6-5

68.

Graph of f

The graph of the function f shown above consists of three line segments. Let h be the function defined by $h(x) = \int_0^x f(t) \, dt$.

At what value of x does h attain its absolute maximum on the interval $[-4, 3]$?

(A) -4 (B) -2 (C) 0 (D) 3 **Answer A**

Correct. Because h is continuous, the Extreme Value Theorem guarantees the existence of an absolute maximum on the closed interval $[-4, 3]$ and that the maximum will occur at a critical value or at one of the endpoints. Since h is an antiderivative of f , $h'(x) = f(x) = 0$ at $x = -2$ and $x = 0$. Therefore, the candidates are $x = -4$, $x = -2$, $x = 0$, and $x = 3$. Evaluate h at each candidate and select the largest.

$$h(-4) = \int_0^{-4} f(t) \, dt = - \int_{-4}^0 f(t) \, dt = - \left(-\frac{1}{2}(2)(2) + \frac{1}{2}(2)(1) \right) = -(-2 + 1) = 1$$

$$h(-2) = \int_0^{-2} f(t) \, dt = - \int_{-2}^0 f(t) \, dt = - \left(\frac{1}{2}(2)(1) \right) = -1$$

$$h(0) = \int_0^0 f(t) \, dt = 0$$

$$h(3) = \int_0^3 f(t) \, dt = - \left(\frac{1}{2}(2)(2) + \frac{1}{2}(2+1) \right) = -(2 + 1.5) = -3.5$$

The maximum occurs at $x = -4$.

6-5

69.  Let g be the function defined by $g(x) = \int_0^x \left(-\frac{3}{4} + t + \cos\left(\frac{\pi}{4}t^2 + t\right) \right) dt$ for $0 < x < 3$. At what value of x does g attain a relative maximum?

(A) 1.025



(B) 1.361

(C) 1.616

(D) 2.310

Answer A

Correct. By the second formulation of the Fundamental Theorem of Calculus,

$g'(x) = \frac{d}{dx} \left(\int_0^x \left(-\frac{3}{4} + t + \cos\left(\frac{\pi}{4}t^2 + t\right) \right) dt \right) = -\frac{3}{4} + x + \cos\left(\frac{\pi}{4}x^2 + x\right)$. Use the calculator to examine the graph of g' on the interval $0 < x < 3$. The function g will have a relative maximum where the derivative of g changes from positive to negative. This happens between 1 and 1.1. The calculator is used to find the zero of g' on this interval.

6-5

70.

x	2	6	7	9	11	13
$h(x)$	16	-8	-9	-5	7	27
$h'(x)$	-10	-2	0	4	8	12

Selected values of the differentiable function h and its first derivative h' are given in the table above. Let f be the function defined by $f(x) = \int_0^x h(t) \, dt$. What is the least possible number of critical points for f in the interval $2 \leq x \leq 13$?

(A) Zero

(B) One

(C) Two

(D) It cannot be determined from the information given.

**Answer C**

Correct. Since $f(x) = \int_0^x h(t) \, dt$, it follows from the Fundamental Theorem of Calculus that $f'(x) = h(x)$.

Since h is differentiable, it is continuous. The Intermediate Value Theorem guarantees that $f'(c_1) = h(c_1) = 0$ for some c_1 in the interval $(2, 6)$ since $h(6) < 0 < h(2)$. The Intermediate Value Theorem also guarantees that $f'(c_2) = h(c_2) = 0$ for some c_2 in the interval $(9, 11)$ since $h(9) < 0 < h(11)$. Therefore, f has at least two critical points in the interval $2 \leq x \leq 13$.

6-5

71.  Let g be the function defined by $g(x) = \int_{-1}^x \left(-\frac{1}{2} + \cos(t^3 + 2t) \right) dt$ for $0 < x < \frac{\pi}{2}$. At what value of x does g attain a relative maximum?

(A) 0.471



(B) 1.028

(C) 1.360

(D) 1.489

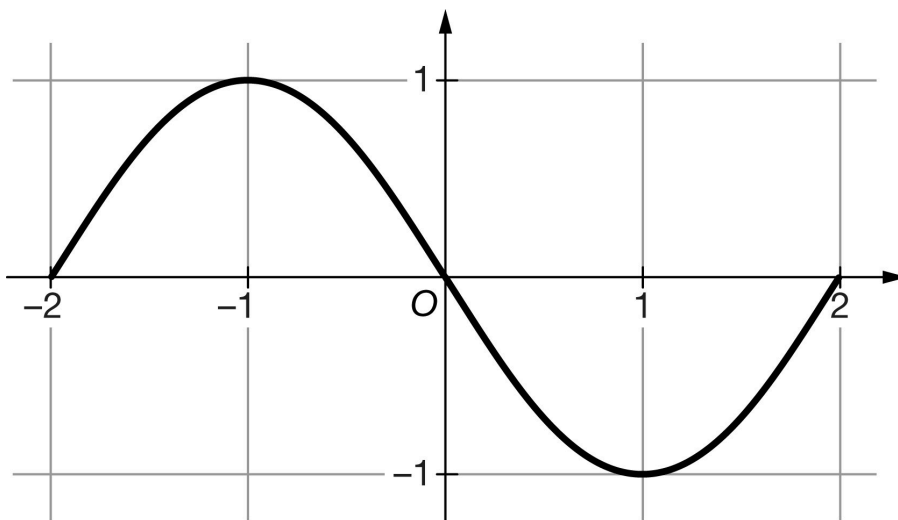
Answer A

Correct. By the second formulation of the Fundamental Theorem of Calculus,

$g'(x) = \frac{d}{dx} \left(\int_{-1}^x \left(-\frac{1}{2} + \cos(t^3 + 2t) \right) dt \right) = -\frac{1}{2} + \cos(x^3 + 2x)$. Use the calculator to examine the graph of g' on the interval $0 < x < \frac{\pi}{2}$. The function g will have a relative maximum, where the derivative of g changes from positive to negative. This happens between 0 and 1. The calculator is used to find the zero of g' on this interval.

6-5

72.

Graph of f

The graph of the function f defined on the closed interval $[-2, 2]$ is shown above. Let g be defined by $g(x) = \int_0^x f(t) \, dt$. On which of the following intervals is the graph of g both decreasing and concave up?

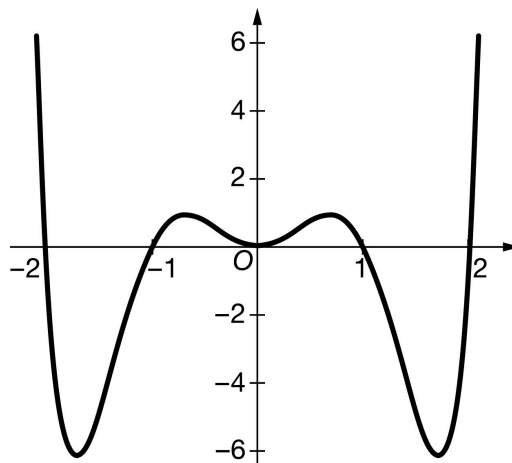
- (A) $(-2, -1)$
- (B) $(-1, 0)$
- (C) $(0, 1)$
- (D) $(1, 2)$

**Answer D**

Correct. The graph of g is decreasing and concave up where g' is negative and increasing. The second formulation of the Fundamental Theorem of Calculus says that $g'(x) = \frac{d}{dx} \left(\int_0^x f(t) \, dt \right) = f(x)$. On the interval $(1, 2)$, $g' = f$ is negative and increasing.

6-5

73.

Graph of f

The graph of the function f is shown above. The graph of f has horizontal tangent lines at $x = -1.7$, $x = -0.7$, $x = 0$, $x = 0.7$, and $x = 1.7$. Let g be the function defined by $g(x) = \int_0^x f(t) \, dt$. Of the following intervals, on which is the graph of g both decreasing and concave up?

(A) $(-1, -0.7)$ (B) $(-0.7, 0)$ (C) $(1, 1.7)$ (D) $(1.7, 2)$ **Answer D**

Correct. The graph of g is decreasing and concave up where g' is negative and increasing. The second formulation of the Fundamental Theorem of Calculus says that $g'(x) = \frac{d}{dx} \left(\int_0^x f(t) \, dt \right) = f(x)$. On the interval $(1.7, 2)$, $g' = f$ is negative and increasing.

6-5

74.

x	0	1	2	3	4	5
$h(x)$	1.08	-0.54	-0.14	1.82	4.56	7.18
$h'(x)$	-2.70	-0.56	1.30	2.49	2.82	2.26

Selected values of the differentiable function h and its continuous first derivative h' are given in the table above. Let f be the function defined by $f(x) = \int_0^x h(t) \, dt$. What is the least possible number of critical points for f in the interval $0 \leq x \leq 5$?

(A) Zero

(B) One

(C) Two

(D) It cannot be determined from the information given.

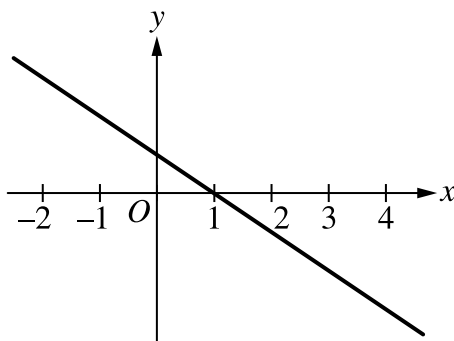
**Answer C**

Correct. Since $f(x) = \int_0^x h(t) \, dt$, it follows from the Fundamental Theorem of Calculus that $f'(x) = h(x)$.

Since h is differentiable, it is continuous. The Intermediate Value Theorem guarantees that $f'(c_1) = h(c_1) = 0$ for some c_1 in the interval $(0, 1)$, since $h(0) > 0 > h(1)$. The Intermediate Value Theorem also guarantees that $f'(c_2) = h(c_2) = 0$ for some c_2 in the interval $(2, 3)$, since $h(2) < 0 < h(3)$. Therefore, f has at least two critical points in the interval $(0, 5)$.

6-5

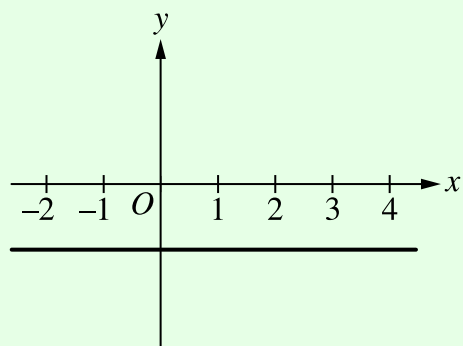
75.

Graph of g

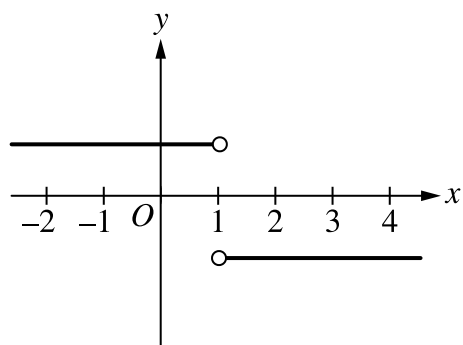
The preceding figure shows the graph of g . If $g(x) = \int_1^x f(t) \, dt$, which of the following could be the graph of $y = f(x)$?

6-5

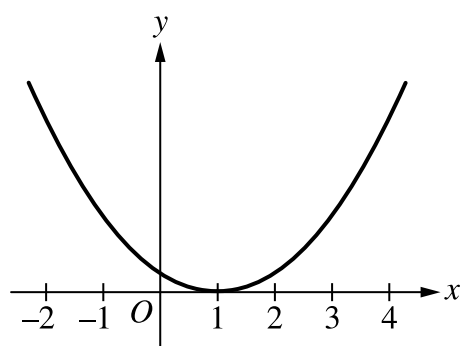
(A)



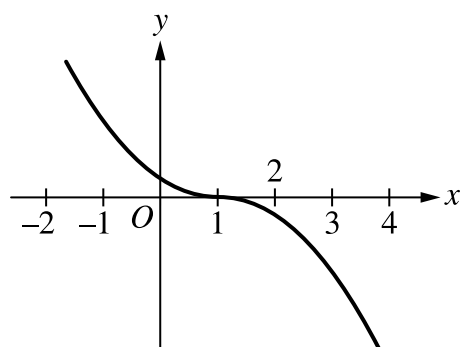
(B)



(C)



(D)

**Answer A**

Correct. Because g is decreasing on the interval $[-2, 4]$, $f(x) = g'(x) < 0$ for $-2 < x < 4$. In addition, the graph of g suggests that g is a linear function. Therefore, $f(x) = g'(x)$ would be constant for all x . This is the only

6-5

graph that satisfies those conditions.

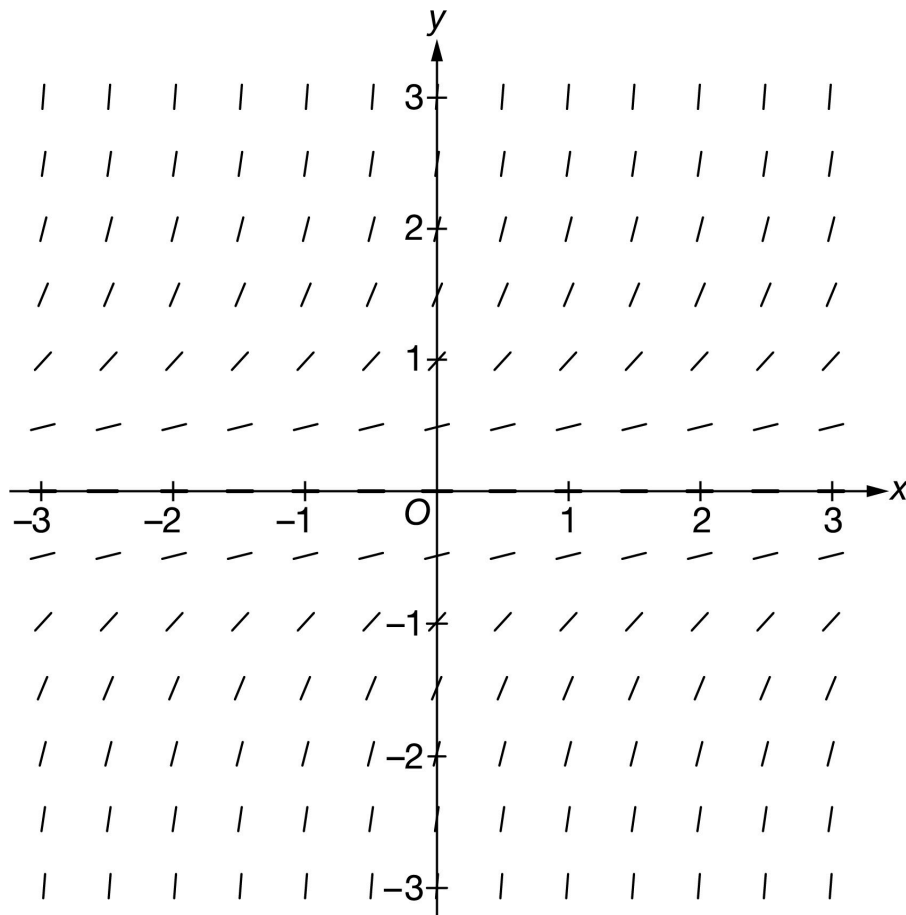
76.  The function f is defined by $f(x) = \int_{-4}^x \left(\frac{t^2}{6} - 2 \sin(e^t) \right) dt$. How many relative extrema does f have on the open interval $-4 < x < 2$?
- (A) One
(B) Two
(C) Three
(D) Four

Answer C

Correct. The derivative of the function f is $f'(x) = \frac{x^2}{6} - 2 \sin(e^x)$. The graph of f' has three zeros in the interval $-4 < x < 2$, where the derivative changes sign. Each of the zeros of f' is the location of a relative extrema of f .

6-5

77.



Shown is a slope field for which of the following differential equations?

(A) $\frac{dy}{dx} = |x + y|$

(B) $\frac{dy}{dx} = x^3$

(C) $\frac{dy}{dx} = y^3$

(D) $\frac{dy}{dx} = y^2$



Answer D

Correct. The segments in the slope field suggest that (1) the slopes depend only on the variable y , (2) the slopes are nonnegative, and (3) the slopes are zero when $y = 0$. The differential equation $\frac{dy}{dx} = y^2$ satisfies all three conditions. The differential equation $\frac{dy}{dx} = |x + y|$ does not satisfy conditions (1) and (3). The differential equation $\frac{dy}{dx} = x^3$ does not satisfy any of the conditions. The differential equation $\frac{dy}{dx} = y^3$ does not satisfy condition (2) when $y < 0$.

6-5

78.



t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

(a) Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

(b) Use the data in the table to evaluate $\int_0^{20} W'(t) dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t) dt$ in the context of this problem.

(c) For $0 \leq t \leq 20$, the average temperature of the water in the tub is $\frac{1}{20} \int_0^{20} W(t) dt$. Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\frac{1}{20} \int_0^{20} W(t) dt$. Does this approximation overestimate or underestimate the average temperature of the water over these 20 minutes? Explain your reasoning.

(d) For $20 \leq t \leq 25$, the function W that models the water temperature has first derivative given by $W'(t) = 0.4\sqrt{t} \cos(0.06t)$. Based on the model, what is the temperature of the water at time $t = 25$?

Part A

1 point is earned for the estimation

1 point is earned for the interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

$$= 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.



0	1	2
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The student response earns all of the following points:

1 point is earned for the estimation

1 point is earned for the interpretation with units

6-5

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

$$= 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.

Part B

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.

Part C

1 point is earned for the left Riemann sum

1 point is earned for the approximation

1 point is earned for underestimate with reason

6-5

$$\begin{aligned}
 \frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\
 &= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\
 &= \frac{1}{20} \cdot 1215.8 = 60.79
 \end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the left Riemann sum

1 point is earned for the approximation

1 point is earned for underestimate with reason

$$\begin{aligned}
 \frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\
 &= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\
 &= \frac{1}{20} \cdot 1215.8 = 60.79
 \end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.

Part D

1 point is earned for the integrand

1 point is earned for the answer

$$W(25) = 71.0 + \int_{20}^{25} W'(t) dt$$

$$= 71.0 + 2.043155 = 73.043$$

6-5



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the answer

$$W(25) = 71.0 + \int_{20}^{25} W'(t) dt$$

$$= 71.0 + 2.043155 = 73.043$$